

REPORT
ASIA 2022 Results
NetworkX modularity: 0.5494

Cluster 1 - Node: 710, Edge: 5961
Cluster 0 - Node : 997, Edge: 8797
Cluster 2 - Node : 411, Edge: 2973
Cluster 3 - Node : 55, Edge: 264
Cluster 4 - Node : 53, Edge: 263
Cluster 5 - Node : 28, Edge: 103

Cluster 0 Full-Stack Web and Mobile Development Competencies Network

Degree Centrality - Top 5:

1. 0: ['JavaScript']: 0.3540
2. 2: ['Web Development']: 0.2708
3. 20: ['Python']: 0.2412
4. 12: ['HTML']: 0.2130
5. 6: ['Web Design']: 0.2003

Betweenness Centrality - Top 5:

1. 0: ['JavaScript']: 0.1630
2. 20: ['Python']: 0.1470
3. 2: ['Web Development']: 0.0793
4. 6: ['Web Design']: 0.0523
5. 21: ['Java']: 0.0482

Closeness Centrality - Top 5:

1. 0: ['JavaScript']: 0.5845
2. 2: ['Web Development']: 0.5379
3. 20: ['Python']: 0.5311
4. 12: ['HTML']: 0.5134
5. 16: ['PHP']: 0.5119

PageRank - Top 5:

1. 0: ['JavaScript']: 0.0177
2. 2: ['Web Development']: 0.0136
3. 20: ['Python']: 0.0136
4. 12: ['HTML']: 0.0104
5. 6: ['Web Design']: 0.0101

Authority (HITS) - Top 5:

1. 4: ['Android']: 0.0588
2. 0: ['JavaScript']: 0.0550
3. 7: ['Mobile App Development']: 0.0531
4. 10: ['Android App Development']: 0.0501
5. 2: ['Web Development']: 0.0498

Eigenvector Centrality - Top 5:

1. 0: ['JavaScript']: 0.2388
2. 2: ['Web Development']: 0.1977
3. 40: ['API Integration']: 0.1659
4. 12: ['HTML']: 0.1641
5. 39: ['Node.js']: 0.1616

Cluster 1 Professional Written Communication and Research Expertise Network

Degree Centrality - Top 5:

1. 1: ['English']: 0.3253
2. 11: ['Researcher']: 0.2098
3. 5: ['Content Writing']: 0.2048
4. 15: ['Data Entry']: 0.1938
5. 9: ['Social Media Marketing']: 0.1627

Betweenness Centrality - Top 5:

1. 1: ['English']: 0.2159
2. 15: ['Data Entry']: 0.0761
3. 11: ['Researcher']: 0.0729
4. 5: ['Content Writing']: 0.0708
5. 23: ['Translation']: 0.0596

Closeness Centrality - Top 5:

1. 1: ['English']: 0.5761
2. 5: ['Content Writing']: 0.5378
3. 11: ['Researcher']: 0.5349
4. 15: ['Data Entry']: 0.5309
5. 37: ['Virtual Assistant']: 0.5137

PageRank - Top 5:

1. 1: ['English']: 0.0179
2. 11: ['Researcher']: 0.0102
3. 5: ['Content Writing']: 0.0101
4. 15: ['Data Entry']: 0.0095
5. 23: ['Translation']: 0.0089

Authority (HITS) - Top 5:

1. 5: ['Content Writing']: 0.0539
2. 1: ['English']: 0.0436
3. 35: ['Article Writing']: 0.0351
4. 11: ['Researcher']: 0.0315
5. 31: ['Writing']: 0.0296

Eigenvector Centrality - Top 5:

1. 1: ['English']: 0.2059
2. 11: ['Researcher']: 0.1991
3. 5: ['Content Writing']: 0.1857
4. 15: ['Data Entry']: 0.1717
5. 9: ['Social Media Marketing']: 0.1591

Cluster 2 Design and Digital Creation Core Skills

Degree Centrality - Top 5:

1. 3: ['Graphic Design']: 0.4634
2. 13: ['Adobe Photoshop']: 0.3366
3. 26: ['Adobe Illustrator']: 0.3220
4. 57: ['3D Modeling']: 0.2268
5. 76: ['3D Design']: 0.1927

Betweenness Centrality - Top 5:

1. 3: ['Graphic Design']: 0.3255
2. 13: ['Adobe Photoshop']: 0.1414
3. 26: ['Adobe Illustrator']: 0.1184

- 4. 57: ['3D Modeling']: 0.0944
- 5. 76: ['3D Design']: 0.0652

Closeness Centrality - Top 5:

- 1. 3: ['Graphic Design']: 0.6337
- 2. 13: ['Adobe Photoshop']: 0.5734
- 3. 26: ['Adobe Illustrator']: 0.5694
- 4. 56: ['Illustration']: 0.5170
- 5. 57: ['3D Modeling']: 0.5151

PageRank - Top 5:

- 1. 3: ['Graphic Design']: 0.0299
- 2. 13: ['Adobe Photoshop']: 0.0199
- 3. 26: ['Adobe Illustrator']: 0.0195
- 4. 57: ['3D Modeling']: 0.0144
- 5. 76: ['3D Design']: 0.0128

Authority (HITS) - Top 5:

- 1. 3: ['Graphic Design']: 0.0833
- 2. 13: ['Adobe Photoshop']: 0.0743
- 3. 26: ['Adobe Illustrator']: 0.0679
- 4. 56: ['Illustration']: 0.0373
- 5. 46: ['Logo Design']: 0.0368

Eigenvector Centrality - Top 5:

- 1. 3: ['Graphic Design']: 0.2935
- 2. 13: ['Adobe Photoshop']: 0.2542
- 3. 26: ['Adobe Illustrator']: 0.2287
- 4. 54: ['Adobe After Effects']: 0.1626
- 5. 32: ['Video Production']: 0.1539

Cluster 3: Electronic Systems and Circuit Design Core Competencies

Degree Centrality - Top 5:

- 1. 213: ['Electronics']: 0.6481
- 2. 153: ['Electrical Engineering']: 0.5741
- 3. 231: ['Electronic Design']: 0.5556
- 4. 217: ['PCB Design']: 0.4815
- 5. 401: ['Embedded System']: 0.4259

Betweenness Centrality - Top 5:

- 1. 213: ['Electronics']: 0.2159
- 2. 231: ['Electronic Design']: 0.1820
- 3. 153: ['Electrical Engineering']: 0.1818
- 4. 217: ['PCB Design']: 0.1235
- 5. 401: ['Embedded System']: 0.1010

Closeness Centrality - Top 5:

- 1. 213: ['Electronics']: 0.7105
- 2. 231: ['Electronic Design']: 0.6923
- 3. 153: ['Electrical Engineering']: 0.6750
- 4. 217: ['PCB Design']: 0.6585
- 5. 401: ['Embedded System']: 0.6136

PageRank - Top 5:

1. 213: ['Electronics']: 0.0607
2. 153: ['Electrical Engineering']: 0.0565
3. 231: ['Electronic Design']: 0.0503
4. 217: ['PCB Design']: 0.0439
5. 401: ['Embedded System']: 0.0402

Authority (HITS) - Top 5:

1. 213: ['Electronics']: 0.0936
2. 153: ['Electrical Engineering']: 0.0926
3. 231: ['Electronic Design']: 0.0843
4. 217: ['PCB Design']: 0.0703
5. 400: ['Circuit Design']: 0.0557

Eigenvector Centrality - Top 5:

1. 213: ['Electronics']: 0.3242
2. 231: ['Electronic Design']: 0.3048
3. 153: ['Electrical Engineering']: 0.2911
4. 217: ['PCB Design']: 0.2787
5. 401: ['Embedded System']: 0.2458

Cluster 4 Software Testing and Quality Assurance Core Competencies

Degree Centrality - Top 5:

1. 263: ['Functional Testing']: 0.5385
2. 305: ['Software QA']: 0.4615
3. 208: ['Mobile App Testing']: 0.4423
4. 268: ['Manual Testing']: 0.4038
5. 209: ['Web Testing']: 0.4038

Betweenness Centrality - Top 5:

1. 263: ['Functional Testing']: 0.1997
2. 360: ['Requirement Analysis']: 0.1813
3. 305: ['Software QA']: 0.1699
4. 268: ['Manual Testing']: 0.1013
5. 393: ['Product Roadmap']: 0.0961

Closeness Centrality - Top 5:

1. 263: ['Functional Testing']: 0.6500
2. 305: ['Software QA']: 0.6500
3. 209: ['Web Testing']: 0.5977
4. 268: ['Manual Testing']: 0.5843
5. 360: ['Requirement Analysis']: 0.5843

PageRank - Top 5:

1. 263: ['Functional Testing']: 0.0453
2. 305: ['Software QA']: 0.0403
3. 208: ['Mobile App Testing']: 0.0385
4. 268: ['Manual Testing']: 0.0381
5. 393: ['Product Roadmap']: 0.0353

Authority (HITS) - Top 5:

1. 209: ['Web Testing']: 0.0901
2. 263: ['Functional Testing']: 0.0870
3. 268: ['Manual Testing']: 0.0823
4. 208: ['Mobile App Testing']: 0.0820
5. 439: ['Test Case Design']: 0.0672

Eigenvector Centrality - Top 5:

1. 263: ['Functional Testing']: 0.3090
2. 305: ['Software QA']: 0.2808
3. 208: ['Mobile App Testing']: 0.2744
4. 209: ['Web Testing']: 0.2705
5. 268: ['Manual Testing']: 0.2561

Cluster 5 Basic Legal Services and Contract Management Centrer

Degree Centrality - Top 5:

1. 267: ['Legal Consulting']: 0.7778
2. 245: ['Legal']: 0.6667
3. 252: ['Contract Law']: 0.5185
4. 365: ['Legal Writing']: 0.4815
5. 841: ['Legal Assistance']: 0.3704

Betweenness Centrality - Top 5:

1. 267: ['Legal Consulting']: 0.3162
2. 245: ['Legal']: 0.2724
3. 365: ['Legal Writing']: 0.1032
4. 657: ['Intellectual Property Law']: 0.0633
5. 252: ['Contract Law']: 0.0560

Closeness Centrality - Top 5:

1. 267: ['Legal Consulting']: 0.8182
2. 245: ['Legal']: 0.7500
3. 252: ['Contract Law']: 0.6750
4. 365: ['Legal Writing']: 0.6585
5. 657: ['Intellectual Property Law']: 0.6000

PageRank - Top 5:

1. 267: ['Legal Consulting']: 0.0931
2. 245: ['Legal']: 0.0864
3. 252: ['Contract Law']: 0.0614
4. 365: ['Legal Writing']: 0.0571
5. 841: ['Legal Assistance']: 0.0462

Authority (HITS) - Top 5:

1. 252: ['Contract Law']: 0.1351
2. 267: ['Legal Consulting']: 0.1277
3. 245: ['Legal']: 0.1056
4. 495: ['Contract Drafting']: 0.0872
5. 365: ['Legal Writing']: 0.0803

Eigenvector Centrality - Top 5:

1. 267: ['Legal Consulting']: 0.3976
2. 252: ['Contract Law']: 0.3254
3. 245: ['Legal']: 0.3176
4. 365: ['Legal Writing']: 0.3034
5. 495: ['Contract Drafting']: 0.2428

America 2022
NetworkX modularity: 0.4530

Cluster 2 - Node : 588, Edge: 4772
Cluster 0 - Node : 1212, Edge: 17481
Cluster 1 - Node : 877, Edge: 7433

Cluster 0 Professional Digital Design and Visualisation Expertise Network

Degree Centrality - Top 5:

1. 0: ['Graphic Design']: 0.3925
2. 15: ['Adobe Photoshop']: 0.3089
3. 24: ['Adobe Illustrator']: 0.2560
4. 38: ['Illustration']: 0.2338
5. 13: ['Video Editing']: 0.1826
6. 70: ['3D Modeling']: 0.1604
7. 26: ['Video Production']: 0.1587
8. 36: ['Logo Design']: 0.1553
9. 90: ['Animation']: 0.1451
10. 59: ['Photo Editing']: 0.1399
11. 141: ['Character Design']: 0.1382
12. 92: ['3D Rendering']: 0.1348
13. 96: ['3D Design']: 0.1331
14. 60: ['Adobe After Effects']: 0.1280
15. 131: ['Drawing']: 0.1229
16. 51: ['Video Post-Editing']: 0.1143
17. 100: ['Audio Editing']: 0.1109
18. 135: ['2D Animation']: 0.1092
19. 66: ['Adobe Premiere Pro']: 0.1024
20. 101: ['Layout Design']: 0.1024

Betweenness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.2613
2. 15: ['Adobe Photoshop']: 0.1450
3. 70: ['3D Modeling']: 0.0846
4. 38: ['Illustration']: 0.0723
5. 24: ['Adobe Illustrator']: 0.0688
6. 100: ['Audio Editing']: 0.0622
7. 13: ['Video Editing']: 0.0600
8. 96: ['3D Design']: 0.0520
9. 301: ['Product Design']: 0.0502
10. 26: ['Video Production']: 0.0479
11. 92: ['3D Rendering']: 0.0433
12. 60: ['Adobe After Effects']: 0.0298
13. 90: ['Animation']: 0.0298
14. 59: ['Photo Editing']: 0.0287
15. 424: ['Construction']: 0.0251
16. 382: ['Concept Design']: 0.0230
17. 223: ['Clothing']: 0.0222
18. 145: ['Autodesk AutoCAD']: 0.0222
19. 141: ['Character Design']: 0.0200
20. 130: ['Architectural Design']: 0.0196

Closeness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.5872
2. 15: ['Adobe Photoshop']: 0.5555

3. 24: ['Adobe Illustrator']: 0.5218
4. 38: ['Illustration']: 0.5158
5. 96: ['3D Design']: 0.4920
6. 92: ['3D Rendering']: 0.4871
7. 13: ['Video Editing']: 0.4867
8. 70: ['3D Modeling']: 0.4851
9. 26: ['Video Production']: 0.4807
10. 141: ['Character Design']: 0.4737
11. 90: ['Animation']: 0.4733
12. 91: ['Motion Graphics']: 0.4684
13. 60: ['Adobe After Effects']: 0.4662
14. 183: ['3D Animation']: 0.4636
15. 51: ['Video Post-Editing']: 0.4632
16. 59: ['Photo Editing']: 0.4632
17. 114: ['Adobe InDesign']: 0.4611
18. 382: ['Concept Design']: 0.4550
19. 134: ['Image Editing']: 0.4539
20. 101: ['Layout Design']: 0.4536

PageRank - Top 5:

1. 0: ['Graphic Design']: 0.0221
2. 15: ['Adobe Photoshop']: 0.0160
3. 24: ['Adobe Illustrator']: 0.0137
4. 38: ['Illustration']: 0.0119
5. 13: ['Video Editing']: 0.0100
6. 70: ['3D Modeling']: 0.0097
7. 26: ['Video Production']: 0.0085
8. 36: ['Logo Design']: 0.0081
9. 96: ['3D Design']: 0.0075
10. 92: ['3D Rendering']: 0.0074
11. 59: ['Photo Editing']: 0.0074
12. 90: ['Animation']: 0.0070
13. 100: ['Audio Editing']: 0.0066
14. 60: ['Adobe After Effects']: 0.0065
15. 141: ['Character Design']: 0.0064
16. 51: ['Video Post-Editing']: 0.0060
17. 145: ['Autodesk AutoCAD']: 0.0058
18. 131: ['Drawing']: 0.0058
19. 101: ['Layout Design']: 0.0057
20. 130: ['Architectural Design']: 0.0055

Authority (HITS) - Top 5:

1. 0: ['Graphic Design']: 0.0814
2. 15: ['Adobe Photoshop']: 0.0652
3. 24: ['Adobe Illustrator']: 0.0640
4. 36: ['Logo Design']: 0.0399
5. 38: ['Illustration']: 0.0342
6. 13: ['Video Editing']: 0.0284
7. 26: ['Video Production']: 0.0242
8. 89: ['Social Media Imagery']: 0.0237
9. 105: ['Brand Identity & Guidelines']: 0.0216
10. 114: ['Adobe InDesign']: 0.0201
11. 60: ['Adobe After Effects']: 0.0195
12. 59: ['Photo Editing']: 0.0190
13. 104: ['Print Design']: 0.0179
14. 116: ['Infographic']: 0.0174

15. 51: ['Video Post-Editing']: 0.0172
16. 66: ['Adobe Premiere Pro']: 0.0157
17. 101: ['Layout Design']: 0.0146
18. 91: ['Motion Graphics']: 0.0125
19. 55: ['Videography']: 0.0117
20. 90: ['Animation']: 0.0107

Eigenvector Centrality - Top 5:

1. 0: ['Graphic Design']: 0.2618
2. 15: ['Adobe Photoshop']: 0.2331
3. 38: ['Illustration']: 0.2105
4. 24: ['Adobe Illustrator']: 0.2083
5. 141: ['Character Design']: 0.1510
6. 90: ['Animation']: 0.1483
7. 131: ['Drawing']: 0.1370
8. 36: ['Logo Design']: 0.1326
9. 135: ['2D Animation']: 0.1301
10. 284: ['Cartoon Art']: 0.1246
11. 348: ['Contemporary Style']: 0.1243
12. 13: ['Video Editing']: 0.1182
13. 59: ['Photo Editing']: 0.1120
14. 580: ['Freestyle Drawing']: 0.1106
15. 26: ['Video Production']: 0.1090
16. 474: ['Colorful Style']: 0.1078
17. 60: ['Adobe After Effects']: 0.1059
18. 518: ['2D Illustration']: 0.1055
19. 116: ['Infographic']: 0.1035
20. 497: ['Freehand Digital Illustration']: 0.1009

Cluster 1 : Network of Digital Content and Administrative Support Specialisations

Degree Centrality - Top 5:

1. 1: ['Content Writing']: 0.3287
2. 4: ['English']: 0.3047
3. 8: ['Researcher']: 0.2998
4. 3: ['Data Entry']: 0.2725
5. 9: ['Virtual Assistant']: 0.2494

Betweenness Centrality - Top 5:

1. 4: ['English']: 0.1233
2. 1: ['Content Writing']: 0.0959
3. 8: ['Researcher']: 0.0722
4. 3: ['Data Entry']: 0.0648
5. 9: ['Virtual Assistant']: 0.0408

Closeness Centrality - Top 5:

1. 1: ['Content Writing']: 0.5881
2. 4: ['English']: 0.5767
3. 8: ['Researcher']: 0.5742
4. 3: ['Data Entry']: 0.5704
5. 9: ['Virtual Assistant']: 0.5545

PageRank - Top 5:

1. 4: ['English']: 0.0110

2. 1: ['Content Writing']: 0.0104
3. 8: ['Researcher']: 0.0096
4. 3: ['Data Entry']: 0.0088
5. 9: ['Virtual Assistant']: 0.0076

Authority (HITS) - Top 5:

1. 1: ['Content Writing']: 0.0336
2. 2: ['Social Media Marketing']: 0.0267
3. 8: ['Researcher']: 0.0234
4. 9: ['Virtual Assistant']: 0.0207
5. 4: ['English']: 0.0204

Eigenvector Centrality - Top 5:

1. 1: ['Content Writing']: 0.1736
2. 8: ['Researcher']: 0.1628
3. 3: ['Data Entry']: 0.1431
4. 4: ['English']: 0.1401
5. 23: ['Administrative Support']: 0.1388

Cluster 2 Web Technologies and Software Development Expertise Network

Degree Centrality - Top 5:

1. 11: ['JavaScript']: 0.3333
2. 5: ['Web Development']: 0.2329
3. 46: ['API']: 0.2169
4. 27: ['HTML']: 0.2123
5. 40: ['Python']: 0.1963

Betweenness Centrality - Top 5:

1. 11: ['JavaScript']: 0.1859
2. 40: ['Python']: 0.1193
3. 10: ['WordPress']: 0.0618
4. 46: ['API']: 0.0617
5. 5: ['Web Development']: 0.0535

Closeness Centrality - Top 5:

1. 11: ['JavaScript']: 0.5786
2. 40: ['Python']: 0.5249
3. 46: ['API']: 0.5144
4. 27: ['HTML']: 0.5141
5. 5: ['Web Development']: 0.5123

PageRank - Top 5:

1. 11: ['JavaScript']: 0.0165
2. 5: ['Web Development']: 0.0111
3. 40: ['Python']: 0.0110
4. 46: ['API']: 0.0107
5. 27: ['HTML']: 0.0099

Authority (HITS) - Top 5:

1. 5: ['Web Development']: 0.0822
2. 6: ['Web Design']: 0.0719
3. 10: ['WordPress']: 0.0675
4. 27: ['HTML']: 0.0598
5. 29: ['CSS']: 0.0566

Eigenvector Centrality - Top 5:

1. 11: ['JavaScript']: 0.2494
2. 5: ['Web Development']: 0.2064
3. 27: ['HTML']: 0.1978
4. 46: ['API']: 0.1845
5. 29: ['CSS']: 0.1686

Europe 2022

NetworkX modularity: 0.5295

Cluster 1: 760 düğüm

Cluster 2: 440 düğüm

Cluster 0: 778 düğüm

Cluster 2

Full-Stack Web Development and API-Centric Software Engineering Skill Cluster

Degree Centrality - Top 5:

1. 5: ['JavaScript']: 0.2833
2. 3: ['Web Development']: 0.2232
3. 28: ['Python']: 0.1945
4. 49: ['API Integration']: 0.1854
5. 34: ['API']: 0.1762
6. 69: ['Java']: 0.1684
7. 18: ['PHP']: 0.1658
8. 14: ['CSS']: 0.1580
9. 19: ['HTML']: 0.1567
10. 7: ['WordPress']: 0.1475
11. 6: ['Web Design']: 0.1371
12. 56: ['Web Application']: 0.1371
13. 98: ['SQL']: 0.1305
14. 93: ['Node.js']: 0.1292
15. 58: ['iOS']: 0.1227
16. 73: ['MySQL']: 0.1227
17. 76: ['Business with 1-9 Employees']: 0.1227
18. 51: ['Android']: 0.1097
19. 65: ['React']: 0.1044
20. 193: ['MongoDB']: 0.0966

Betweenness Centrality - Top 5:

1. 28: ['Python']: 0.1293
2. 5: ['JavaScript']: 0.1195
3. 3: ['Web Development']: 0.0770
4. 69: ['Java']: 0.0755
5. 173: ['C++']: 0.0712
6. 49: ['API Integration']: 0.0537
7. 7: ['WordPress']: 0.0463
8. 34: ['API']: 0.0421
9. 19: ['HTML']: 0.0399
10. 18: ['PHP']: 0.0370
11. 76: ['Business with 1-9 Employees']: 0.0368
12. 175: ['Linux']: 0.0364
13. 98: ['SQL']: 0.0339
14. 142: ['C#']: 0.0293
15. 93: ['Node.js']: 0.0288
16. 109: ['Data Analysis']: 0.0287

17. 104: ['Project Management']: 0.0274
18. 58: ['iOS']: 0.0269
19. 169: ['Microsoft Windows']: 0.0267
20. 14: ['CSS']: 0.0244

Closeness Centrality - Top 5:

1. 5: ['JavaScript']: 0.5394
2. 28: ['Python']: 0.5214
3. 3: ['Web Development']: 0.5197
4. 34: ['API']: 0.4987
5. 69: ['Java']: 0.4926
6. 18: ['PHP']: 0.4904
7. 49: ['API Integration']: 0.4882
8. 73: ['MySQL']: 0.4812
9. 7: ['WordPress']: 0.4803
10. 14: ['CSS']: 0.4770
11. 56: ['Web Application']: 0.4743
12. 93: ['Node.js']: 0.4740
13. 76: ['Business with 1-9 Employees']: 0.4688
14. 193: ['MongoDB']: 0.4614
15. 142: ['C#']: 0.4606
16. 19: ['HTML']: 0.4603
17. 98: ['SQL']: 0.4603
18. 236: ['Git']: 0.4595
19. 58: ['iOS']: 0.4584
20. 65: ['React']: 0.4573

PageRank - Top 5:

1. 5: ['JavaScript']: 0.0148
2. 28: ['Python']: 0.0118
3. 3: ['Web Development']: 0.0115
4. 49: ['API Integration']: 0.0095
5. 34: ['API']: 0.0091
6. 69: ['Java']: 0.0090
7. 19: ['HTML']: 0.0085
8. 18: ['PHP']: 0.0085
9. 7: ['WordPress']: 0.0082
10. 14: ['CSS']: 0.0080
11. 56: ['Web Application']: 0.0070
12. 98: ['SQL']: 0.0070
13. 6: ['Web Design']: 0.0068
14. 76: ['Business with 1-9 Employees']: 0.0067
15. 93: ['Node.js']: 0.0064
16. 58: ['iOS']: 0.0061
17. 73: ['MySQL']: 0.0061
18. 173: ['C++']: 0.0059
19. 51: ['Android']: 0.0056
20. 109: ['Data Analysis']: 0.0054

Authority (HITS) - Top 5:

1. 3: ['Web Development']: 0.0764
2. 14: ['CSS']: 0.0672
3. 5: ['JavaScript']: 0.0662
4. 19: ['HTML']: 0.0642
5. 7: ['WordPress']: 0.0634
6. 6: ['Web Design']: 0.0540

7. 18: ['PHP']: 0.0535
8. 64: ['WooCommerce']: 0.0213
9. 81: ['HTML5']: 0.0205
10. 34: ['API']: 0.0201
11. 56: ['Web Application']: 0.0176
12. 94: ['WordPress Plugin']: 0.0162
13. 73: ['MySQL']: 0.0161
14. 32: ['Shopify']: 0.0160
15. 55: ['Landing Page']: 0.0156
16. 49: ['API Integration']: 0.0139
17. 95: ['Website']: 0.0125
18. 65: ['React']: 0.0109
19. 28: ['Python']: 0.0089
20. 189: ['Custom PHP']: 0.0086

Eigenvector Centrality - Top 5:

1. 5: ['JavaScript']: 0.2388
2. 3: ['Web Development']: 0.2088
3. 18: ['PHP']: 0.1788
4. 34: ['API']: 0.1787
5. 49: ['API Integration']: 0.1751
6. 14: ['CSS']: 0.1719
7. 19: ['HTML']: 0.1632
8. 56: ['Web Application']: 0.1535
9. 7: ['WordPress']: 0.1486
10. 73: ['MySQL']: 0.1480
11. 6: ['Web Design']: 0.1381
12. 93: ['Node.js']: 0.1376
13. 28: ['Python']: 0.1375
14. 65: ['React']: 0.1308
15. 76: ['Business with 1-9 Employees']: 0.1308
16. 69: ['Java']: 0.1273
17. 81: ['HTML5']: 0.1234
18. 98: ['SQL']: 0.1197
19. 236: ['Git']: 0.1174
20. 58: ['iOS']: 0.1137

Cluster : 0 Digital Communication and Content Strategy Skill Cluster

Degree Centrality - Top 5:

1. 1: ['English']: 0.3900
2. 0: ['Content Writing']: 0.3478
3. 9: ['Researcher']: 0.2609
4. 4: ['Social Media Marketing']: 0.2424
5. 8: ['Lead Generation']: 0.2240
6. 25: ['Marketing Strategy']: 0.2200
7. 10: ['Copywriting']: 0.2082
8. 13: ['Search Engine Optimization']: 0.1950
9. 35: ['Market Research']: 0.1765
10. 24: ['Virtual Assistant']: 0.1739
11. 11: ['Facebook']: 0.1713
12. 15: ['Writing']: 0.1673
13. 23: ['Creative Writing']: 0.1621

14. 29: ['Translation']: 0.1607
15. 12: ['Data Entry']: 0.1594
16. 17: ['Instagram']: 0.1476
17. 44: ['German']: 0.1476
18. 20: ['Article Writing']: 0.1436
19. 37: ['Google Ads']: 0.1436
20. 21: ['Social Media Management']: 0.1410

Betweenness Centrality - Top 5:

1. 1: ['English']: 0.2100
2. 0: ['Content Writing']: 0.1156
3. 9: ['Researcher']: 0.0724
4. 8: ['Lead Generation']: 0.0587
5. 4: ['Social Media Marketing']: 0.0445
6. 35: ['Market Research']: 0.0411
7. 29: ['Translation']: 0.0401
8. 13: ['Search Engine Optimization']: 0.0355
9. 10: ['Copywriting']: 0.0334
10. 25: ['Marketing Strategy']: 0.0308
11. 24: ['Virtual Assistant']: 0.0306
12. 44: ['German']: 0.0299
13. 12: ['Data Entry']: 0.0223
14. 37: ['Google Ads']: 0.0217
15. 99: ['Native']: 0.0211
16. 63: ['Email Communication']: 0.0209
17. 15: ['Writing']: 0.0188
18. 53: ['Communications']: 0.0186
19. 11: ['Facebook']: 0.0181
20. 23: ['Creative Writing']: 0.0175

Closeness Centrality - Top 5:

1. 1: ['English']: 0.6053
2. 0: ['Content Writing']: 0.5920
3. 9: ['Researcher']: 0.5597
4. 8: ['Lead Generation']: 0.5500
5. 4: ['Social Media Marketing']: 0.5480
6. 10: ['Copywriting']: 0.5425
7. 25: ['Marketing Strategy']: 0.5379
8. 13: ['Search Engine Optimization']: 0.5356
9. 35: ['Market Research']: 0.5304
10. 24: ['Virtual Assistant']: 0.5282
11. 11: ['Facebook']: 0.5245
12. 44: ['German']: 0.5224
13. 12: ['Data Entry']: 0.5206
14. 23: ['Creative Writing']: 0.5199
15. 15: ['Writing']: 0.5188
16. 37: ['Google Ads']: 0.5153
17. 63: ['Email Communication']: 0.5111
18. 20: ['Article Writing']: 0.5101
19. 21: ['Social Media Management']: 0.5097
20. 38: ['Social Media Content']: 0.5094

PageRank - Top 5:

1. 1: ['English']: 0.0199
2. 0: ['Content Writing']: 0.0156
3. 9: ['Researcher']: 0.0117

4. 4: ['Social Media Marketing']: 0.0097
5. 8: ['Lead Generation']: 0.0094
6. 10: ['Copywriting']: 0.0090
7. 29: ['Translation']: 0.0087
8. 25: ['Marketing Strategy']: 0.0086
9. 13: ['Search Engine Optimization']: 0.0078
10. 35: ['Market Research']: 0.0077
11. 15: ['Writing']: 0.0075
12. 24: ['Virtual Assistant']: 0.0074
13. 23: ['Creative Writing']: 0.0070
14. 12: ['Data Entry']: 0.0068
15. 44: ['German']: 0.0068
16. 11: ['Facebook']: 0.0066
17. 20: ['Article Writing']: 0.0063
18. 63: ['Email Communication']: 0.0059
19. 99: ['Native']: 0.0058
20. 17: ['Instagram']: 0.0057

Authority (HITS) - Top 5:

1. 0: ['Content Writing']: 0.0678
2. 10: ['Copywriting']: 0.0445
3. 15: ['Writing']: 0.0421
4. 20: ['Article Writing']: 0.0420
5. 1: ['English']: 0.0367
6. 23: ['Creative Writing']: 0.0350
7. 9: ['Researcher']: 0.0338
8. 33: ['Blog Content']: 0.0315
9. 26: ['SEO Writing']: 0.0315
10. 40: ['Blog Writing']: 0.0290
11. 4: ['Social Media Marketing']: 0.0253
12. 13: ['Search Engine Optimization']: 0.0245
13. 54: ['Article']: 0.0229
14. 17: ['Instagram']: 0.0187
15. 11: ['Facebook']: 0.0184
16. 21: ['Social Media Management']: 0.0165
17. 75: ['Website Content']: 0.0130
18. 25: ['Marketing Strategy']: 0.0130
19. 38: ['Social Media Content']: 0.0126
20. 42: ['Social Media Content Creation']: 0.0108

Eigenvector Centrality - Top 5:

1. 0: ['Content Writing']: 0.1992
2. 1: ['English']: 0.1784
3. 4: ['Social Media Marketing']: 0.1768
4. 25: ['Marketing Strategy']: 0.1712
5. 9: ['Researcher']: 0.1622
6. 8: ['Lead Generation']: 0.1585
7. 13: ['Search Engine Optimization']: 0.1498
8. 10: ['Copywriting']: 0.1489
9. 11: ['Facebook']: 0.1444
10. 24: ['Virtual Assistant']: 0.1291
11. 17: ['Instagram']: 0.1257
12. 38: ['Social Media Content']: 0.1236
13. 12: ['Data Entry']: 0.1227
14. 23: ['Creative Writing']: 0.1224
15. 35: ['Market Research']: 0.1214

16. 37: ['Google Ads']: 0.1203
17. 21: ['Social Media Management']: 0.1149
18. 15: ['Writing']: 0.1148
19. 44: ['German']: 0.1148
20. 45: ['SEO Keyword Research']: 0.1119

Cluster 1 Creative Media Production and Visual Design Ecosystem

Degree Centrality - Top 5:

1. 2: ['Graphic Design']: 0.4351
2. 22: ['Adobe Photoshop']: 0.2893
3. 27: ['Adobe Illustrator']: 0.2642
4. 43: ['Illustration']: 0.2278
5. 16: ['Video Editing']: 0.2005
6. 36: ['Logo Design']: 0.1845
7. 82: ['3D Design']: 0.1686
8. 88: ['3D Modeling']: 0.1572
9. 52: ['Adobe After Effects']: 0.1503
10. 72: ['Photo Editing']: 0.1503
11. 31: ['Video Production']: 0.1458
12. 91: ['3D Rendering']: 0.1412
13. 177: ['Infographic']: 0.1298
14. 113: ['Audio Editing']: 0.1276
15. 110: ['Social Media Imagery']: 0.1253
16. 50: ['Video Post-Editing']: 0.1230
17. 86: ['Motion Graphics']: 0.1207
18. 141: ['Adobe InDesign']: 0.1207
19. 48: ['Adobe Premiere Pro']: 0.1162
20. 127: ['Layout Design']: 0.1162

Betweenness Centrality - Top 5:

1. 2: ['Graphic Design']: 0.3123
2. 22: ['Adobe Photoshop']: 0.1051
3. 16: ['Video Editing']: 0.0848
4. 88: ['3D Modeling']: 0.0774
5. 82: ['3D Design']: 0.0719
6. 100: ['Voice-Over']: 0.0642
7. 36: ['Logo Design']: 0.0574
8. 43: ['Illustration']: 0.0569
9. 91: ['3D Rendering']: 0.0493
10. 113: ['Audio Editing']: 0.0489
11. 27: ['Adobe Illustrator']: 0.0464
12. 144: ['Education']: 0.0399
13. 52: ['Adobe After Effects']: 0.0373
14. 303: ['Audio Production']: 0.0352
15. 322: ['Autodesk AutoCAD']: 0.0341
16. 31: ['Video Production']: 0.0326
17. 106: ['Female']: 0.0311
18. 285: ['Elearning']: 0.0296
19. 457: ['Elearning Design']: 0.0283
20. 220: ['CAD']: 0.0275

Closeness Centrality - Top 5:

1. 2: ['Graphic Design']: 0.5997

2. 22: ['Adobe Photoshop']: 0.5354
3. 16: ['Video Editing']: 0.5189
4. 27: ['Adobe Illustrator']: 0.5058
5. 36: ['Logo Design']: 0.5046
6. 43: ['Illustration']: 0.4994
7. 52: ['Adobe After Effects']: 0.4983
8. 31: ['Video Production']: 0.4938
9. 82: ['3D Design']: 0.4927
10. 88: ['3D Modeling']: 0.4883
11. 86: ['Motion Graphics']: 0.4835
12. 72: ['Photo Editing']: 0.4793
13. 91: ['3D Rendering']: 0.4761
14. 130: ['Animation']: 0.4690
15. 113: ['Audio Editing']: 0.4650
16. 152: ['2D Animation']: 0.4607
17. 50: ['Video Post-Editing']: 0.4568
18. 48: ['Adobe Premiere Pro']: 0.4559
19. 92: ['Video Commercial']: 0.4559
20. 116: ['Videography']: 0.4549

PageRank - Top 5:

1. 2: ['Graphic Design']: 0.0270
2. 22: ['Adobe Photoshop']: 0.0165
3. 27: ['Adobe Illustrator']: 0.0147
4. 43: ['Illustration']: 0.0131
5. 16: ['Video Editing']: 0.0122
6. 36: ['Logo Design']: 0.0111
7. 88: ['3D Modeling']: 0.0111
8. 82: ['3D Design']: 0.0107
9. 91: ['3D Rendering']: 0.0096
10. 72: ['Photo Editing']: 0.0089
11. 52: ['Adobe After Effects']: 0.0087
12. 31: ['Video Production']: 0.0084
13. 113: ['Audio Editing']: 0.0076
14. 177: ['Infographic']: 0.0069
15. 50: ['Video Post-Editing']: 0.0069
16. 110: ['Social Media Imagery']: 0.0068
17. 127: ['Layout Design']: 0.0068
18. 86: ['Motion Graphics']: 0.0068
19. 141: ['Adobe InDesign']: 0.0066
20. 48: ['Adobe Premiere Pro']: 0.0065

Authority (HITS) - Top 5:

1. 2: ['Graphic Design']: 0.0609
2. 16: ['Video Editing']: 0.0556
3. 22: ['Adobe Photoshop']: 0.0463
4. 31: ['Video Production']: 0.0441
5. 48: ['Adobe Premiere Pro']: 0.0429
6. 27: ['Adobe Illustrator']: 0.0424
7. 52: ['Adobe After Effects']: 0.0417
8. 50: ['Video Post-Editing']: 0.0408
9. 36: ['Logo Design']: 0.0292
10. 43: ['Illustration']: 0.0277
11. 126: ['Final Cut Pro X']: 0.0235
12. 89: ['Explainer Video']: 0.0218
13. 86: ['Motion Graphics']: 0.0211

14. 92: ['Video Commercial']: 0.0208
15. 84: ['Brand Identity & Guidelines']: 0.0187
16. 116: ['Videography']: 0.0181
17. 72: ['Photo Editing']: 0.0163
18. 113: ['Audio Editing']: 0.0154
19. 141: ['Adobe InDesign']: 0.0137
20. 110: ['Social Media Imagery']: 0.0131

Eigenvector Centrality - Top 5:

1. 2: ['Graphic Design']: 0.3002
2. 22: ['Adobe Photoshop']: 0.2533
3. 27: ['Adobe Illustrator']: 0.2324
4. 43: ['Illustration']: 0.1971
5. 36: ['Logo Design']: 0.1620
6. 16: ['Video Editing']: 0.1478
7. 72: ['Photo Editing']: 0.1452
8. 177: ['Infographic']: 0.1419
9. 52: ['Adobe After Effects']: 0.1390
10. 110: ['Social Media Imagery']: 0.1387
11. 141: ['Adobe InDesign']: 0.1313
12. 82: ['3D Design']: 0.1256
13. 86: ['Motion Graphics']: 0.1248
14. 127: ['Layout Design']: 0.1179
15. 31: ['Video Production']: 0.1151
16. 48: ['Adobe Premiere Pro']: 0.1092
17. 84: ['Brand Identity & Guidelines']: 0.1080
18. 50: ['Video Post-Editing']: 0.1055
19. 88: ['3D Modeling']: 0.0994
20. 130: ['Animation']: 0.0960

America 2025

Cluster 2 - Node : 83, Edge: 338

Cluster 1 - Node : 119, Edge: 529

Cluster 0 - Node : 146, Edge: 524

Cluster 3 - Node : 75, Edge: 214

Cluster 0 Visual Communication and Multimedia Production Skill Cluster

Degree Centrality - Top 5:

1. 0: ['Graphic Design']: 0.5488
2. 6: ['Adobe Photoshop']: 0.5122
3. 5: ['Adobe Illustrator']: 0.4024
4. 4: ['Video Editing']: 0.2927
5. 11: ['Adobe Premiere Pro']: 0.2561
6. 16: ['Adobe After Effects']: 0.2561
7. 37: ['Videography']: 0.2561
8. 13: ['Video Production']: 0.2195
9. 47: ['Logo Design']: 0.2073
10. 55: ['Social Media Imagery']: 0.1829
11. 18: ['Video Post-Editing']: 0.1707
12. 63: ['Photo Editing']: 0.1707
13. 41: ['Canva']: 0.1463
14. 60: ['Illustration']: 0.1463
15. 68: ['Video Commercial']: 0.1463
16. 54: ['Brand Identity & Guidelines']: 0.1341

17. 71: ['Adobe InDesign']: 0.1341
18. 72: ['Infographic']: 0.1341
19. 263: ['Print Design']: 0.1220
20. 42: ['Figma']: 0.1098

Betweenness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.3352
2. 6: ['Adobe Photoshop']: 0.2284
3. 37: ['Videography']: 0.1097
4. 5: ['Adobe Illustrator']: 0.0989
5. 4: ['Video Editing']: 0.0981
6. 63: ['Photo Editing']: 0.0785
7. 13: ['Video Production']: 0.0761
8. 47: ['Logo Design']: 0.0610
9. 16: ['Adobe After Effects']: 0.0569
10. 11: ['Adobe Premiere Pro']: 0.0552
11. 68: ['Video Commercial']: 0.0338
12. 269: ['Photography']: 0.0244
13. 133: ['Testimonial Video']: 0.0211
14. 332: ['People']: 0.0146
15. 18: ['Video Post-Editing']: 0.0129
16. 60: ['Illustration']: 0.0114
17. 41: ['Canva']: 0.0111
18. 55: ['Social Media Imagery']: 0.0109
19. 124: ['Product Photography']: 0.0108
20. 101: ['UGC']: 0.0073

Closeness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.6667
2. 6: ['Adobe Photoshop']: 0.6357
3. 5: ['Adobe Illustrator']: 0.5899
4. 4: ['Video Editing']: 0.5734
5. 16: ['Adobe After Effects']: 0.5616
6. 13: ['Video Production']: 0.5395
7. 11: ['Adobe Premiere Pro']: 0.5256
8. 63: ['Photo Editing']: 0.5125
9. 68: ['Video Commercial']: 0.5093
10. 41: ['Canva']: 0.5031
11. 253: ['Explainer Video']: 0.5000
12. 18: ['Video Post-Editing']: 0.4940
13. 216: ['TikTok Video']: 0.4824
14. 60: ['Illustration']: 0.4795
15. 71: ['Adobe InDesign']: 0.4767
16. 47: ['Logo Design']: 0.4740
17. 332: ['People']: 0.4740
18. 55: ['Social Media Imagery']: 0.4633
19. 37: ['Videography']: 0.4607
20. 534: ['2D Animation']: 0.4581

PageRank - Top 5:

1. 0: ['Graphic Design']: 0.0607
2. 6: ['Adobe Photoshop']: 0.0552
3. 5: ['Adobe Illustrator']: 0.0433
4. 4: ['Video Editing']: 0.0314
5. 37: ['Videography']: 0.0303
6. 11: ['Adobe Premiere Pro']: 0.0276

7. 16: ['Adobe After Effects']: 0.0269
8. 47: ['Logo Design']: 0.0254
9. 13: ['Video Production']: 0.0240
10. 63: ['Photo Editing']: 0.0209
11. 55: ['Social Media Imagery']: 0.0201
12. 18: ['Video Post-Editing']: 0.0185
13. 60: ['Illustration']: 0.0174
14. 68: ['Video Commercial']: 0.0164
15. 41: ['Canva']: 0.0156
16. 54: ['Brand Identity & Guidelines']: 0.0153
17. 72: ['Infographic']: 0.0152
18. 71: ['Adobe InDesign']: 0.0152
19. 263: ['Print Design']: 0.0141
20. 332: ['People']: 0.0133

Authority (HITS) - Top 5:

1. 4: ['Video Editing']: 0.1220
2. 11: ['Adobe Premiere Pro']: 0.1201
3. 13: ['Video Production']: 0.1064
4. 18: ['Video Post-Editing']: 0.1056
5. 16: ['Adobe After Effects']: 0.1034
6. 41: ['Canva']: 0.0529
7. 6: ['Adobe Photoshop']: 0.0384
8. 0: ['Graphic Design']: 0.0350
9. 5: ['Adobe Illustrator']: 0.0346
10. 37: ['Videography']: 0.0329
11. 68: ['Video Commercial']: 0.0232
12. 244: ['Instagram Reels']: 0.0120
13. 47: ['Logo Design']: 0.0112
14. 71: ['Adobe InDesign']: 0.0104
15. 54: ['Brand Identity & Guidelines']: 0.0103
16. 42: ['Figma']: 0.0095
17. 60: ['Illustration']: 0.0090
18. 366: ['CapCut']: 0.0090
19. 267: ['YouTube Video']: 0.0083
20. 102: ['Social Media Content Creation']: 0.0082

Eigenvector Centrality - Top 5:

1. 6: ['Adobe Photoshop']: 0.3574
2. 0: ['Graphic Design']: 0.3494
3. 5: ['Adobe Illustrator']: 0.2983
4. 16: ['Adobe After Effects']: 0.2184
5. 4: ['Video Editing']: 0.2079
6. 11: ['Adobe Premiere Pro']: 0.1892
7. 41: ['Canva']: 0.1698
8. 55: ['Social Media Imagery']: 0.1584
9. 13: ['Video Production']: 0.1549
10. 47: ['Logo Design']: 0.1510
11. 18: ['Video Post-Editing']: 0.1443
12. 71: ['Adobe InDesign']: 0.1378
13. 253: ['Explainer Video']: 0.1354
14. 60: ['Illustration']: 0.1329
15. 54: ['Brand Identity & Guidelines']: 0.1315
16. 72: ['Infographic']: 0.1273
17. 63: ['Photo Editing']: 0.1272
18. 42: ['Figma']: 0.1267

19. 263: ['Print Design']: 0.1266
20. 68: ['Video Commercial']: 0.1216

Cluster 1 Modern Web Ecosystem and Full-Stack Interface Development Skill Cluster

Degree Centrality - Top 5:

1. 1: ['Web Development']: 0.4576
2. 2: ['Web Design']: 0.4407
3. 3: ['JavaScript']: 0.3559
4. 7: ['WordPress']: 0.3305
5. 35: ['React']: 0.2458
6. 19: ['HTML']: 0.2373
7. 50: ['Node.js']: 0.2119
8. 28: ['PHP']: 0.2034
9. 38: ['API']: 0.2034
10. 17: ['Search Engine Optimization']: 0.1780
11. 32: ['CSS']: 0.1780
12. 69: ['API Integration']: 0.1695
13. 45: ['Shopify']: 0.1525
14. 39: ['Landing Page']: 0.1441
15. 74: ['Automation']: 0.1356
16. 172: ['PostgreSQL']: 0.1271
17. 62: ['SEO Keyword Research']: 0.1102
18. 94: ['Next.js']: 0.1102
19. 106: ['WooCommerce']: 0.1102
20. 95: ['MySQL']: 0.1017

Betweenness Centrality - Top 5:

1. 2: ['Web Design']: 0.2791
2. 1: ['Web Development']: 0.2220
3. 17: ['Search Engine Optimization']: 0.1753
4. 7: ['WordPress']: 0.1063
5. 3: ['JavaScript']: 0.1061
6. 38: ['API']: 0.0831
7. 50: ['Node.js']: 0.0625
8. 35: ['React']: 0.0467
9. 45: ['Shopify']: 0.0414
10. 19: ['HTML']: 0.0399
11. 69: ['API Integration']: 0.0364
12. 91: ['Gemini']: 0.0362
13. 172: ['PostgreSQL']: 0.0349
14. 184: ['Microsoft Azure']: 0.0336
15. 28: ['PHP']: 0.0331
16. 62: ['SEO Keyword Research']: 0.0252
17. 245: ['API Testing']: 0.0227
18. 251: ['Shopify Apps']: 0.0218
19. 74: ['Automation']: 0.0190
20. 342: ['Usability Testing']: 0.0190

Closeness Centrality - Top 5:

1. 1: ['Web Development']: 0.6211
2. 2: ['Web Design']: 0.5990
3. 7: ['WordPress']: 0.5592
4. 3: ['JavaScript']: 0.5540
5. 19: ['HTML']: 0.5413

6. 28: ['PHP']: 0.5315
7. 35: ['React']: 0.5198
8. 45: ['Shopify']: 0.5153
9. 32: ['CSS']: 0.4958
10. 38: ['API']: 0.4958
11. 69: ['API Integration']: 0.4937
12. 17: ['Search Engine Optimization']: 0.4816
13. 50: ['Node.js']: 0.4777
14. 94: ['Next.js']: 0.4739
15. 39: ['Landing Page']: 0.4701
16. 172: ['PostgreSQL']: 0.4609
17. 116: ['Web Application']: 0.4591
18. 106: ['WooCommerce']: 0.4538
19. 95: ['MySQL']: 0.4521
20. 104: ['HTML5']: 0.4521

PageRank - Top 5:

1. 2: ['Web Design']: 0.0449
2. 1: ['Web Development']: 0.0427
3. 3: ['JavaScript']: 0.0327
4. 7: ['WordPress']: 0.0313
5. 35: ['React']: 0.0224
6. 17: ['Search Engine Optimization']: 0.0220
7. 19: ['HTML']: 0.0216
8. 50: ['Node.js']: 0.0204
9. 38: ['API']: 0.0189
10. 28: ['PHP']: 0.0184
11. 32: ['CSS']: 0.0162
12. 69: ['API Integration']: 0.0159
13. 45: ['Shopify']: 0.0156
14. 39: ['Landing Page']: 0.0147
15. 74: ['Automation']: 0.0134
16. 62: ['SEO Keyword Research']: 0.0129
17. 172: ['PostgreSQL']: 0.0127
18. 245: ['API Testing']: 0.0112
19. 106: ['WooCommerce']: 0.0109
20. 94: ['Next.js']: 0.0106

Authority (HITS) - Top 5:

1. 1: ['Web Development']: 0.1033
2. 2: ['Web Design']: 0.0827
3. 3: ['JavaScript']: 0.0775
4. 19: ['HTML']: 0.0634
5. 7: ['WordPress']: 0.0626
6. 28: ['PHP']: 0.0524
7. 32: ['CSS']: 0.0482
8. 35: ['React']: 0.0441
9. 38: ['API']: 0.0253
10. 50: ['Node.js']: 0.0249
11. 95: ['MySQL']: 0.0243
12. 39: ['Landing Page']: 0.0211
13. 104: ['HTML5']: 0.0186
14. 94: ['Next.js']: 0.0176
15. 45: ['Shopify']: 0.0166
16. 196: ['Full-Stack Development']: 0.0156
17. 17: ['Search Engine Optimization']: 0.0144

- 18. 69: ['API Integration']: 0.0141
- 19. 158: ['Tailwind CSS']: 0.0133
- 20. 174: ['Responsive Design']: 0.0122

Eigenvector Centrality - Top 5:

- 1. 1: ['Web Development']: 0.3319
- 2. 3: ['JavaScript']: 0.2888
- 3. 2: ['Web Design']: 0.2693
- 4. 7: ['WordPress']: 0.2671
- 5. 19: ['HTML']: 0.2348
- 6. 35: ['React']: 0.2237
- 7. 28: ['PHP']: 0.2170
- 8. 32: ['CSS']: 0.1963
- 9. 38: ['API']: 0.1796
- 10. 50: ['Node.js']: 0.1689
- 11. 69: ['API Integration']: 0.1559
- 12. 95: ['MySQL']: 0.1401
- 13. 45: ['Shopify']: 0.1339
- 14. 104: ['HTML5']: 0.1320
- 15. 106: ['WooCommerce']: 0.1272
- 16. 39: ['Landing Page']: 0.1260
- 17. 229: ['CSS 3']: 0.1190
- 18. 311: ['Bootstrap']: 0.1190
- 19. 196: ['Full-Stack Development']: 0.1143
- 20. 94: ['Next.js']: 0.1133
- []:

Cluster 2 Administrative Intelligence and Data-Driven Support Skill Cluster

Degree Centrality - Top 5:

- 1. 9: ['Data Entry']: 0.2552
- 2. 8: ['Lead Generation']: 0.2207
- 3. 31: ['Microsoft Excel']: 0.2069
- 4. 15: ['Administrative Support']: 0.1931
- 5. 29: ['Python']: 0.1793
- 6. 27: ['Email Communication']: 0.1655
- 7. 46: ['Communications']: 0.1586
- 8. 33: ['Customer Service']: 0.1379
- 9. 92: ['Accounting']: 0.1241
- 10. 66: ['Bookkeeping']: 0.1172
- 11. 51: ['Virtual Assistance']: 0.1103
- 12. 70: ['Scheduling']: 0.1103
- 13. 186: ['Real Estate']: 0.1103
- 14. 43: ['Market Research']: 0.1034
- 15. 36: ['Cold Calling']: 0.0966
- 16. 40: ['Email Marketing']: 0.0966
- 17. 93: ['Financial Analysis']: 0.0966
- 18. 48: ['Artificial Intelligence']: 0.0897
- 19. 56: ['Sales']: 0.0897
- 20. 80: ['Intuit QuickBooks']: 0.0897

Betweenness Centrality - Top 5:

- 1. 29: ['Python']: 0.2244
- 2. 9: ['Data Entry']: 0.1982
- 3. 31: ['Microsoft Excel']: 0.1588

4. 15: ['Administrative Support']: 0.1538
5. 8: ['Lead Generation']: 0.1527
6. 46: ['Communications']: 0.1190
7. 27: ['Email Communication']: 0.0978
8. 256: ['Microsoft Windows']: 0.0827
9. 119: ['System Administration']: 0.0676
10. 92: ['Accounting']: 0.0650
11. 87: ['Customer Support']: 0.0634
12. 43: ['Market Research']: 0.0627
13. 66: ['Bookkeeping']: 0.0499
14. 36: ['Cold Calling']: 0.0495
15. 186: ['Real Estate']: 0.0494
16. 93: ['Financial Analysis']: 0.0410
17. 33: ['Customer Service']: 0.0407
18. 96: ['Embedded System']: 0.0395
19. 48: ['Artificial Intelligence']: 0.0314
20. 64: ['Data Mining']: 0.0271

Closeness Centrality - Top 5:

1. 9: ['Data Entry']: 0.5160
2. 8: ['Lead Generation']: 0.4866
3. 15: ['Administrative Support']: 0.4754
4. 31: ['Microsoft Excel']: 0.4754
5. 46: ['Communications']: 0.4739
6. 27: ['Email Communication']: 0.4475
7. 29: ['Python']: 0.4448
8. 33: ['Customer Service']: 0.4341
9. 43: ['Market Research']: 0.4315
10. 64: ['Data Mining']: 0.4240
11. 92: ['Accounting']: 0.4155
12. 66: ['Bookkeeping']: 0.4119
13. 51: ['Virtual Assistance']: 0.4039
14. 70: ['Scheduling']: 0.4039
15. 130: ['Digital Marketing']: 0.4017
16. 40: ['Email Marketing']: 0.4006
17. 87: ['Customer Support']: 0.4006
18. 186: ['Real Estate']: 0.3973
19. 65: ['Data Scraping']: 0.3962
20. 132: ['List Building']: 0.3940

PageRank - Top 5:

1. 9: ['Data Entry']: 0.0293
2. 8: ['Lead Generation']: 0.0259
3. 31: ['Microsoft Excel']: 0.0247
4. 29: ['Python']: 0.0229
5. 15: ['Administrative Support']: 0.0229
6. 27: ['Email Communication']: 0.0195
7. 46: ['Communications']: 0.0182
8. 33: ['Customer Service']: 0.0161
9. 92: ['Accounting']: 0.0155
10. 66: ['Bookkeeping']: 0.0141
11. 43: ['Market Research']: 0.0137
12. 93: ['Financial Analysis']: 0.0133
13. 186: ['Real Estate']: 0.0133
14. 51: ['Virtual Assistance']: 0.0132
15. 70: ['Scheduling']: 0.0128

16. 119: ['System Administration']: 0.0122
17. 48: ['Artificial Intelligence']: 0.0122
18. 36: ['Cold Calling']: 0.0122
19. 40: ['Email Marketing']: 0.0120
20. 80: ['Intuit QuickBooks']: 0.0118

Authority (HITS) - Top 5:

1. 9: ['Data Entry']: 0.0808
2. 27: ['Email Communication']: 0.0683
3. 33: ['Customer Service']: 0.0597
4. 15: ['Administrative Support']: 0.0588
5. 8: ['Lead Generation']: 0.0572
6. 31: ['Microsoft Excel']: 0.0405
7. 40: ['Email Marketing']: 0.0340
8. 51: ['Virtual Assistance']: 0.0320
9. 46: ['Communications']: 0.0303
10. 122: ['Product Listings']: 0.0288
11. 120: ['Amazon FBA']: 0.0233
12. 97: ['Email Campaign Setup']: 0.0221
13. 36: ['Cold Calling']: 0.0190
14. 87: ['Customer Support']: 0.0186
15. 70: ['Scheduling']: 0.0165
16. 157: ['Interpersonal Skills']: 0.0159
17. 56: ['Sales']: 0.0159
18. 66: ['Bookkeeping']: 0.0156
19. 147: ['Online Chat Support']: 0.0155
20. 64: ['Data Mining']: 0.0148

Eigenvector Centrality - Top 5:

1. 9: ['Data Entry']: 0.3334
2. 8: ['Lead Generation']: 0.2844
3. 15: ['Administrative Support']: 0.2597
4. 27: ['Email Communication']: 0.2526
5. 46: ['Communications']: 0.2453
6. 31: ['Microsoft Excel']: 0.2348
7. 33: ['Customer Service']: 0.2313
8. 70: ['Scheduling']: 0.1858
9. 130: ['Digital Marketing']: 0.1760
10. 51: ['Virtual Assistance']: 0.1737
11. 66: ['Bookkeeping']: 0.1471
12. 40: ['Email Marketing']: 0.1406
13. 92: ['Accounting']: 0.1340
14. 186: ['Real Estate']: 0.1311
15. 43: ['Market Research']: 0.1296
16. 122: ['Product Listings']: 0.1242
17. 177: ['Prospect List']: 0.1228
18. 29: ['Python']: 0.1220
19. 132: ['List Building']: 0.1212
20. 56: ['Sales']: 0.1170
[]:

Cluster 3 Digital Marketing and Social Media Advertising Skill Cluster

Degree Centrality - Top 5:

1. 10: ['Marketing Strategy']: 0.2703
2. 23: ['Instagram']: 0.2703

3. 14: ['Facebook']: 0.2568
4. 12: ['Social Media Marketing']: 0.2162
5. 61: ['Google Analytics']: 0.2027
6. 52: ['Google Ads']: 0.1757
7. 34: ['Social Media Management']: 0.1486
8. 128: ['Facebook Ads Manager']: 0.1351
9. 165: ['Influencer Marketing']: 0.1351
10. 118: ['Facebook Advertising']: 0.1351
11. 83: ['Ecommerce']: 0.1216
12. 109: ['Internet Marketing']: 0.1216
13. 121: ['TikTok']: 0.1216
14. 143: ['Campaign Management']: 0.1081
15. 78: ['Sales Analytics']: 0.1081
16. 79: ['Marketing Analytics']: 0.1081
17. 131: ['Branding']: 0.0946
18. 161: ['Sales & Marketing']: 0.0946
19. 181: ['Presentation Design']: 0.0946
20. 77: ['Data Visualization']: 0.0946

Betweenness Centrality - Top 5:

1. 10: ['Marketing Strategy']: 0.3739
2. 61: ['Google Analytics']: 0.2915
3. 23: ['Instagram']: 0.2061
4. 131: ['Branding']: 0.1736
5. 14: ['Facebook']: 0.1549
6. 165: ['Influencer Marketing']: 0.1141
7. 52: ['Google Ads']: 0.1061
8. 181: ['Presentation Design']: 0.1037
9. 109: ['Internet Marketing']: 0.1017
10. 12: ['Social Media Marketing']: 0.0831
11. 143: ['Campaign Management']: 0.0811
12. 77: ['Data Visualization']: 0.0533
13. 121: ['TikTok']: 0.0440
14. 128: ['Facebook Ads Manager']: 0.0425
15. 118: ['Facebook Advertising']: 0.0404
16. 78: ['Sales Analytics']: 0.0389
17. 79: ['Marketing Analytics']: 0.0389
18. 83: ['Ecommerce']: 0.0388
19. 34: ['Social Media Management']: 0.0385
20. 161: ['Sales & Marketing']: 0.0327

Closeness Centrality - Top 5:

1. 10: ['Marketing Strategy']: 0.5034
2. 23: ['Instagram']: 0.4540
3. 14: ['Facebook']: 0.4512
4. 109: ['Internet Marketing']: 0.4431
5. 12: ['Social Media Marketing']: 0.4302
6. 61: ['Google Analytics']: 0.4277
7. 34: ['Social Media Management']: 0.4205
8. 128: ['Facebook Ads Manager']: 0.4157
9. 52: ['Google Ads']: 0.4111
10. 118: ['Facebook Advertising']: 0.4111
11. 161: ['Sales & Marketing']: 0.4088
12. 214: ['Social Media Content']: 0.3978
13. 271: ['Retail & Consumer Goods']: 0.3895
14. 406: ['Social Media Marketing Plan']: 0.3895

15. 83: ['Ecommerce']: 0.3895
16. 131: ['Branding']: 0.3737
17. 165: ['Influencer Marketing']: 0.3737
18. 121: ['TikTok']: 0.3719
19. 206: ['Fashion & Apparel']: 0.3610
20. 159: ['Growth Hacking']: 0.3558

PageRank - Top 5:

1. 23: ['Instagram']: 0.0408
2. 10: ['Marketing Strategy']: 0.0390
3. 14: ['Facebook']: 0.0370
4. 12: ['Social Media Marketing']: 0.0320
5. 61: ['Google Analytics']: 0.0309
6. 52: ['Google Ads']: 0.0269
7. 165: ['Influencer Marketing']: 0.0227
8. 34: ['Social Media Management']: 0.0222
9. 128: ['Facebook Ads Manager']: 0.0207
10. 118: ['Facebook Advertising']: 0.0206
11. 121: ['TikTok']: 0.0199
12. 83: ['Ecommerce']: 0.0193
13. 181: ['Presentation Design']: 0.0192
14. 143: ['Campaign Management']: 0.0191
15. 109: ['Internet Marketing']: 0.0188
16. 78: ['Sales Analytics']: 0.0185
17. 79: ['Marketing Analytics']: 0.0185
18. 77: ['Data Visualization']: 0.0179
19. 131: ['Branding']: 0.0166
20. 161: ['Sales & Marketing']: 0.0150

Authority (HITS) - Top 5:

1. 12: ['Social Media Marketing']: 0.1444
2. 14: ['Facebook']: 0.1375
3. 23: ['Instagram']: 0.1339
4. 34: ['Social Media Management']: 0.1150
5. 10: ['Marketing Strategy']: 0.0998
6. 118: ['Facebook Advertising']: 0.0301
7. 214: ['Social Media Content']: 0.0245
8. 159: ['Growth Hacking']: 0.0210
9. 52: ['Google Ads']: 0.0206
10. 121: ['TikTok']: 0.0205
11. 109: ['Internet Marketing']: 0.0165
12. 83: ['Ecommerce']: 0.0145
13. 143: ['Campaign Management']: 0.0133
14. 128: ['Facebook Ads Manager']: 0.0126
15. 406: ['Social Media Marketing Plan']: 0.0125
16. 161: ['Sales & Marketing']: 0.0115
17. 165: ['Influencer Marketing']: 0.0113
18. 308: ['Social Media Strategy']: 0.0111
19. 348: ['Book Marketing']: 0.0106
20. 131: ['Branding']: 0.0096

Eigenvector Centrality - Top 5:

1. 10: ['Marketing Strategy']: 0.3516
2. 14: ['Facebook']: 0.3438
3. 23: ['Instagram']: 0.3363
4. 12: ['Social Media Marketing']: 0.3125

5. 34: ['Social Media Management']: 0.2531
6. 118: ['Facebook Advertising']: 0.2183
7. 214: ['Social Media Content']: 0.1988
8. 121: ['TikTok']: 0.1874
9. 109: ['Internet Marketing']: 0.1770
10. 128: ['Facebook Ads Manager']: 0.1608
11. 52: ['Google Ads']: 0.1568
12. 161: ['Sales & Marketing']: 0.1485
13. 61: ['Google Analytics']: 0.1402
14. 83: ['Ecommerce']: 0.1364
15. 406: ['Social Media Marketing Plan']: 0.1324
16. 630: ['Pinterest']: 0.1270
17. 143: ['Campaign Management']: 0.1267
18. 165: ['Influencer Marketing']: 0.1251
19. 516: ['LinkedIn']: 0.1184
20. 308: ['Social Media Strategy']: 0.0970

Asia 2025

NetworkX modularity: 0.7180

Cluster 1 - Node : 98, Edge: 481
 Cluster 3 - Node : 56, Edge: 188
 Cluster 0 - Node : 121, Edge: 408
 Cluster 2 - Node : 66, Edge: 209

Cluster 0: Modern Web and Mobile Development Ecosystem

Degree Centrality - Top 5:

1. 8: ['JavaScript']: 0.4330
2. 0: ['Web Development']: 0.3814
3. 10: ['CSS']: 0.3608
4. 3: ['HTML']: 0.3402
5. 4: ['WordPress']: 0.3196
6. 6: ['Web Design']: 0.3093
7. 17: ['PHP']: 0.2680
8. 56: ['React']: 0.2577
9. 67: ['Node.js']: 0.1959
10. 87: ['HTML5']: 0.1959
11. 16: ['Shopify']: 0.1856
12. 76: ['RESTful API']: 0.1856
13. 64: ['Web Application']: 0.1753
14. 77: ['jQuery']: 0.1753
15. 102: ['Website']: 0.1753
16. 107: ['TypeScript']: 0.1753
17. 118: ['CSS 3']: 0.1753
18. 278: ['Laravel']: 0.1753
19. 97: ['Landing Page']: 0.1546
20. 119: ['Mobile App Development']: 0.1546

Betweenness Centrality - Top 5:

1. 8: ['JavaScript']: 0.1858
2. 6: ['Web Design']: 0.1447

3. 4: ['WordPress']: 0.1261
4. 0: ['Web Development']: 0.1182
5. 3: ['HTML']: 0.0849
6. 195: ['React Native']: 0.0845
7. 10: ['CSS']: 0.0842
8. 16: ['Shopify']: 0.0680
9. 199: ['Figma']: 0.0635
10. 119: ['Mobile App Development']: 0.0631
11. 275: ['Flutter']: 0.0595
12. 75: ['Android']: 0.0560
13. 56: ['React']: 0.0505
14. 17: ['PHP']: 0.0419
15. 64: ['Web Application']: 0.0414
16. 115: ['iOS']: 0.0369
17. 50: ['Shopify Theme']: 0.0263
18. 104: ['Front-End Development']: 0.0231
19. 76: ['RESTful API']: 0.0226
20. 73: ['DevOps']: 0.0206

Closeness Centrality - Top 5:

1. 8: ['JavaScript']: 0.6025
2. 6: ['Web Design']: 0.5774
3. 10: ['CSS']: 0.5774
4. 0: ['Web Development']: 0.5575
5. 4: ['WordPress']: 0.5511
6. 3: ['HTML']: 0.5480
7. 17: ['PHP']: 0.5243
8. 56: ['React']: 0.5079
9. 87: ['HTML5']: 0.5052
10. 67: ['Node.js']: 0.4924
11. 102: ['Website']: 0.4924
12. 118: ['CSS 3']: 0.4924
13. 16: ['Shopify']: 0.4874
14. 278: ['Laravel']: 0.4802
15. 107: ['TypeScript']: 0.4778
16. 64: ['Web Application']: 0.4755
17. 199: ['Figma']: 0.4755
18. 77: ['jQuery']: 0.4732
19. 195: ['React Native']: 0.4686
20. 97: ['Landing Page']: 0.4663

PageRank - Top 5:

1. 8: ['JavaScript']: 0.0369
2. 0: ['Web Development']: 0.0347
3. 3: ['HTML']: 0.0308
4. 4: ['WordPress']: 0.0301
5. 10: ['CSS']: 0.0301
6. 6: ['Web Design']: 0.0279
7. 56: ['React']: 0.0227
8. 17: ['PHP']: 0.0227
9. 16: ['Shopify']: 0.0184
10. 119: ['Mobile App Development']: 0.0183
11. 67: ['Node.js']: 0.0169
12. 76: ['RESTful API']: 0.0165
13. 64: ['Web Application']: 0.0165
14. 87: ['HTML5']: 0.0164

15. 199: ['Figma']: 0.0162
16. 102: ['Website']: 0.0156
17. 75: ['Android']: 0.0150
18. 107: ['TypeScript']: 0.0150
19. 77: ['jQuery']: 0.0149
20. 278: ['Laravel']: 0.0149

Authority (HITS) - Top 5:

1. 0: ['Web Development']: 0.0921
2. 3: ['HTML']: 0.0865
3. 10: ['CSS']: 0.0773
4. 8: ['JavaScript']: 0.0717
5. 6: ['Web Design']: 0.0698
6. 17: ['PHP']: 0.0597
7. 4: ['WordPress']: 0.0565
8. 16: ['Shopify']: 0.0291
9. 57: ['Shopify Templates']: 0.0243
10. 87: ['HTML5']: 0.0233
11. 102: ['Website']: 0.0231
12. 118: ['CSS 3']: 0.0203
13. 97: ['Landing Page']: 0.0191
14. 77: ['jQuery']: 0.0183
15. 50: ['Shopify Theme']: 0.0144
16. 129: ['Bootstrap']: 0.0136
17. 76: ['RESTful API']: 0.0135
18. 56: ['React']: 0.0133
19. 139: ['Ecommerce Website']: 0.0127
20. 67: ['Node.js']: 0.0127

Eigenvector Centrality - Top 5:

1. 8: ['JavaScript']: 0.3037
2. 10: ['CSS']: 0.2899
3. 3: ['HTML']: 0.2544
4. 0: ['Web Development']: 0.2475
5. 6: ['Web Design']: 0.2236
6. 17: ['PHP']: 0.2166
7. 4: ['WordPress']: 0.2099
8. 87: ['HTML5']: 0.1984
9. 118: ['CSS 3']: 0.1811
10. 107: ['TypeScript']: 0.1650
11. 102: ['Website']: 0.1634
12. 56: ['React']: 0.1625
13. 77: ['jQuery']: 0.1547
14. 97: ['Landing Page']: 0.1503
15. 64: ['Web Application']: 0.1491
16. 278: ['Laravel']: 0.1481
17. 76: ['RESTful API']: 0.1459
18. 67: ['Node.js']: 0.1442
19. 16: ['Shopify']: 0.1354
20. 199: ['Figma']: 0.1248

Cluster 1 Professional Visual Content Production and Digital Design Network

Degree Centrality - Top 5:

1. 1: ['Video Editing']: 0.4727
2. 12: ['Adobe Photoshop']: 0.3818
3. 9: ['Graphic Design']: 0.2545
4. 18: ['Adobe Premiere Pro']: 0.2545
5. 20: ['Video Production']: 0.2545
6. 30: ['Adobe Illustrator']: 0.2545
7. 19: ['Adobe After Effects']: 0.2182
8. 25: ['Video Post-Editing']: 0.2182
9. 52: ['Motion Graphics']: 0.2182
10. 88: ['Explainer Video']: 0.2182
11. 66: ['Videography']: 0.2000
12. 53: ['Illustration']: 0.1818
13. 45: ['Logo Design']: 0.1636
14. 109: ['Brand Identity & Guidelines']: 0.1455
15. 110: ['Finance & Accounting']: 0.1455
16. 150: ['Product Photography']: 0.1273
17. 106: ['Photography']: 0.1273
18. 256: ['Bottle']: 0.0909
19. 261: ['Layout Design']: 0.0909
20. 157: ['Photo Retouching']: 0.0909

Betweenness Centrality - Top 5:

1. 1: ['Video Editing']: 0.3700
2. 12: ['Adobe Photoshop']: 0.3344
3. 52: ['Motion Graphics']: 0.1495
4. 45: ['Logo Design']: 0.1152
5. 110: ['Finance & Accounting']: 0.1051
6. 20: ['Video Production']: 0.0789
7. 9: ['Graphic Design']: 0.0682
8. 30: ['Adobe Illustrator']: 0.0682
9. 88: ['Explainer Video']: 0.0565
10. 53: ['Illustration']: 0.0539
11. 18: ['Adobe Premiere Pro']: 0.0528
12. 150: ['Product Photography']: 0.0337
13. 19: ['Adobe After Effects']: 0.0315
14. 25: ['Video Post-Editing']: 0.0248
15. 109: ['Brand Identity & Guidelines']: 0.0239
16. 66: ['Videography']: 0.0224
17. 83: ['Character Design']: 0.0073
18. 388: ['Video Commercial']: 0.0040
19. 106: ['Photography']: 0.0011
20. 256: ['Bottle']: 0.0000

Closeness Centrality - Top 5:

1. 1: ['Video Editing']: 0.6044
2. 12: ['Adobe Photoshop']: 0.5914
3. 52: ['Motion Graphics']: 0.5238
4. 18: ['Adobe Premiere Pro']: 0.5189
5. 19: ['Adobe After Effects']: 0.5093
6. 9: ['Graphic Design']: 0.4741
7. 30: ['Adobe Illustrator']: 0.4741
8. 20: ['Video Production']: 0.4701

9. 110: ['Finance & Accounting']: 0.4701
10. 25: ['Video Post-Editing']: 0.4622
11. 222: ['Infographic']: 0.4583
12. 225: ['Banner Ad Design']: 0.4583
13. 229: ['Advertising Design']: 0.4583
14. 150: ['Product Photography']: 0.4508
15. 226: ['YouTube Shorts']: 0.4508
16. 231: ['YouTube Video']: 0.4508
17. 88: ['Explainer Video']: 0.4435
18. 53: ['Illustration']: 0.4365
19. 66: ['Videography']: 0.4264
20. 45: ['Logo Design']: 0.4135

PageRank - Top 5:

1. 1: ['Video Editing']: 0.0623
2. 12: ['Adobe Photoshop']: 0.0499
3. 9: ['Graphic Design']: 0.0346
4. 30: ['Adobe Illustrator']: 0.0346
5. 20: ['Video Production']: 0.0330
6. 18: ['Adobe Premiere Pro']: 0.0325
7. 52: ['Motion Graphics']: 0.0289
8. 88: ['Explainer Video']: 0.0289
9. 25: ['Video Post-Editing']: 0.0283
10. 19: ['Adobe After Effects']: 0.0278
11. 53: ['Illustration']: 0.0265
12. 66: ['Videography']: 0.0264
13. 110: ['Finance & Accounting']: 0.0242
14. 45: ['Logo Design']: 0.0236
15. 109: ['Brand Identity & Guidelines']: 0.0217
16. 106: ['Photography']: 0.0175
17. 150: ['Product Photography']: 0.0172
18. 83: ['Character Design']: 0.0154
19. 224: ['Corporate Brand Identity']: 0.0147
20. 228: ['Brand Guidelines']: 0.0147

Authority (HITS) - Top 5:

1. 1: ['Video Editing']: 0.1319
2. 19: ['Adobe After Effects']: 0.1088
3. 18: ['Adobe Premiere Pro']: 0.1012
4. 25: ['Video Post-Editing']: 0.0937
5. 20: ['Video Production']: 0.0748
6. 12: ['Adobe Photoshop']: 0.0470
7. 52: ['Motion Graphics']: 0.0337
8. 66: ['Videography']: 0.0314
9. 106: ['Photography']: 0.0274
10. 9: ['Graphic Design']: 0.0242
11. 158: ['Photo Editing']: 0.0238
12. 157: ['Photo Retouching']: 0.0238
13. 226: ['YouTube Shorts']: 0.0226
14. 231: ['YouTube Video']: 0.0226
15. 30: ['Adobe Illustrator']: 0.0206
16. 88: ['Explainer Video']: 0.0177
17. 150: ['Product Photography']: 0.0141
18. 110: ['Finance & Accounting']: 0.0118
19. 225: ['Banner Ad Design']: 0.0118
20. 229: ['Advertising Design']: 0.0118

Eigenvector Centrality - Top 5:

1. 1: ['Video Editing']: 0.3812
2. 12: ['Adobe Photoshop']: 0.3091
3. 18: ['Adobe Premiere Pro']: 0.2775
4. 19: ['Adobe After Effects']: 0.2586
5. 20: ['Video Production']: 0.2449
6. 52: ['Motion Graphics']: 0.2364
7. 25: ['Video Post-Editing']: 0.2306
8. 88: ['Explainer Video']: 0.2022
9. 66: ['Videography']: 0.1792
10. 9: ['Graphic Design']: 0.1681
11. 30: ['Adobe Illustrator']: 0.1681
12. 150: ['Product Photography']: 0.1512
13. 226: ['YouTube Shorts']: 0.1401
14. 231: ['YouTube Video']: 0.1401
15. 106: ['Photography']: 0.1379
16. 53: ['Illustration']: 0.1249
17. 434: ['DaVinci Resolve']: 0.1177
18. 110: ['Finance & Accounting']: 0.1066
19. 157: ['Photo Retouching']: 0.1061
20. 158: ['Photo Editing']: 0.1061

Cluster 2 Digital Marketing and Content Strategies Network

Degree Centrality - Top 5:

1. 7: ['Lead Generation']: 0.2667
2. 14: ['Content Writing']: 0.2417
3. 2: ['Social Media Marketing']: 0.2167
4. 15: ['Marketing Strategy']: 0.2167
5. 24: ['Sales']: 0.1833
6. 11: ['Search Engine Optimization']: 0.1750
7. 21: ['English']: 0.1750
8. 126: ['Email Marketing']: 0.1667
9. 23: ['Social Media Management']: 0.1417
10. 27: ['SEO Backlinking']: 0.1167
11. 38: ['Writing']: 0.1167
12. 22: ['Data Entry']: 0.1083
13. 51: ['Communications']: 0.1083
14. 59: ['Copywriting']: 0.1083
15. 90: ['Administrative Support']: 0.1000
16. 34: ['Instagram']: 0.0917
17. 70: ['Marketing Automation']: 0.0917
18. 101: ['Customer Service']: 0.0917
19. 13: ['Facebook']: 0.0833
20. 37: ['SEO Keyword Research']: 0.0833

Betweenness Centrality - Top 5:

1. 14: ['Content Writing']: 0.2139
2. 7: ['Lead Generation']: 0.1808
3. 21: ['English']: 0.1512
4. 15: ['Marketing Strategy']: 0.1497
5. 2: ['Social Media Marketing']: 0.1383
6. 126: ['Email Marketing']: 0.1118
7. 11: ['Search Engine Optimization']: 0.1117

8. 24: ['Sales']: 0.0786
9. 23: ['Social Media Management']: 0.0694
10. 54: ['Market Analysis']: 0.0670
11. 22: ['Data Entry']: 0.0423
12. 27: ['SEO Backlinking']: 0.0398
13. 38: ['Writing']: 0.0383
14. 145: ['Online Research']: 0.0383
15. 59: ['Copywriting']: 0.0324
16. 101: ['Customer Service']: 0.0321
17. 55: ['Virtual Assistance']: 0.0316
18. 324: ['Project Management']: 0.0310
19. 70: ['Marketing Automation']: 0.0309
20. 44: ['Google Analytics']: 0.0284

Closeness Centrality - Top 5:

1. 15: ['Marketing Strategy']: 0.5106
2. 7: ['Lead Generation']: 0.5063
3. 2: ['Social Media Marketing']: 0.5042
4. 14: ['Content Writing']: 0.5042
5. 24: ['Sales']: 0.4781
6. 11: ['Search Engine Optimization']: 0.4688
7. 59: ['Copywriting']: 0.4633
8. 23: ['Social Media Management']: 0.4598
9. 21: ['English']: 0.4580
10. 126: ['Email Marketing']: 0.4478
11. 101: ['Customer Service']: 0.4364
12. 70: ['Marketing Automation']: 0.4255
13. 55: ['Virtual Assistance']: 0.4196
14. 13: ['Facebook']: 0.4152
15. 90: ['Administrative Support']: 0.4138
16. 324: ['Project Management']: 0.4138
17. 22: ['Data Entry']: 0.4110
18. 34: ['Instagram']: 0.4110
19. 37: ['SEO Keyword Research']: 0.4096
20. 464: ['Affiliate Marketing']: 0.4054

PageRank - Top 5:

1. 7: ['Lead Generation']: 0.0341
2. 14: ['Content Writing']: 0.0308
3. 2: ['Social Media Marketing']: 0.0277
4. 15: ['Marketing Strategy']: 0.0276
5. 21: ['English']: 0.0247
6. 11: ['Search Engine Optimization']: 0.0232
7. 24: ['Sales']: 0.0232
8. 126: ['Email Marketing']: 0.0224
9. 23: ['Social Media Management']: 0.0185
10. 27: ['SEO Backlinking']: 0.0175
11. 38: ['Writing']: 0.0163
12. 51: ['Communications']: 0.0152
13. 22: ['Data Entry']: 0.0146
14. 59: ['Copywriting']: 0.0135
15. 70: ['Marketing Automation']: 0.0132
16. 90: ['Administrative Support']: 0.0132
17. 44: ['Google Analytics']: 0.0128
18. 101: ['Customer Service']: 0.0124
19. 34: ['Instagram']: 0.0119

20. 145: ['Online Research']: 0.0117

Authority (HITS) - Top 5:

1. 2: ['Social Media Marketing']: 0.0873
2. 15: ['Marketing Strategy']: 0.0634
3. 13: ['Facebook']: 0.0596
4. 11: ['Search Engine Optimization']: 0.0542
5. 23: ['Social Media Management']: 0.0438
6. 14: ['Content Writing']: 0.0436
7. 34: ['Instagram']: 0.0420
8. 59: ['Copywriting']: 0.0296
9. 24: ['Sales']: 0.0293
10. 7: ['Lead Generation']: 0.0284
11. 37: ['SEO Keyword Research']: 0.0265
12. 27: ['SEO Backlinking']: 0.0241
13. 84: ['Influencer Marketing']: 0.0195
14. 108: ['Social Media Content Creation']: 0.0176
15. 98: ['Media Relations']: 0.0171
16. 71: ['On-Page SEO']: 0.0171
17. 40: ['PPC Campaign Setup & Management']: 0.0170
18. 160: ['Public Relations']: 0.0158
19. 162: ['Press Release']: 0.0158
20. 154: ['Amazon Webstore']: 0.0144

Eigenvector Centrality - Top 5:

1. 7: ['Lead Generation']: 0.3037
2. 15: ['Marketing Strategy']: 0.3021
3. 2: ['Social Media Marketing']: 0.3007
4. 14: ['Content Writing']: 0.2568
5. 24: ['Sales']: 0.2529
6. 23: ['Social Media Management']: 0.2041
7. 59: ['Copywriting']: 0.2041
8. 126: ['Email Marketing']: 0.2037
9. 11: ['Search Engine Optimization']: 0.2028
10. 13: ['Facebook']: 0.1477
11. 34: ['Instagram']: 0.1449
12. 101: ['Customer Service']: 0.1315
13. 21: ['English']: 0.1308
14. 22: ['Data Entry']: 0.1123
15. 70: ['Marketing Automation']: 0.1098
16. 447: ['Direct Marketing']: 0.1092
17. 453: ['Internet Marketing']: 0.1092
18. 464: ['Affiliate Marketing']: 0.1089
19. 55: ['Virtual Assistance']: 0.1045
20. 37: ['SEO Keyword Research']: 0.0999

Cluster 3 Data-Driven Artificial Intelligence and Financial Applications Ecosystem

Degree Centrality - Top 5:

1. 5: ['Microsoft Excel']: 0.3846
2. 39: ['Python']: 0.3385
3. 28: ['API']: 0.2154
4. 33: ['Artificial Intelligence']: 0.2154
5. 49: ['Accounting']: 0.2154
6. 29: ['Machine Learning']: 0.2000
7. 31: ['Deep Learning']: 0.2000

8. 91: ['C#']: 0.1692
9. 42: ['Financial Analysis']: 0.1538
10. 94: ['SQL']: 0.1538
11. 32: ['Data Analysis']: 0.1385
12. 86: ['AI Agent Development']: 0.1385
13. 148: ['Generative AI']: 0.1231
14. 152: ['Bookkeeping']: 0.1231
15. 161: ['API Integration']: 0.1231
16. 81: ['TensorFlow']: 0.1231
17. 190: ['Financial Accounting']: 0.1077
18. 113: ['Microsoft Power BI']: 0.1077
19. 114: ['Data Visualization']: 0.1077
20. 263: ['Windows Forms']: 0.0923

Betweenness Centrality - Top 5:

1. 5: ['Microsoft Excel']: 0.4251
2. 39: ['Python']: 0.4176
3. 86: ['AI Agent Development']: 0.2192
4. 28: ['API']: 0.1533
5. 94: ['SQL']: 0.1222
6. 91: ['C#']: 0.0958
7. 161: ['API Integration']: 0.0894
8. 58: ['Mathematics']: 0.0606
9. 29: ['Machine Learning']: 0.0395
10. 31: ['Deep Learning']: 0.0395
11. 49: ['Accounting']: 0.0369
12. 33: ['Artificial Intelligence']: 0.0358
13. 353: ['Ruby on Rails']: 0.0262
14. 32: ['Data Analysis']: 0.0219
15. 148: ['Generative AI']: 0.0190
16. 152: ['Bookkeeping']: 0.0176
17. 190: ['Financial Accounting']: 0.0167
18. 136: ['MySQL']: 0.0156
19. 137: ['Linux System Administration']: 0.0156
20. 113: ['Microsoft Power BI']: 0.0151

Closeness Centrality - Top 5:

1. 39: ['Python']: 0.5603
2. 5: ['Microsoft Excel']: 0.5372
3. 91: ['C#']: 0.4643
4. 29: ['Machine Learning']: 0.4545
5. 31: ['Deep Learning']: 0.4545
6. 94: ['SQL']: 0.4483
7. 32: ['Data Analysis']: 0.4392
8. 81: ['TensorFlow']: 0.4392
9. 263: ['Windows Forms']: 0.4333
10. 113: ['Microsoft Power BI']: 0.4333
11. 114: ['Data Visualization']: 0.4333
12. 331: ['Statistics']: 0.4305
13. 28: ['API']: 0.4276
14. 33: ['Artificial Intelligence']: 0.4194
15. 86: ['AI Agent Development']: 0.4062
16. 148: ['Generative AI']: 0.4012
17. 350: ['OpenAI API']: 0.3963
18. 352: ['Twilio API']: 0.3963
19. 356: ['Chatbot Development']: 0.3963

20. 49: ['Accounting']: 0.3869

PageRank - Top 5:

1. 5: ['Microsoft Excel']: 0.0509
2. 39: ['Python']: 0.0438
3. 28: ['API']: 0.0336
4. 49: ['Accounting']: 0.0311
5. 33: ['Artificial Intelligence']: 0.0284
6. 29: ['Machine Learning']: 0.0257
7. 31: ['Deep Learning']: 0.0257
8. 91: ['C#']: 0.0231
9. 161: ['API Integration']: 0.0230
10. 94: ['SQL']: 0.0227
11. 42: ['Financial Analysis']: 0.0223
12. 86: ['AI Agent Development']: 0.0219
13. 32: ['Data Analysis']: 0.0194
14. 152: ['Bookkeeping']: 0.0183
15. 148: ['Generative AI']: 0.0173
16. 58: ['Mathematics']: 0.0168
17. 136: ['MySQL']: 0.0165
18. 137: ['Linux System Administration']: 0.0165
19. 81: ['TensorFlow']: 0.0162
20. 190: ['Financial Accounting']: 0.0161

Authority (HITS) - Top 5:

1. 29: ['Machine Learning']: 0.1241
2. 31: ['Deep Learning']: 0.1220
3. 33: ['Artificial Intelligence']: 0.1085
4. 68: ['Neural Network']: 0.0754
5. 81: ['TensorFlow']: 0.0650
6. 39: ['Python']: 0.0510
7. 5: ['Microsoft Excel']: 0.0499
8. 178: ['Artificial Neural Network']: 0.0418
9. 91: ['C#']: 0.0316
10. 148: ['Generative AI']: 0.0307
11. 263: ['Windows Forms']: 0.0296
12. 242: ['Stable Diffusion']: 0.0284
13. 42: ['Financial Analysis']: 0.0219
14. 49: ['Accounting']: 0.0183
15. 32: ['Data Analysis']: 0.0177
16. 360: ['Recommendation System']: 0.0138
17. 266: ['Data Science']: 0.0138
18. 58: ['Mathematics']: 0.0119
19. 138: ['Financial Modeling']: 0.0118
20. 28: ['API']: 0.0109

Eigenvector Centrality - Top 5:

1. 39: ['Python']: 0.3727
2. 5: ['Microsoft Excel']: 0.3645
3. 29: ['Machine Learning']: 0.2681
4. 31: ['Deep Learning']: 0.2681
5. 33: ['Artificial Intelligence']: 0.2394
6. 91: ['C#']: 0.2119
7. 81: ['TensorFlow']: 0.2104
8. 263: ['Windows Forms']: 0.1771
9. 32: ['Data Analysis']: 0.1670

10. 148: ['Generative AI']: 0.1528
11. 94: ['SQL']: 0.1494
12. 28: ['API']: 0.1484
13. 49: ['Accounting']: 0.1447
14. 113: ['Microsoft Power BI']: 0.1427
15. 114: ['Data Visualization']: 0.1427
16. 331: ['Statistics']: 0.1398
17. 266: ['Data Science']: 0.1339
18. 360: ['Recommendation System']: 0.1339
19. 42: ['Financial Analysis']: 0.1322
20. 68: ['Neural Network']: 0.1127

Cluster 4 Product and Space Orientated Design Engineering Ecosystem

Degree Centrality - Top 5:

1. 26: ['Product Design']: 0.3182
2. 174: ['Mechanical Engineering']: 0.2500
3. 69: ['3D Design']: 0.2500
4. 78: ['Autodesk AutoCAD']: 0.2500
5. 79: ['Architectural Design']: 0.2500
6. 264: ['CAD']: 0.2273
7. 100: ['HVAC System Design']: 0.2045
8. 103: ['LEED Consulting']: 0.2045
9. 35: ['Fashion Design']: 0.1818
10. 36: ['Accessories']: 0.1818
11. 243: ['Drafting']: 0.1818
12. 172: ['Print Design']: 0.1364
13. 184: ['Clothing']: 0.1364
14. 185: ['Tech Pack']: 0.1364
15. 186: ['Fashion & Apparel']: 0.1364
16. 321: ['Floor Plan Design']: 0.1364
17. 204: ['2D Design']: 0.1364
18. 355: ['Design for Manufacturing']: 0.1364
19. 357: ['SolidWorks']: 0.1364
20. 358: ['CATIA']: 0.1364

Betweenness Centrality - Top 5:

1. 264: ['CAD']: 0.4736
2. 174: ['Mechanical Engineering']: 0.4440
3. 103: ['LEED Consulting']: 0.3330
4. 26: ['Product Design']: 0.3073
5. 69: ['3D Design']: 0.2082
6. 100: ['HVAC System Design']: 0.2061
7. 78: ['Autodesk AutoCAD']: 0.1564
8. 79: ['Architectural Design']: 0.1564
9. 204: ['2D Design']: 0.0706
10. 321: ['Floor Plan Design']: 0.0181
11. 243: ['Drafting']: 0.0103
12. 35: ['Fashion Design']: 0.0028
13. 36: ['Accessories']: 0.0028
14. 315: ['3D Rendering']: 0.0007
15. 316: ['Autodesk 3ds Max']: 0.0007
16. 257: ['Energy Engineering']: 0.0000
17. 258: ['eQUEST']: 0.0000
18. 259: ['Energy Optimization']: 0.0000

- 19. 260: ['LEED Standard']: 0.0000
- 20. 397: ['3D Modeling']: 0.0000

Closeness Centrality - Top 5:

- 1. 264: ['CAD']: 0.4536
- 2. 174: ['Mechanical Engineering']: 0.4444
- 3. 26: ['Product Design']: 0.4190
- 4. 78: ['Autodesk AutoCAD']: 0.3894
- 5. 79: ['Architectural Design']: 0.3894
- 6. 355: ['Design for Manufacturing']: 0.3894
- 7. 357: ['SolidWorks']: 0.3894
- 8. 358: ['CATIA']: 0.3894
- 9. 359: ['KeyShot']: 0.3894
- 10. 204: ['2D Design']: 0.3729
- 11. 103: ['LEED Consulting']: 0.3667
- 12. 321: ['Floor Plan Design']: 0.3577
- 13. 257: ['Energy Engineering']: 0.3411
- 14. 258: ['eQUEST']: 0.3411
- 15. 259: ['Energy Optimization']: 0.3411
- 16. 260: ['LEED Standard']: 0.3411
- 17. 35: ['Fashion Design']: 0.3121
- 18. 36: ['Accessories']: 0.3121
- 19. 69: ['3D Design']: 0.3121
- 20. 172: ['Print Design']: 0.3077

PageRank - Top 5:

- 1. 26: ['Product Design']: 0.0433
- 2. 69: ['3D Design']: 0.0399
- 3. 78: ['Autodesk AutoCAD']: 0.0369
- 4. 79: ['Architectural Design']: 0.0369
- 5. 174: ['Mechanical Engineering']: 0.0358
- 6. 100: ['HVAC System Design']: 0.0348
- 7. 103: ['LEED Consulting']: 0.0330
- 8. 264: ['CAD']: 0.0316
- 9. 243: ['Drafting']: 0.0276
- 10. 35: ['Fashion Design']: 0.0264
- 11. 36: ['Accessories']: 0.0264
- 12. 315: ['3D Rendering']: 0.0213
- 13. 316: ['Autodesk 3ds Max']: 0.0213
- 14. 321: ['Floor Plan Design']: 0.0208
- 15. 245: ['IES']: 0.0207
- 16. 247: ['Architecture']: 0.0207
- 17. 249: ['EnergyPlus']: 0.0207
- 18. 251: ['Energy Simulation']: 0.0207
- 19. 252: ['AutoCAD HVAC Loads Calculations']: 0.0207
- 20. 204: ['2D Design']: 0.0207

Authority (HITS) - Top 5:

- 1. 26: ['Product Design']: 0.1649
- 2. 35: ['Fashion Design']: 0.1641
- 3. 36: ['Accessories']: 0.1641
- 4. 47: ['Activewear']: 0.1274
- 5. 48: ['Made-to-Measure']: 0.1274
- 6. 186: ['Fashion & Apparel']: 0.0534
- 7. 172: ['Print Design']: 0.0534
- 8. 184: ['Clothing']: 0.0534

9. 185: ['Tech Pack']: 0.0534
10. 174: ['Mechanical Engineering']: 0.0060
11. 264: ['CAD']: 0.0059
12. 358: ['CATIA']: 0.0058
13. 359: ['KeyShot']: 0.0058
14. 355: ['Design for Manufacturing']: 0.0058
15. 357: ['SolidWorks']: 0.0058
16. 103: ['LEED Consulting']: 0.0005
17. 258: ['eQUEST']: 0.0005
18. 260: ['LEED Standard']: 0.0005
19. 259: ['Energy Optimization']: 0.0005
20. 257: ['Energy Engineering']: 0.0005

Eigenvector Centrality - Top 5:

1. 26: ['Product Design']: 0.3740
2. 264: ['CAD']: 0.3007
3. 174: ['Mechanical Engineering']: 0.2589
4. 78: ['Autodesk AutoCAD']: 0.2184
5. 79: ['Architectural Design']: 0.2184
6. 355: ['Design for Manufacturing']: 0.2110
7. 357: ['SolidWorks']: 0.2110
8. 358: ['CATIA']: 0.2110
9. 359: ['KeyShot']: 0.2110
10. 35: ['Fashion Design']: 0.2074
11. 36: ['Accessories']: 0.2074
12. 172: ['Print Design']: 0.1783
13. 184: ['Clothing']: 0.1783
14. 185: ['Tech Pack']: 0.1783
15. 186: ['Fashion & Apparel']: 0.1783
16. 204: ['2D Design']: 0.1625
17. 243: ['Drafting']: 0.1610
18. 321: ['Floor Plan Design']: 0.1566
19. 69: ['3D Design']: 0.1516
20. 47: ['Activewear']: 0.1228

Africa 2025

NetworkX modularity: 0.7374

Cluster 0 - Node : 49, Edge: 214

Cluster 1 - Node : 47, Edge: 125

Cluster 3 - Node : 32, Edge: 88

Cluster 2 - Node : 33, Edge: 117

Cluster 0: Modern Web and Application Development Ecosystem

Degree Centrality - Top 5:

1. 0: ['API']: 0.5283
2. 1: ['JavaScript']: 0.4906
3. 3: ['Web Development']: 0.4528
4. 14: ['HTML']: 0.4528
5. 2: ['PHP']: 0.4151
6. 22: ['CSS']: 0.3962
7. 7: ['WordPress']: 0.3585
8. 13: ['Adobe Photoshop']: 0.2830

9. 18: ['Web Design']: 0.2453
10. 41: ['MySQL']: 0.2453
11. 42: ['Custom PHP']: 0.2453
12. 31: ['API Integration']: 0.2264
13. 20: ['Shopify']: 0.2075
14. 34: ['Wix']: 0.1887
15. 37: ['cPanel']: 0.1887
16. 21: ['Search Engine Optimization']: 0.1698
17. 84: ['MySQL Programming']: 0.1698
18. 85: ['PHP Script']: 0.1698
19. 86: ['SQL']: 0.1698
20. 174: ['Laravel']: 0.1509

Betweenness Centrality - Top 5:

1. 0: ['API']: 0.1904
2. 13: ['Adobe Photoshop']: 0.1879
3. 3: ['Web Development']: 0.1574
4. 1: ['JavaScript']: 0.1503
5. 14: ['HTML']: 0.1197
6. 7: ['WordPress']: 0.0896
7. 2: ['PHP']: 0.0863
8. 21: ['Search Engine Optimization']: 0.0824
9. 51: ['Visual Basic for Applications']: 0.0740
10. 18: ['Web Design']: 0.0673
11. 22: ['CSS']: 0.0669
12. 23: ['SEO Keyword Research']: 0.0260
13. 20: ['Shopify']: 0.0260
14. 34: ['Wix']: 0.0178
15. 31: ['API Integration']: 0.0044
16. 41: ['MySQL']: 0.0044
17. 42: ['Custom PHP']: 0.0044
18. 15: ['Python']: 0.0038
19. 174: ['Laravel']: 0.0020
20. 43: ['Adobe Illustrator']: 0.0018

Closeness Centrality - Top 5:

1. 1: ['JavaScript']: 0.6386
2. 3: ['Web Development']: 0.6310
3. 0: ['API']: 0.6163
4. 2: ['PHP']: 0.6092
5. 14: ['HTML']: 0.6092
6. 7: ['WordPress']: 0.5955
7. 18: ['Web Design']: 0.5579
8. 22: ['CSS']: 0.5579
9. 13: ['Adobe Photoshop']: 0.5408
10. 41: ['MySQL']: 0.5096
11. 42: ['Custom PHP']: 0.5096
12. 31: ['API Integration']: 0.5048
13. 37: ['cPanel']: 0.4862
14. 174: ['Laravel']: 0.4862
15. 199: ['Back-End Development']: 0.4818
16. 20: ['Shopify']: 0.4690
17. 15: ['Python']: 0.4609
18. 34: ['Wix']: 0.4569
19. 84: ['MySQL Programming']: 0.4492
20. 85: ['PHP Script']: 0.4492

PageRank - Top 5:

1. 0: ['API']: 0.0569
2. 1: ['JavaScript']: 0.0506
3. 3: ['Web Development']: 0.0471
4. 14: ['HTML']: 0.0462
5. 2: ['PHP']: 0.0420
6. 22: ['CSS']: 0.0410
7. 7: ['WordPress']: 0.0367
8. 13: ['Adobe Photoshop']: 0.0365
9. 18: ['Web Design']: 0.0264
10. 20: ['Shopify']: 0.0243
11. 41: ['MySQL']: 0.0242
12. 42: ['Custom PHP']: 0.0242
13. 31: ['API Integration']: 0.0231
14. 21: ['Search Engine Optimization']: 0.0221
15. 34: ['Wix']: 0.0221
16. 37: ['cPanel']: 0.0192
17. 23: ['SEO Keyword Research']: 0.0186
18. 51: ['Visual Basic for Applications']: 0.0178
19. 43: ['Adobe Illustrator']: 0.0177
20. 84: ['MySQL Programming']: 0.0175

Authority (HITS) - Top 5:

1. 1: ['JavaScript']: 0.1093
2. 2: ['PHP']: 0.1010
3. 0: ['API']: 0.1002
4. 3: ['Web Development']: 0.0601
5. 14: ['HTML']: 0.0533
6. 7: ['WordPress']: 0.0511
7. 15: ['Python']: 0.0401
8. 42: ['Custom PHP']: 0.0394
9. 41: ['MySQL']: 0.0394
10. 31: ['API Integration']: 0.0391
11. 37: ['cPanel']: 0.0353
12. 86: ['SQL']: 0.0307
13. 85: ['PHP Script']: 0.0307
14. 84: ['MySQL Programming']: 0.0307
15. 22: ['CSS']: 0.0274
16. 18: ['Web Design']: 0.0211
17. 13: ['Adobe Photoshop']: 0.0160
18. 51: ['Visual Basic for Applications']: 0.0152
19. 101: ['Visual Basic']: 0.0152
20. 65: ['Chatbot Development']: 0.0142

Eigenvector Centrality - Top 5:

1. 1: ['JavaScript']: 0.3057
2. 0: ['API']: 0.3051
3. 2: ['PHP']: 0.2890
4. 14: ['HTML']: 0.2873
5. 3: ['Web Development']: 0.2688
6. 22: ['CSS']: 0.2518
7. 7: ['WordPress']: 0.2485
8. 41: ['MySQL']: 0.2210
9. 42: ['Custom PHP']: 0.2210
10. 31: ['API Integration']: 0.2010

11. 18: ['Web Design']: 0.1822
12. 37: ['cPanel']: 0.1717
13. 84: ['MySQL Programming']: 0.1547
14. 85: ['PHP Script']: 0.1547
15. 86: ['SQL']: 0.1547
16. 13: ['Adobe Photoshop']: 0.1412
17. 174: ['Laravel']: 0.1347
18. 199: ['Back-End Development']: 0.1287
19. 20: ['Shopify']: 0.1017
20. 15: ['Python']: 0.0987

Cluster 1: Digital Content and Social Media Design Ecosystem

Degree Centrality - Top 5:

1. 8: ['Graphic Design']: 0.3696
2. 10: ['Instagram']: 0.3043
3. 5: ['Facebook']: 0.2391
4. 4: ['Social Media Marketing']: 0.1957
5. 27: ['Content Writing']: 0.1957
6. 38: ['LinkedIn']: 0.1739
7. 45: ['Influencer Marketing']: 0.1739
8. 43: ['Adobe Illustrator']: 0.1522
9. 50: ['Brand Identity & Guidelines']: 0.1522
10. 16: ['Marketing Strategy']: 0.1304
11. 24: ['Social Media Management']: 0.1304
12. 48: ['Illustration']: 0.1304
13. 49: ['Animation']: 0.1304
14. 98: ['Social Media Content']: 0.1304
15. 129: ['Page Layout Design']: 0.1087
16. 145: ['Instagram Story']: 0.1087
17. 146: ['Pinterest']: 0.1087
18. 188: ['Health & Wellness']: 0.1087
19. 190: ['Fashion & Beauty']: 0.1087
20. 193: ['Sports & Recreation']: 0.1087

Betweenness Centrality - Top 5:

1. 8: ['Graphic Design']: 0.6419
2. 48: ['Illustration']: 0.2319
3. 27: ['Content Writing']: 0.1832
4. 38: ['LinkedIn']: 0.1623
5. 50: ['Brand Identity & Guidelines']: 0.1623
6. 10: ['Instagram']: 0.1291
7. 5: ['Facebook']: 0.1262
8. 49: ['Animation']: 0.1246
9. 4: ['Social Media Marketing']: 0.1215
10. 45: ['Influencer Marketing']: 0.0532
11. 16: ['Marketing Strategy']: 0.0454
12. 43: ['Adobe Illustrator']: 0.0367
13. 98: ['Social Media Content']: 0.0019
14. 24: ['Social Media Management']: 0.0016
15. 32: ['Social Media Content Creation']: 0.0003
16. 129: ['Page Layout Design']: 0.0000
17. 135: ['Packaging Design']: 0.0000
18. 137: ['Covers & Packaging']: 0.0000
19. 145: ['Instagram Story']: 0.0000
20. 146: ['Pinterest']: 0.0000

Closeness Centrality - Top 5:

1. 8: ['Graphic Design']: 0.5610
2. 5: ['Facebook']: 0.4894
3. 4: ['Social Media Marketing']: 0.4646
4. 27: ['Content Writing']: 0.4600
5. 16: ['Marketing Strategy']: 0.4510
6. 38: ['LinkedIn']: 0.4423
7. 43: ['Adobe Illustrator']: 0.4144
8. 48: ['Illustration']: 0.4071
9. 10: ['Instagram']: 0.3932
10. 50: ['Brand Identity & Guidelines']: 0.3932
11. 194: ['Drawing']: 0.3866
12. 183: ['Corporate Brand Identity']: 0.3802
13. 129: ['Page Layout Design']: 0.3740
14. 116: ['Book']: 0.3740
15. 119: ['Layout Design']: 0.3740
16. 123: ['Adobe InDesign']: 0.3740
17. 126: ['Print Design']: 0.3740
18. 135: ['Packaging Design']: 0.3680
19. 137: ['Covers & Packaging']: 0.3680
20. 98: ['Social Media Content']: 0.3680

PageRank - Top 5:

1. 8: ['Graphic Design']: 0.0609
2. 10: ['Instagram']: 0.0455
3. 5: ['Facebook']: 0.0362
4. 27: ['Content Writing']: 0.0333
5. 49: ['Animation']: 0.0309
6. 4: ['Social Media Marketing']: 0.0301
7. 38: ['LinkedIn']: 0.0297
8. 50: ['Brand Identity & Guidelines']: 0.0294
9. 43: ['Adobe Illustrator']: 0.0290
10. 48: ['Illustration']: 0.0270
11. 45: ['Influencer Marketing']: 0.0269
12. 98: ['Social Media Content']: 0.0211
13. 16: ['Marketing Strategy']: 0.0208
14. 24: ['Social Media Management']: 0.0207
15. 129: ['Page Layout Design']: 0.0195
16. 116: ['Book']: 0.0195
17. 119: ['Layout Design']: 0.0195
18. 123: ['Adobe InDesign']: 0.0195
19. 126: ['Print Design']: 0.0195
20. 115: ['Catalog']: 0.0186

Authority (HITS) - Top 5:

1. 5: ['Facebook']: 0.1523
2. 4: ['Social Media Marketing']: 0.1508
3. 10: ['Instagram']: 0.1229
4. 32: ['Social Media Content Creation']: 0.0781
5. 16: ['Marketing Strategy']: 0.0774
6. 24: ['Social Media Management']: 0.0691
7. 8: ['Graphic Design']: 0.0532
8. 38: ['LinkedIn']: 0.0480
9. 98: ['Social Media Content']: 0.0340
10. 45: ['Influencer Marketing']: 0.0339

11. 27: ['Content Writing']: 0.0284
12. 145: ['Instagram Story']: 0.0250
13. 146: ['Pinterest']: 0.0250
14. 43: ['Adobe Illustrator']: 0.0085
15. 119: ['Layout Design']: 0.0065
16. 126: ['Print Design']: 0.0065
17. 116: ['Book']: 0.0065
18. 129: ['Page Layout Design']: 0.0065
19. 123: ['Adobe InDesign']: 0.0065
20. 135: ['Packaging Design']: 0.0052

Eigenvector Centrality - Top 5:

1. 10: ['Instagram']: 0.3748
2. 5: ['Facebook']: 0.3506
3. 8: ['Graphic Design']: 0.3050
4. 4: ['Social Media Marketing']: 0.2948
5. 27: ['Content Writing']: 0.2374
6. 45: ['Influencer Marketing']: 0.2264
7. 24: ['Social Media Management']: 0.2162
8. 16: ['Marketing Strategy']: 0.2151
9. 98: ['Social Media Content']: 0.2048
10. 145: ['Instagram Story']: 0.1847
11. 146: ['Pinterest']: 0.1847
12. 38: ['LinkedIn']: 0.1822
13. 32: ['Social Media Content Creation']: 0.1673
14. 188: ['Health & Wellness']: 0.1391
15. 190: ['Fashion & Beauty']: 0.1391
16. 193: ['Sports & Recreation']: 0.1391
17. 198: ['TikTok']: 0.1391
18. 224: ['Ethical Hacking']: 0.1210
19. 43: ['Adobe Illustrator']: 0.0960
20. 129: ['Page Layout Design']: 0.0918

Cluster 2: Data-Driven Support Services and Statistical Applications Ecosystem

Degree Centrality - Top 5:

1. 12: ['Data Entry']: 0.4839
2. 26: ['Microsoft Excel']: 0.3871
3. 6: ['Lead Generation']: 0.3548
4. 17: ['List Building']: 0.2903
5. 60: ['R']: 0.2903
6. 46: ['Statistics']: 0.2581
7. 47: ['Data Science']: 0.2581
8. 28: ['Administrative Support']: 0.2258
9. 130: ['Quantitative Analysis']: 0.1935
10. 136: ['Communications']: 0.1935
11. 25: ['Data Scraping']: 0.1935
12. 93: ['Virtual Assistance']: 0.1935
13. 124: ['Data Analysis']: 0.1935
14. 127: ['SAS']: 0.1935
15. 35: ['Customer Service']: 0.1613
16. 80: ['Data Mining']: 0.1613
17. 191: ['Document Conversion']: 0.1290
18. 192: ['Research Proposals']: 0.1290
19. 195: ['Molecular Biology']: 0.1290

20. 196: ['Social Media Lead Generation']: 0.1290

Betweenness Centrality - Top 5:

1. 12: ['Data Entry']: 0.6265
2. 26: ['Microsoft Excel']: 0.4957
3. 6: ['Lead Generation']: 0.1767
4. 46: ['Statistics']: 0.0602
5. 47: ['Data Science']: 0.0602
6. 17: ['List Building']: 0.0520
7. 35: ['Customer Service']: 0.0337
8. 136: ['Communications']: 0.0308
9. 28: ['Administrative Support']: 0.0229
10. 60: ['R']: 0.0226
11. 25: ['Data Scraping']: 0.0165
12. 93: ['Virtual Assistance']: 0.0043
13. 130: ['Quantitative Analysis']: 0.0000
14. 139: ['Marketing']: 0.0000
15. 172: ['Sales']: 0.0000
16. 173: ['Relationship Management']: 0.0000
17. 180: ['IBM SPSS']: 0.0000
18. 181: ['Analytics']: 0.0000
19. 191: ['Document Conversion']: 0.0000
20. 192: ['Research Proposals']: 0.0000

Closeness Centrality - Top 5:

1. 12: ['Data Entry']: 0.6327
2. 26: ['Microsoft Excel']: 0.5636
3. 6: ['Lead Generation']: 0.4769
4. 94: ['Spreadsheet Software']: 0.4697
5. 95: ['Google Docs']: 0.4697
6. 17: ['List Building']: 0.4627
7. 28: ['Administrative Support']: 0.4559
8. 25: ['Data Scraping']: 0.4429
9. 80: ['Data Mining']: 0.4366
10. 191: ['Document Conversion']: 0.4306
11. 196: ['Social Media Lead Generation']: 0.4306
12. 136: ['Communications']: 0.4247
13. 93: ['Virtual Assistance']: 0.4247
14. 35: ['Customer Service']: 0.4189
15. 60: ['R']: 0.4133
16. 215: ['Email Communication']: 0.4133
17. 216: ['Castilian Spanish']: 0.4133
18. 46: ['Statistics']: 0.4079
19. 47: ['Data Science']: 0.4079
20. 130: ['Quantitative Analysis']: 0.3974

PageRank - Top 5:

1. 12: ['Data Entry']: 0.0792
2. 6: ['Lead Generation']: 0.0642
3. 26: ['Microsoft Excel']: 0.0609
4. 17: ['List Building']: 0.0501
5. 60: ['R']: 0.0457
6. 46: ['Statistics']: 0.0416
7. 47: ['Data Science']: 0.0416
8. 28: ['Administrative Support']: 0.0385
9. 136: ['Communications']: 0.0358

10. 93: ['Virtual Assistance']: 0.0343
11. 25: ['Data Scraping']: 0.0338
12. 130: ['Quantitative Analysis']: 0.0309
13. 124: ['Data Analysis']: 0.0309
14. 127: ['SAS']: 0.0309
15. 35: ['Customer Service']: 0.0301
16. 80: ['Data Mining']: 0.0283
17. 215: ['Email Communication']: 0.0243
18. 216: ['Castilian Spanish']: 0.0243
19. 191: ['Document Conversion']: 0.0240
20. 196: ['Social Media Lead Generation']: 0.0240

Authority (HITS) - Top 5:

1. 6: ['Lead Generation']: 0.1366
2. 17: ['List Building']: 0.1357
3. 25: ['Data Scraping']: 0.1242
4. 12: ['Data Entry']: 0.1206
5. 28: ['Administrative Support']: 0.1054
6. 80: ['Data Mining']: 0.0850
7. 71: ['Prospect List']: 0.0541
8. 93: ['Virtual Assistance']: 0.0330
9. 35: ['Customer Service']: 0.0329
10. 26: ['Microsoft Excel']: 0.0248
11. 95: ['Google Docs']: 0.0214
12. 94: ['Spreadsheet Software']: 0.0214
13. 191: ['Document Conversion']: 0.0140
14. 196: ['Social Media Lead Generation']: 0.0140
15. 172: ['Sales']: 0.0094
16. 173: ['Relationship Management']: 0.0094
17. 202: ['Market Research']: 0.0092
18. 60: ['R']: 0.0064
19. 47: ['Data Science']: 0.0059
20. 46: ['Statistics']: 0.0059

Eigenvector Centrality - Top 5:

1. 12: ['Data Entry']: 0.3677
2. 26: ['Microsoft Excel']: 0.3280
3. 60: ['R']: 0.2526
4. 6: ['Lead Generation']: 0.2497
5. 17: ['List Building']: 0.2392
6. 46: ['Statistics']: 0.2278
7. 47: ['Data Science']: 0.2278
8. 28: ['Administrative Support']: 0.2175
9. 130: ['Quantitative Analysis']: 0.2086
10. 124: ['Data Analysis']: 0.2086
11. 127: ['SAS']: 0.2086
12. 25: ['Data Scraping']: 0.1944
13. 80: ['Data Mining']: 0.1821
14. 93: ['Virtual Assistance']: 0.1526
15. 191: ['Document Conversion']: 0.1436
16. 196: ['Social Media Lead Generation']: 0.1436
17. 136: ['Communications']: 0.1299
18. 35: ['Customer Service']: 0.1299
19. 192: ['Research Proposals']: 0.1169
20. 195: ['Molecular Biology']: 0.1169

Cluster 3 Mobile Application Development and Cross Platform Ecosystem

Degree Centrality - Top 5:

1. 9: ['Mobile App Development']: 0.8125
2. 11: ['Flutter']: 0.6250
3. 19: ['English']: 0.3438
4. 39: ['iOS']: 0.3125
5. 96: ['React Native']: 0.2812
6. 97: ['Swift']: 0.2812
7. 132: ['Dart']: 0.2500
8. 128: ['iOS Development']: 0.2188
9. 131: ['Kotlin']: 0.2188
10. 44: ['Translation']: 0.2188
11. 122: ['Hybrid App']: 0.2188
12. 138: ['Mobile App Localization']: 0.1875
13. 140: ['Proofreading']: 0.1875
14. 141: ['Urdu']: 0.1875
15. 142: ['JSON']: 0.1875
16. 106: ['Firebase']: 0.1875
17. 107: ['Firebase Cloud Firestore']: 0.1875
18. 108: ['Google Cloud Platform']: 0.1875
19. 133: ['Tablet']: 0.1562
20. 33: ['Android']: 0.1562

Betweenness Centrality - Top 5:

1. 9: ['Mobile App Development']: 0.6022
2. 19: ['English']: 0.2722
3. 11: ['Flutter']: 0.1183
4. 44: ['Translation']: 0.0625
5. 39: ['iOS']: 0.0101
6. 132: ['Dart']: 0.0060
7. 96: ['React Native']: 0.0040
8. 97: ['Swift']: 0.0040
9. 106: ['Firebase']: 0.0004
10. 107: ['Firebase Cloud Firestore']: 0.0004
11. 108: ['Google Cloud Platform']: 0.0004
12. 128: ['iOS Development']: 0.0000
13. 131: ['Kotlin']: 0.0000
14. 133: ['Tablet']: 0.0000
15. 138: ['Mobile App Localization']: 0.0000
16. 140: ['Proofreading']: 0.0000
17. 141: ['Urdu']: 0.0000
18. 142: ['JSON']: 0.0000
19. 156: ['Kannada']: 0.0000
20. 33: ['Android']: 0.0000

Closeness Centrality - Top 5:

1. 9: ['Mobile App Development']: 0.8421
2. 11: ['Flutter']: 0.6400
3. 19: ['English']: 0.6038
4. 44: ['Translation']: 0.5614
5. 138: ['Mobile App Localization']: 0.5517
6. 140: ['Proofreading']: 0.5517
7. 141: ['Urdu']: 0.5517
8. 142: ['JSON']: 0.5517
9. 39: ['iOS']: 0.5333

10. 96: ['React Native']: 0.5246
11. 97: ['Swift']: 0.5246
12. 132: ['Dart']: 0.5161
13. 128: ['iOS Development']: 0.5079
14. 131: ['Kotlin']: 0.5079
15. 122: ['Hybrid App']: 0.5079
16. 106: ['Firebase']: 0.5000
17. 107: ['Firebase Cloud Firestore']: 0.5000
18. 108: ['Google Cloud Platform']: 0.5000
19. 133: ['Tablet']: 0.4923
20. 33: ['Android']: 0.4923

PageRank - Top 5:

1. 9: ['Mobile App Development']: 0.0984
2. 11: ['Flutter']: 0.0748
3. 19: ['English']: 0.0499
4. 39: ['iOS']: 0.0380
5. 44: ['Translation']: 0.0347
6. 96: ['React Native']: 0.0346
7. 97: ['Swift']: 0.0346
8. 132: ['Dart']: 0.0321
9. 138: ['Mobile App Localization']: 0.0275
10. 140: ['Proofreading']: 0.0275
11. 141: ['Urdu']: 0.0275
12. 142: ['JSON']: 0.0275
13. 128: ['iOS Development']: 0.0273
14. 131: ['Kotlin']: 0.0273
15. 122: ['Hybrid App']: 0.0273
16. 54: ['German']: 0.0263
17. 68: ['Male']: 0.0263
18. 74: ['Female']: 0.0263
19. 78: ['Voice Acting']: 0.0263
20. 83: ['Voice-Over']: 0.0263

Authority (HITS) - Top 5:

1. 9: ['Mobile App Development']: 0.1192
2. 11: ['Flutter']: 0.1142
3. 39: ['iOS']: 0.0650
4. 96: ['React Native']: 0.0463
5. 97: ['Swift']: 0.0463
6. 55: ['In-App Purchases']: 0.0408
7. 79: ['TV Set']: 0.0408
8. 63: ['Offline Functionality']: 0.0408
9. 69: ['Android App Development']: 0.0408
10. 122: ['Hybrid App']: 0.0405
11. 128: ['iOS Development']: 0.0405
12. 131: ['Kotlin']: 0.0405
13. 132: ['Dart']: 0.0300
14. 133: ['Tablet']: 0.0295
15. 33: ['Android']: 0.0295
16. 106: ['Firebase']: 0.0279
17. 108: ['Google Cloud Platform']: 0.0279
18. 107: ['Firebase Cloud Firestore']: 0.0279
19. 114: ['Google Maps']: 0.0274
20. 19: ['English']: 0.0168

Eigenvector Centrality - Top 5:

1. 9: ['Mobile App Development']: 0.4455
2. 11: ['Flutter']: 0.3874
3. 39: ['iOS']: 0.2533
4. 96: ['React Native']: 0.2355
5. 97: ['Swift']: 0.2355
6. 128: ['iOS Development']: 0.2065
7. 131: ['Kotlin']: 0.2065
8. 122: ['Hybrid App']: 0.2065
9. 132: ['Dart']: 0.1937
10. 106: ['Firebase']: 0.1542
11. 107: ['Firebase Cloud Firestore']: 0.1542
12. 108: ['Google Cloud Platform']: 0.1542
13. 209: ['Ionic Framework']: 0.1527
14. 210: ['Java']: 0.1527
15. 133: ['Tablet']: 0.1499
16. 33: ['Android']: 0.1499
17. 114: ['Google Maps']: 0.1358
18. 55: ['In-App Purchases']: 0.1273
19. 63: ['Offline Functionality']: 0.1273
20. 69: ['Android App Development']: 0.1273

Europe 2025

NetworkX modularity: 0.6582

Cluster 2 - Node : 116, Edge: 406

Cluster 1 - Node : 159, Edge: 666

Cluster 0 - Node : 232, Edge: 988

Cluster 3 - Node : 40, Edge: 136

Cluster 4 - Node : 38, Edge: 102

Cluster 0: Visual Communication and Design Oriented Creative Production Ecosystem

Degree Centrality - Top 5:

1. 0: ['Graphic Design']: 0.4696
2. 8: ['Adobe Illustrator']: 0.3826
3. 6: ['Adobe Photoshop']: 0.3565
4. 54: ['Print Design']: 0.1565
5. 72: ['3D Modeling']: 0.1565
6. 25: ['Illustration']: 0.1478
7. 52: ['Brand Identity & Guidelines']: 0.1478
8. 84: ['Product Design']: 0.1478
9. 113: ['3D Design']: 0.1217
10. 58: ['Logo Design']: 0.1130
11. 80: ['Adobe InDesign']: 0.1043
12. 87: ['Layout Design']: 0.1043
13. 187: ['3D Rendering']: 0.1043
14. 51: ['Photo Editing']: 0.0957
15. 162: ['Ebook']: 0.0957
16. 70: ['Image Editing']: 0.0870
17. 75: ['Canva']: 0.0870
18. 158: ['T-Shirt Design']: 0.0870
19. 171: ['Mechanical Engineering']: 0.0870
20. 93: ['Presentation Design']: 0.0783

Betweenness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.4257
2. 8: ['Adobe Illustrator']: 0.2477
3. 6: ['Adobe Photoshop']: 0.2134
4. 72: ['3D Modeling']: 0.1535
5. 220: ['Sourcing']: 0.1317
6. 113: ['3D Design']: 0.1174
7. 84: ['Product Design']: 0.0996
8. 187: ['3D Rendering']: 0.0743
9. 338: ['Alibaba Sourcing']: 0.0677
10. 221: ['Architectural Design']: 0.0563
11. 70: ['Image Editing']: 0.0529
12. 54: ['Print Design']: 0.0435
13. 75: ['Canva']: 0.0417
14. 52: ['Brand Identity & Guidelines']: 0.0237
15. 87: ['Layout Design']: 0.0194
16. 134: ['UX & UI']: 0.0174
17. 25: ['Illustration']: 0.0156
18. 58: ['Logo Design']: 0.0111
19. 162: ['Ebook']: 0.0097
20. 51: ['Photo Editing']: 0.0094

Closeness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.5867
2. 8: ['Adobe Illustrator']: 0.5502
3. 6: ['Adobe Photoshop']: 0.5022
4. 72: ['3D Modeling']: 0.4600
5. 113: ['3D Design']: 0.4528
6. 84: ['Product Design']: 0.4440
7. 52: ['Brand Identity & Guidelines']: 0.4406
8. 54: ['Print Design']: 0.4406
9. 25: ['Illustration']: 0.4356
10. 70: ['Image Editing']: 0.4356
11. 80: ['Adobe InDesign']: 0.4307
12. 87: ['Layout Design']: 0.4307
13. 107: ['Infographic']: 0.4307
14. 58: ['Logo Design']: 0.4291
15. 158: ['T-Shirt Design']: 0.4275
16. 51: ['Photo Editing']: 0.4244
17. 140: ['Clothing']: 0.4212
18. 289: ['Etsy']: 0.4182
19. 359: ['Adobe XD']: 0.4182
20. 363: ['Catalog']: 0.4182

PageRank - Top 5:

1. 0: ['Graphic Design']: 0.0592
2. 8: ['Adobe Illustrator']: 0.0448
3. 6: ['Adobe Photoshop']: 0.0445
4. 72: ['3D Modeling']: 0.0201
5. 84: ['Product Design']: 0.0191
6. 54: ['Print Design']: 0.0189
7. 52: ['Brand Identity & Guidelines']: 0.0178
8. 25: ['Illustration']: 0.0178
9. 113: ['3D Design']: 0.0157
10. 187: ['3D Rendering']: 0.0147
11. 58: ['Logo Design']: 0.0136
12. 220: ['Sourcing']: 0.0127

13. 87: ['Layout Design']: 0.0126
14. 338: ['Alibaba Sourcing']: 0.0126
15. 51: ['Photo Editing']: 0.0125
16. 80: ['Adobe InDesign']: 0.0125
17. 75: ['Canva']: 0.0123
18. 176: ['Interior Design Consultation']: 0.0123
19. 171: ['Mechanical Engineering']: 0.0118
20. 162: ['Ebook']: 0.0118

Authority (HITS) - Top 5:

1. 0: ['Graphic Design']: 0.1340
2. 6: ['Adobe Photoshop']: 0.1219
3. 8: ['Adobe Illustrator']: 0.1158
4. 25: ['Illustration']: 0.0654
5. 51: ['Photo Editing']: 0.0386
6. 54: ['Print Design']: 0.0333
7. 52: ['Brand Identity & Guidelines']: 0.0298
8. 80: ['Adobe InDesign']: 0.0296
9. 70: ['Image Editing']: 0.0280
10. 58: ['Logo Design']: 0.0253
11. 87: ['Layout Design']: 0.0251
12. 107: ['Infographic']: 0.0162
13. 140: ['Clothing']: 0.0148
14. 226: ['AI-Generated Image']: 0.0145
15. 225: ['AI-Generated Art']: 0.0145
16. 158: ['T-Shirt Design']: 0.0143
17. 289: ['Etsy']: 0.0113
18. 84: ['Product Design']: 0.0109
19. 93: ['Presentation Design']: 0.0109
20. 359: ['Adobe XD']: 0.0102

Eigenvector Centrality - Top 5:

1. 0: ['Graphic Design']: 0.4004
2. 8: ['Adobe Illustrator']: 0.3714
3. 6: ['Adobe Photoshop']: 0.3294
4. 25: ['Illustration']: 0.1888
5. 52: ['Brand Identity & Guidelines']: 0.1841
6. 54: ['Print Design']: 0.1773
7. 80: ['Adobe InDesign']: 0.1671
8. 58: ['Logo Design']: 0.1598
9. 87: ['Layout Design']: 0.1465
10. 158: ['T-Shirt Design']: 0.1420
11. 70: ['Image Editing']: 0.1342
12. 51: ['Photo Editing']: 0.1334
13. 107: ['Infographic']: 0.1231
14. 140: ['Clothing']: 0.1194
15. 289: ['Etsy']: 0.1137
16. 359: ['Adobe XD']: 0.1128
17. 363: ['Catalog']: 0.1128
18. 241: ['Brochure']: 0.1068
19. 225: ['AI-Generated Art']: 0.1029
20. 226: ['AI-Generated Image']: 0.1029

Cluster 1: Web Technologies and Application Development Ecosystem

Degree Centrality - Top 5:

1. 7: ['JavaScript']: 0.3165
2. 1: ['Web Design']: 0.2468
3. 2: ['WordPress']: 0.2405
4. 24: ['Python']: 0.2278
5. 27: ['API']: 0.2278
6. 3: ['Web Development']: 0.2215
7. 10: ['HTML']: 0.2215
8. 14: ['CSS']: 0.2025
9. 20: ['PHP']: 0.1899
10. 33: ['iOS']: 0.1519
11. 67: ['Java']: 0.1456
12. 85: ['API Integration']: 0.1392
13. 88: ['Web Application']: 0.1392
14. 36: ['Android']: 0.1329
15. 62: ['React']: 0.1266
16. 17: ['Shopify']: 0.1203
17. 89: ['PostgreSQL']: 0.1139
18. 30: ['Figma']: 0.1076
19. 49: ['Mobile App Development']: 0.1076
20. 53: ['Android App Development']: 0.1013

Betweenness Centrality - Top 5:

1. 7: ['JavaScript']: 0.2246
2. 24: ['Python']: 0.1515
3. 27: ['API']: 0.1332
4. 2: ['WordPress']: 0.0973
5. 33: ['iOS']: 0.0790
6. 1: ['Web Design']: 0.0778
7. 10: ['HTML']: 0.0753
8. 67: ['Java']: 0.0714
9. 85: ['API Integration']: 0.0708
10. 89: ['PostgreSQL']: 0.0652
11. 3: ['Web Development']: 0.0619
12. 14: ['CSS']: 0.0592
13. 20: ['PHP']: 0.0574
14. 275: ['Shopify Theme']: 0.0497
15. 88: ['Web Application']: 0.0490
16. 109: ['n8n']: 0.0383
17. 183: ['Make.com']: 0.0289
18. 17: ['Shopify']: 0.0281
19. 81: ['Node.js']: 0.0254
20. 247: ['DevOps']: 0.0252

Closeness Centrality - Top 5:

1. 7: ['JavaScript']: 0.5725
2. 27: ['API']: 0.5392
3. 24: ['Python']: 0.5215
4. 3: ['Web Development']: 0.5130
5. 10: ['HTML']: 0.5113
6. 20: ['PHP']: 0.5048
7. 14: ['CSS']: 0.5016
8. 2: ['WordPress']: 0.4984
9. 1: ['Web Design']: 0.4817

10. 88: ['Web Application']: 0.4788
11. 85: ['API Integration']: 0.4759
12. 67: ['Java']: 0.4675
13. 33: ['iOS']: 0.4580
14. 62: ['React']: 0.4580
15. 17: ['Shopify']: 0.4566
16. 81: ['Node.js']: 0.4553
17. 89: ['PostgreSQL']: 0.4463
18. 30: ['Figma']: 0.4413
19. 133: ['Next.js']: 0.4353
20. 163: ['Prototyping']: 0.4329

PageRank - Top 5:

1. 7: ['JavaScript']: 0.0335
2. 24: ['Python']: 0.0254
3. 2: ['WordPress']: 0.0250
4. 1: ['Web Design']: 0.0246
5. 27: ['API']: 0.0235
6. 10: ['HTML']: 0.0222
7. 3: ['Web Development']: 0.0215
8. 14: ['CSS']: 0.0199
9. 20: ['PHP']: 0.0192
10. 33: ['iOS']: 0.0154
11. 67: ['Java']: 0.0153
12. 85: ['API Integration']: 0.0149
13. 88: ['Web Application']: 0.0140
14. 36: ['Android']: 0.0137
15. 89: ['PostgreSQL']: 0.0129
16. 62: ['React']: 0.0126
17. 17: ['Shopify']: 0.0126
18. 81: ['Node.js']: 0.0110
19. 49: ['Mobile App Development']: 0.0109
20. 30: ['Figma']: 0.0107

Authority (HITS) - Top 5:

1. 3: ['Web Development']: 0.1206
2. 2: ['WordPress']: 0.1098
3. 1: ['Web Design']: 0.1095
4. 10: ['HTML']: 0.0884
5. 14: ['CSS']: 0.0783
6. 7: ['JavaScript']: 0.0591
7. 20: ['PHP']: 0.0529
8. 50: ['WooCommerce']: 0.0312
9. 17: ['Shopify']: 0.0283
10. 64: ['HTML5']: 0.0203
11. 42: ['Landing Page']: 0.0179
12. 97: ['WordPress Plugin']: 0.0175
13. 90: ['Webflow']: 0.0149
14. 78: ['Mockup']: 0.0141
15. 30: ['Figma']: 0.0111
16. 104: ['Elementor']: 0.0110
17. 88: ['Web Application']: 0.0094
18. 27: ['API']: 0.0091
19. 147: ['WordPress Development']: 0.0091
20. 121: ['Website']: 0.0070

Eigenvector Centrality - Top 5:

1. 7: ['JavaScript']: 0.2862
2. 10: ['HTML']: 0.2645
3. 3: ['Web Development']: 0.2591
4. 2: ['WordPress']: 0.2500
5. 14: ['CSS']: 0.2487
6. 1: ['Web Design']: 0.2465
7. 20: ['PHP']: 0.2273
8. 27: ['API']: 0.2146
9. 88: ['Web Application']: 0.1810
10. 24: ['Python']: 0.1761
11. 121: ['Website']: 0.1622
12. 17: ['Shopify']: 0.1616
13. 85: ['API Integration']: 0.1600
14. 62: ['React']: 0.1525
15. 64: ['HTML5']: 0.1473
16. 50: ['WooCommerce']: 0.1368
17. 67: ['Java']: 0.1303
18. 30: ['Figma']: 0.1280
19. 42: ['Landing Page']: 0.1182
20. 81: ['Node.js']: 0.1174

Cluster 2: Digital Marketing and Content Driven Growth Ecosystem

Degree Centrality - Top 5:

1. 4: ['Lead Generation']: 0.3160
2. 11: ['English']: 0.2641
3. 12: ['Social Media Marketing']: 0.2165
4. 21: ['Marketing Strategy']: 0.1775
5. 19: ['Data Entry']: 0.1645
6. 13: ['Content Writing']: 0.1602
7. 9: ['Search Engine Optimization']: 0.1558
8. 23: ['SEO Keyword Research']: 0.1299
9. 38: ['Google Ads']: 0.1255
10. 31: ['Email Marketing']: 0.1169
11. 59: ['Sales']: 0.1039
12. 28: ['Copywriting']: 0.0952
13. 34: ['Customer Service']: 0.0909
14. 37: ['Instagram']: 0.0909
15. 48: ['Communications']: 0.0909
16. 26: ['Facebook']: 0.0866
17. 71: ['Administrative Support']: 0.0866
18. 74: ['Blog Content']: 0.0866
19. 43: ['Article Writing']: 0.0779
20. 103: ['Blog Writing']: 0.0779

Betweenness Centrality - Top 5:

1. 11: ['English']: 0.2524
2. 4: ['Lead Generation']: 0.2440
3. 12: ['Social Media Marketing']: 0.1482
4. 9: ['Search Engine Optimization']: 0.1305
5. 21: ['Marketing Strategy']: 0.0891
6. 13: ['Content Writing']: 0.0831
7. 19: ['Data Entry']: 0.0768
8. 23: ['SEO Keyword Research']: 0.0501

9. 38: ['Google Ads']: 0.0501
10. 31: ['Email Marketing']: 0.0436
11. 48: ['Communications']: 0.0345
12. 71: ['Administrative Support']: 0.0334
13. 46: ['Market Research']: 0.0332
14. 74: ['Blog Content']: 0.0281
15. 166: ['Proofreading']: 0.0262
16. 63: ['Email Campaign Setup']: 0.0237
17. 26: ['Facebook']: 0.0219
18. 28: ['Copywriting']: 0.0218
19. 55: ['Email Support']: 0.0212
20. 59: ['Sales']: 0.0211

Closeness Centrality - Top 5:

1. 4: ['Lead Generation']: 0.5593
2. 11: ['English']: 0.5323
3. 12: ['Social Media Marketing']: 0.5033
4. 21: ['Marketing Strategy']: 0.4989
5. 13: ['Content Writing']: 0.4968
6. 9: ['Search Engine Optimization']: 0.4946
7. 38: ['Google Ads']: 0.4853
8. 19: ['Data Entry']: 0.4793
9. 48: ['Communications']: 0.4620
10. 28: ['Copywriting']: 0.4602
11. 59: ['Sales']: 0.4547
12. 31: ['Email Marketing']: 0.4529
13. 34: ['Customer Service']: 0.4503
14. 26: ['Facebook']: 0.4434
15. 46: ['Market Research']: 0.4400
16. 37: ['Instagram']: 0.4392
17. 65: ['SEO Writing']: 0.4383
18. 76: ['Social Media Management']: 0.4358
19. 47: ['German']: 0.4254
20. 23: ['SEO Keyword Research']: 0.4239

PageRank - Top 5:

1. 4: ['Lead Generation']: 0.0322
2. 11: ['English']: 0.0265
3. 12: ['Social Media Marketing']: 0.0234
4. 21: ['Marketing Strategy']: 0.0183
5. 9: ['Search Engine Optimization']: 0.0177
6. 19: ['Data Entry']: 0.0166
7. 13: ['Content Writing']: 0.0157
8. 23: ['SEO Keyword Research']: 0.0140
9. 38: ['Google Ads']: 0.0131
10. 31: ['Email Marketing']: 0.0122
11. 59: ['Sales']: 0.0103
12. 71: ['Administrative Support']: 0.0098
13. 37: ['Instagram']: 0.0097
14. 28: ['Copywriting']: 0.0095
15. 48: ['Communications']: 0.0094
16. 26: ['Facebook']: 0.0092
17. 34: ['Customer Service']: 0.0091
18. 74: ['Blog Content']: 0.0089
19. 166: ['Proofreading']: 0.0089
20. 73: ['Translation']: 0.0087

Authority (HITS) - Top 5:

1. 4: ['Lead Generation']: 0.0542
2. 12: ['Social Media Marketing']: 0.0437
3. 21: ['Marketing Strategy']: 0.0381
4. 13: ['Content Writing']: 0.0346
5. 19: ['Data Entry']: 0.0314
6. 26: ['Facebook']: 0.0280
7. 11: ['English']: 0.0247
8. 40: ['List Building']: 0.0247
9. 39: ['Prospect List']: 0.0243
10. 9: ['Search Engine Optimization']: 0.0232
11. 37: ['Instagram']: 0.0208
12. 31: ['Email Marketing']: 0.0178
13. 28: ['Copywriting']: 0.0175
14. 29: ['Data Scraping']: 0.0169
15. 38: ['Google Ads']: 0.0166
16. 23: ['SEO Keyword Research']: 0.0152
17. 34: ['Customer Service']: 0.0147
18. 76: ['Social Media Management']: 0.0145
19. 43: ['Article Writing']: 0.0144
20. 46: ['Market Research']: 0.0137

Eigenvector Centrality - Top 5:

1. 4: ['Lead Generation']: 0.3217
2. 11: ['English']: 0.2525
3. 21: ['Marketing Strategy']: 0.2310
4. 12: ['Social Media Marketing']: 0.2268
5. 13: ['Content Writing']: 0.2167
6. 19: ['Data Entry']: 0.2040
7. 38: ['Google Ads']: 0.1809
8. 9: ['Search Engine Optimization']: 0.1684
9. 28: ['Copywriting']: 0.1558
10. 31: ['Email Marketing']: 0.1526
11. 59: ['Sales']: 0.1443
12. 48: ['Communications']: 0.1328
13. 26: ['Facebook']: 0.1322
14. 34: ['Customer Service']: 0.1299
15. 37: ['Instagram']: 0.1269
16. 23: ['SEO Keyword Research']: 0.1233
17. 65: ['SEO Writing']: 0.1214
18. 43: ['Article Writing']: 0.1116
19. 98: ['Outbound Sales']: 0.1083
20. 39: ['Prospect List']: 0.1079

Cluster 3: The Digital Video Production and Post-Production Ecosystem

This cluster constitutes a highly interactive digital content production network encompassing creative processes that lie at the core of contemporary media production, including video editing, visual effects, audio engineering, and animation. Video Editing, which holds the highest centrality measures, occupies the technical and structural core of the cluster. Following closely are Video Production, Audio Editing, Adobe After Effects, and Adobe Premiere Pro, all of which reinforce both the visual and auditory dimensions of professional video production workflows. Specializations such as Motion Graphics, 2D/3D Animation, Explainer Video, Final Cut Pro, and Visual Effects significantly enhance the visual narrative capacity of content creation, while nodes such as Sound Mixing and Audio Production complement this experience on the auditory plane. Elevated PageRank and

Eigenvector centrality values underscore the sectoral significance and diffusion capacity of these domains across the network, whereas high Authority scores further indicate that these competencies function as knowledge-generating hubs requiring advanced expertise. In sum, this cluster represents an integrated and creative digital production ecosystem built around professional video production, innovative media content development, and sophisticated post-production skill sets.

Cluster 4: The Business Intelligence and Data-Driven Decision Support Ecosystem

This cluster represents an ecosystem structured around decision support systems in which data analytics, business intelligence tools, and reporting techniques are highly integrated and interactive. At the center, Microsoft Excel ranks first in degree, betweenness, and eigenvector centrality, underscoring its indispensable role in data processing, analysis, and presentation functions. Nodes such as Data Analysis, Data Visualization, and Microsoft Power BI provide the technical infrastructure for generating data-driven insights, while SAP and its related modules (SAP HANA, ABAP, SAP Cloud Platform Integration) reflect the integration dimension of enterprise data management. Prominent competencies including Qualitative Research, Report Writing, Survey Design, and Statistics contribute an analytical layer that combines qualitative and quantitative research methodologies, enabling the interpretation of results within a business context. Platforms such as Zoho Analytics, Looker Studio, Business Intelligence, and Pitchbook furnish decision-makers with advanced dashboards and strategic data visualizations that offer broad, multidimensional perspectives. Overall, this structure can be characterized as a multilayered and interactive decision support network focused on data-driven strategic management, operational reporting, and enterprise business intelligence solutions.

Global 2025

NetworkX modularity: 0.6580

Cluster 2 - Node : 179, Edge: 817
 Cluster 1 - Node : 264, Edge: 1439
 Cluster 0 - Node : 440, Edge: 1979
 Cluster 3 - Node : 122, Edge: 404
 Cluster 5 - Node : 45, Edge: 188
 Cluster 4 - Node : 101, Edge: 370

Cluster 0 : Visual Content Production and Multimedia Design Ecosystem

Degree Centrality - Top 5:

1. 0: ['Graphic Design']: 0.4326
2. 9: ['Adobe Photoshop']: 0.4157
3. 18: ['Adobe Illustrator']: 0.3708
4. 7: ['Video Editing']: 0.2697
5. 35: ['Illustration']: 0.1910
6. 23: ['Adobe After Effects']: 0.1854
7. 20: ['Adobe Premiere Pro']: 0.1742
8. 25: ['Video Production']: 0.1742
9. 83: ['Print Design']: 0.1685
10. 44: ['Logo Design']: 0.1573
11. 48: ['Brand Identity & Guidelines']: 0.1461
12. 85: ['Videography']: 0.1461
13. 29: ['Video Post-Editing']: 0.1348
14. 101: ['Canva']: 0.1124
15. 124: ['Explainer Video']: 0.1124
16. 56: ['Animation']: 0.1067
17. 93: ['Photo Editing']: 0.1067
18. 88: ['Adobe InDesign']: 0.1011

- 19. 97: ['Video Commercial']: 0.1011
- 20. 132: ['Infographic']: 0.1011

Betweenness Centrality - Top 5:

- 1. 0: ['Graphic Design']: 0.3187
- 2. 9: ['Adobe Photoshop']: 0.2050
- 3. 18: ['Adobe Illustrator']: 0.1269
- 4. 126: ['Project Management']: 0.1082
- 5. 7: ['Video Editing']: 0.1053
- 6. 25: ['Video Production']: 0.0578
- 7. 125: ['Photography']: 0.0533
- 8. 56: ['Animation']: 0.0471
- 9. 44: ['Logo Design']: 0.0408
- 10. 23: ['Adobe After Effects']: 0.0401
- 11. 124: ['Explainer Video']: 0.0367
- 12. 185: ['Product Photography']: 0.0344
- 13. 35: ['Illustration']: 0.0323
- 14. 83: ['Print Design']: 0.0302
- 15. 20: ['Adobe Premiere Pro']: 0.0293
- 16. 101: ['Canva']: 0.0281
- 17. 87: ['Product Design']: 0.0262
- 18. 85: ['Videography']: 0.0256
- 19. 149: ['DaVinci Resolve']: 0.0224
- 20. 93: ['Photo Editing']: 0.0176

Closeness Centrality - Top 5:

- 1. 0: ['Graphic Design']: 0.6312
- 2. 9: ['Adobe Photoshop']: 0.6054
- 3. 18: ['Adobe Illustrator']: 0.5687
- 4. 7: ['Video Editing']: 0.5361
- 5. 23: ['Adobe After Effects']: 0.5174
- 6. 25: ['Video Production']: 0.5028
- 7. 20: ['Adobe Premiere Pro']: 0.5014
- 8. 35: ['Illustration']: 0.4917
- 9. 29: ['Video Post-Editing']: 0.4877
- 10. 124: ['Explainer Video']: 0.4798
- 11. 93: ['Photo Editing']: 0.4785
- 12. 99: ['Motion Graphics']: 0.4785
- 13. 44: ['Logo Design']: 0.4759
- 14. 56: ['Animation']: 0.4747
- 15. 101: ['Canva']: 0.4734
- 16. 94: ['2D Animation']: 0.4721
- 17. 185: ['Product Photography']: 0.4721
- 18. 132: ['Infographic']: 0.4672
- 19. 83: ['Print Design']: 0.4623
- 20. 196: ['YouTube']: 0.4611

PageRank - Top 5:

- 1. 0: ['Graphic Design']: 0.0421
- 2. 9: ['Adobe Photoshop']: 0.0391
- 3. 18: ['Adobe Illustrator']: 0.0343
- 4. 7: ['Video Editing']: 0.0260
- 5. 35: ['Illustration']: 0.0179
- 6. 23: ['Adobe After Effects']: 0.0168
- 7. 25: ['Video Production']: 0.0167
- 8. 83: ['Print Design']: 0.0160

9. 20: ['Adobe Premiere Pro']: 0.0159
10. 44: ['Logo Design']: 0.0153
11. 85: ['Videography']: 0.0148
12. 48: ['Brand Identity & Guidelines']: 0.0138
13. 126: ['Project Management']: 0.0130
14. 29: ['Video Post-Editing']: 0.0125
15. 124: ['Explainer Video']: 0.0115
16. 101: ['Canva']: 0.0110
17. 125: ['Photography']: 0.0110
18. 56: ['Animation']: 0.0109
19. 93: ['Photo Editing']: 0.0106
20. 97: ['Video Commercial']: 0.0100

Authority (HITS) - Top 5:

1. 7: ['Video Editing']: 0.1203
2. 20: ['Adobe Premiere Pro']: 0.1100
3. 23: ['Adobe After Effects']: 0.1001
4. 25: ['Video Production']: 0.0905
5. 29: ['Video Post-Editing']: 0.0811
6. 9: ['Adobe Photoshop']: 0.0474
7. 0: ['Graphic Design']: 0.0403
8. 18: ['Adobe Illustrator']: 0.0331
9. 97: ['Video Commercial']: 0.0244
10. 143: ['Final Cut Pro']: 0.0200
11. 153: ['CapCut']: 0.0194
12. 149: ['DaVinci Resolve']: 0.0179
13. 85: ['Videography']: 0.0162
14. 101: ['Canva']: 0.0147
15. 99: ['Motion Graphics']: 0.0141
16. 35: ['Illustration']: 0.0139
17. 124: ['Explainer Video']: 0.0137
18. 196: ['YouTube']: 0.0114
19. 93: ['Photo Editing']: 0.0101
20. 44: ['Logo Design']: 0.0091

Eigenvector Centrality - Top 5:

1. 9: ['Adobe Photoshop']: 0.3282
2. 0: ['Graphic Design']: 0.3204
3. 18: ['Adobe Illustrator']: 0.2905
4. 7: ['Video Editing']: 0.2032
5. 23: ['Adobe After Effects']: 0.1962
6. 35: ['Illustration']: 0.1854
7. 20: ['Adobe Premiere Pro']: 0.1726
8. 25: ['Video Production']: 0.1516
9. 44: ['Logo Design']: 0.1458
10. 29: ['Video Post-Editing']: 0.1425
11. 83: ['Print Design']: 0.1378
12. 99: ['Motion Graphics']: 0.1356
13. 48: ['Brand Identity & Guidelines']: 0.1315
14. 101: ['Canva']: 0.1315
15. 124: ['Explainer Video']: 0.1267
16. 94: ['2D Animation']: 0.1254
17. 56: ['Animation']: 0.1199
18. 93: ['Photo Editing']: 0.1169
19. 88: ['Adobe InDesign']: 0.1160
20. 132: ['Infographic']: 0.1118

Cluster 1: Modern Web Development and Integration Technologies Ecosystem

Degree Centrality - Top 5:

1. 2: ['JavaScript']: 0.3764
2. 1: ['Web Development']: 0.3308
3. 3: ['Web Design']: 0.2852
4. 10: ['API']: 0.2624
5. 4: ['WordPress']: 0.2586
6. 8: ['PHP']: 0.2510
7. 11: ['HTML']: 0.2471
8. 24: ['Python']: 0.2205
9. 17: ['CSS']: 0.2167
10. 61: ['React']: 0.2053
11. 43: ['API Integration']: 0.1749
12. 52: ['Node.js']: 0.1559
13. 104: ['Web Application']: 0.1445
14. 151: ['PostgreSQL']: 0.1369
15. 242: ['RESTful API']: 0.1255
16. 21: ['Shopify']: 0.1179
17. 40: ['Landing Page']: 0.1141
18. 96: ['SQL']: 0.1065
19. 109: ['Java']: 0.1065
20. 129: ['Website']: 0.1065

Betweenness Centrality - Top 5:

1. 2: ['JavaScript']: 0.2030
2. 3: ['Web Design']: 0.1380
3. 1: ['Web Development']: 0.1151
4. 10: ['API']: 0.1046
5. 24: ['Python']: 0.0898
6. 4: ['WordPress']: 0.0714
7. 8: ['PHP']: 0.0664
8. 11: ['HTML']: 0.0661
9. 61: ['React']: 0.0481
10. 43: ['API Integration']: 0.0446
11. 96: ['SQL']: 0.0397
12. 604: ['Usability Testing']: 0.0371
13. 17: ['CSS']: 0.0355
14. 151: ['PostgreSQL']: 0.0329
15. 82: ['Shopify Theme']: 0.0319
16. 60: ['Automation']: 0.0272
17. 76: ['Artificial Intelligence']: 0.0266
18. 551: ['Microsoft Azure']: 0.0240
19. 104: ['Web Application']: 0.0236
20. 874: ['Character Animation']: 0.0226

Closeness Centrality - Top 5:

1. 2: ['JavaScript']: 0.5950
2. 1: ['Web Development']: 0.5644
3. 10: ['API']: 0.5479
4. 8: ['PHP']: 0.5389
5. 3: ['Web Design']: 0.5302
6. 4: ['WordPress']: 0.5271
7. 11: ['HTML']: 0.5271
8. 61: ['React']: 0.5250

9. 17: ['CSS']: 0.5137
10. 24: ['Python']: 0.5058
11. 43: ['API Integration']: 0.5058
12. 104: ['Web Application']: 0.4972
13. 52: ['Node.js']: 0.4962
14. 151: ['PostgreSQL']: 0.4962
15. 21: ['Shopify']: 0.4739
16. 109: ['Java']: 0.4680
17. 242: ['RESTful API']: 0.4680
18. 145: ['Next.js']: 0.4671
19. 91: ['HTML5']: 0.4647
20. 60: ['Automation']: 0.4622

PageRank - Top 5:

1. 2: ['JavaScript']: 0.0296
2. 1: ['Web Development']: 0.0248
3. 3: ['Web Design']: 0.0239
4. 10: ['API']: 0.0204
5. 4: ['WordPress']: 0.0204
6. 11: ['HTML']: 0.0193
7. 8: ['PHP']: 0.0191
8. 24: ['Python']: 0.0185
9. 17: ['CSS']: 0.0160
10. 61: ['React']: 0.0155
11. 43: ['API Integration']: 0.0136
12. 52: ['Node.js']: 0.0116
13. 104: ['Web Application']: 0.0110
14. 151: ['PostgreSQL']: 0.0110
15. 242: ['RESTful API']: 0.0096
16. 21: ['Shopify']: 0.0094
17. 96: ['SQL']: 0.0092
18. 40: ['Landing Page']: 0.0089
19. 76: ['Artificial Intelligence']: 0.0087
20. 60: ['Automation']: 0.0083

Authority (HITS) - Top 5:

1. 1: ['Web Development']: 0.0902
2. 2: ['JavaScript']: 0.0788
3. 8: ['PHP']: 0.0712
4. 4: ['WordPress']: 0.0695
5. 11: ['HTML']: 0.0689
6. 3: ['Web Design']: 0.0640
7. 17: ['CSS']: 0.0561
8. 10: ['API']: 0.0463
9. 21: ['Shopify']: 0.0233
10. 43: ['API Integration']: 0.0206
11. 55: ['MySQL']: 0.0198
12. 24: ['Python']: 0.0184
13. 40: ['Landing Page']: 0.0156
14. 113: ['Custom PHP']: 0.0149
15. 91: ['HTML5']: 0.0135
16. 61: ['React']: 0.0129
17. 74: ['Shopify Templates']: 0.0120
18. 103: ['cPanel']: 0.0117
19. 52: ['Node.js']: 0.0114
20. 96: ['SQL']: 0.0098

Eigenvector Centrality - Top 5:

1. 2: ['JavaScript']: 0.2687
2. 1: ['Web Development']: 0.2605
3. 11: ['HTML']: 0.2185
4. 8: ['PHP']: 0.2160
5. 17: ['CSS']: 0.2099
6. 4: ['WordPress']: 0.2098
7. 3: ['Web Design']: 0.2021
8. 10: ['API']: 0.2009
9. 61: ['React']: 0.1894
10. 52: ['Node.js']: 0.1620
11. 24: ['Python']: 0.1609
12. 43: ['API Integration']: 0.1588
13. 104: ['Web Application']: 0.1558
14. 91: ['HTML5']: 0.1263
15. 129: ['Website']: 0.1248
16. 109: ['Java']: 0.1207
17. 242: ['RESTful API']: 0.1157
18. 21: ['Shopify']: 0.1154
19. 194: ['CSS 3']: 0.1133
20. 55: ['MySQL']: 0.1131

Cluster 2 : Digital Marketing and Communication Focused Service Ecosystem

Degree Centrality - Top 5:

1. 16: ['English']: 0.2528
2. 5: ['Lead Generation']: 0.2096
3. 13: ['Data Entry']: 0.1595
4. 19: ['Content Writing']: 0.1526
5. 6: ['Social Media Marketing']: 0.1458
6. 14: ['Search Engine Optimization']: 0.1276
7. 15: ['Marketing Strategy']: 0.1276
8. 58: ['Email Marketing']: 0.1025
9. 31: ['Administrative Support']: 0.1002
10. 41: ['Sales']: 0.0979
11. 22: ['Instagram']: 0.0888
12. 33: ['Customer Service']: 0.0866
13. 65: ['Google Ads']: 0.0866
14. 45: ['Communications']: 0.0843
15. 12: ['Facebook']: 0.0820
16. 28: ['SEO Keyword Research']: 0.0797
17. 30: ['Social Media Management']: 0.0706
18. 53: ['Virtual Assistance']: 0.0706
19. 77: ['German']: 0.0706
20. 71: ['Email Communication']: 0.0683

Betweenness Centrality - Top 5:

1. 16: ['English']: 0.3085
2. 5: ['Lead Generation']: 0.1337
3. 13: ['Data Entry']: 0.1160
4. 19: ['Content Writing']: 0.1124
5. 14: ['Search Engine Optimization']: 0.0896
6. 6: ['Social Media Marketing']: 0.0675

7. 31: ['Administrative Support']: 0.0620
8. 58: ['Email Marketing']: 0.0497
9. 77: ['German']: 0.0474
10. 15: ['Marketing Strategy']: 0.0404
11. 65: ['Google Ads']: 0.0338
12. 41: ['Sales']: 0.0317
13. 32: ['Data Scraping']: 0.0313
14. 520: ['Price & Quote Negotiation']: 0.0300
15. 45: ['Communications']: 0.0293
16. 49: ['Market Research']: 0.0279
17. 349: ['Candidate Sourcing']: 0.0279
18. 12: ['Facebook']: 0.0276
19. 322: ['Podcast']: 0.0273
20. 33: ['Customer Service']: 0.0270

Closeness Centrality - Top 5:

1. 16: ['English']: 0.5105
2. 5: ['Lead Generation']: 0.5000
3. 13: ['Data Entry']: 0.4878
4. 19: ['Content Writing']: 0.4862
5. 14: ['Search Engine Optimization']: 0.4670
6. 41: ['Sales']: 0.4559
7. 6: ['Social Media Marketing']: 0.4521
8. 45: ['Communications']: 0.4461
9. 31: ['Administrative Support']: 0.4439
10. 33: ['Customer Service']: 0.4403
11. 65: ['Google Ads']: 0.4386
12. 15: ['Marketing Strategy']: 0.4342
13. 49: ['Market Research']: 0.4338
14. 72: ['Copywriting']: 0.4317
15. 58: ['Email Marketing']: 0.4308
16. 12: ['Facebook']: 0.4266
17. 226: ['Sales & Marketing']: 0.4193
18. 154: ['SEO Writing']: 0.4126
19. 77: ['German']: 0.4107
20. 78: ['Article Writing']: 0.4084

PageRank - Top 5:

1. 16: ['English']: 0.0249
2. 5: ['Lead Generation']: 0.0189
3. 13: ['Data Entry']: 0.0145
4. 19: ['Content Writing']: 0.0139
5. 6: ['Social Media Marketing']: 0.0135
6. 14: ['Search Engine Optimization']: 0.0120
7. 15: ['Marketing Strategy']: 0.0113
8. 58: ['Email Marketing']: 0.0099
9. 31: ['Administrative Support']: 0.0098
10. 41: ['Sales']: 0.0087
11. 22: ['Instagram']: 0.0081
12. 65: ['Google Ads']: 0.0080
13. 45: ['Communications']: 0.0079
14. 33: ['Customer Service']: 0.0078
15. 12: ['Facebook']: 0.0074
16. 28: ['SEO Keyword Research']: 0.0074
17. 68: ['Translation']: 0.0072
18. 77: ['German']: 0.0070

- 19. 53: ['Virtual Assistance']: 0.0069
- 20. 30: ['Social Media Management']: 0.0068

Authority (HITS) - Top 5:

- 1. 6: ['Social Media Marketing']: 0.1038
- 2. 12: ['Facebook']: 0.0930
- 3. 22: ['Instagram']: 0.0766
- 4. 15: ['Marketing Strategy']: 0.0646
- 5. 30: ['Social Media Management']: 0.0578
- 6. 5: ['Lead Generation']: 0.0368
- 7. 54: ['Social Media Content Creation']: 0.0321
- 8. 19: ['Content Writing']: 0.0301
- 9. 14: ['Search Engine Optimization']: 0.0206
- 10. 81: ['Influencer Marketing']: 0.0196
- 11. 69: ['LinkedIn']: 0.0172
- 12. 13: ['Data Entry']: 0.0164
- 13. 37: ['List Building']: 0.0164
- 14. 127: ['Social Media Content']: 0.0154
- 15. 72: ['Copywriting']: 0.0127
- 16. 32: ['Data Scraping']: 0.0127
- 17. 28: ['SEO Keyword Research']: 0.0114
- 18. 41: ['Sales']: 0.0112
- 19. 216: ['Pinterest']: 0.0104
- 20. 65: ['Google Ads']: 0.0094

Eigenvector Centrality - Top 5:

- 1. 5: ['Lead Generation']: 0.2820
- 2. 13: ['Data Entry']: 0.2341
- 3. 6: ['Social Media Marketing']: 0.2191
- 4. 16: ['English']: 0.2140
- 5. 19: ['Content Writing']: 0.2090
- 6. 15: ['Marketing Strategy']: 0.2027
- 7. 14: ['Search Engine Optimization']: 0.1827
- 8. 41: ['Sales']: 0.1694
- 9. 58: ['Email Marketing']: 0.1619
- 10. 45: ['Communications']: 0.1492
- 11. 65: ['Google Ads']: 0.1474
- 12. 12: ['Facebook']: 0.1451
- 13. 33: ['Customer Service']: 0.1420
- 14. 72: ['Copywriting']: 0.1382
- 15. 31: ['Administrative Support']: 0.1354
- 16. 30: ['Social Media Management']: 0.1262
- 17. 22: ['Instagram']: 0.1246
- 18. 28: ['SEO Keyword Research']: 0.1178
- 19. 71: ['Email Communication']: 0.1116
- 20. 49: ['Market Research']: 0.1108

Cluster 3: Financial and Legal Data Management Ecosystem

Degree Centrality - Top 5:

- 1. 26: ['Microsoft Excel']: 0.4215
- 2. 107: ['Accounting']: 0.2231
- 3. 46: ['Data Analysis']: 0.1901
- 4. 119: ['Financial Analysis']: 0.1570
- 5. 128: ['Bookkeeping']: 0.1570
- 6. 135: ['Legal']: 0.1570

7. 258: ['Financial Accounting']: 0.1488
8. 147: ['Legal Consulting']: 0.1322
9. 115: ['Data Visualization']: 0.1240
10. 293: ['Intuit QuickBooks']: 0.1157
11. 105: ['Statistics']: 0.0992
12. 264: ['Financial Modeling']: 0.0992
13. 329: ['Bank Reconciliation']: 0.0992
14. 112: ['Tax Law']: 0.0909
15. 222: ['Legal Research']: 0.0909
16. 393: ['SAP']: 0.0826
17. 396: ['Tax Preparation']: 0.0826
18. 609: ['Immigration Law']: 0.0744
19. 613: ['Accounting Basics']: 0.0744
20. 141: ['R']: 0.0744

Betweenness Centrality - Top 5:

1. 26: ['Microsoft Excel']: 0.5456
2. 135: ['Legal']: 0.2906
3. 258: ['Financial Accounting']: 0.2355
4. 107: ['Accounting']: 0.2152
5. 46: ['Data Analysis']: 0.0908
6. 393: ['SAP']: 0.0799
7. 731: ['AI Chatbot']: 0.0645
8. 609: ['Immigration Law']: 0.0642
9. 119: ['Financial Analysis']: 0.0637
10. 128: ['Bookkeeping']: 0.0479
11. 112: ['Tax Law']: 0.0408
12. 284: ['Analytics']: 0.0388
13. 293: ['Intuit QuickBooks']: 0.0348
14. 687: ['Romanian']: 0.0328
15. 378: ['Inventory Management']: 0.0328
16. 618: ['Employer-Sponsored Visa']: 0.0314
17. 115: ['Data Visualization']: 0.0305
18. 335: ['Intellectual Property Protection']: 0.0246
19. 291: ['Intellectual Property Law']: 0.0239
20. 147: ['Legal Consulting']: 0.0199

Closeness Centrality - Top 5:

1. 26: ['Microsoft Excel']: 0.5193
2. 107: ['Accounting']: 0.4840
3. 258: ['Financial Accounting']: 0.4783
4. 119: ['Financial Analysis']: 0.4384
5. 128: ['Bookkeeping']: 0.4337
6. 293: ['Intuit QuickBooks']: 0.4172
7. 329: ['Bank Reconciliation']: 0.4144
8. 438: ['Business Plan']: 0.4074
9. 1175: ['Budget Management']: 0.4033
10. 264: ['Financial Modeling']: 0.4007
11. 613: ['Accounting Basics']: 0.3993
12. 46: ['Data Analysis']: 0.3967
13. 135: ['Legal']: 0.3954
14. 1178: ['Excel Formula']: 0.3891
15. 1180: ['Excel Macros']: 0.3891
16. 1181: ['Finance']: 0.3891
17. 1184: ['Budget Planning']: 0.3891
18. 260: ['Mathematics']: 0.3891

- 19. 112: ['Tax Law']: 0.3781
- 20. 396: ['Tax Preparation']: 0.3769

PageRank - Top 5:

- 1. 26: ['Microsoft Excel']: 0.0570
- 2. 107: ['Accounting']: 0.0275
- 3. 46: ['Data Analysis']: 0.0259
- 4. 135: ['Legal']: 0.0206
- 5. 119: ['Financial Analysis']: 0.0199
- 6. 128: ['Bookkeeping']: 0.0190
- 7. 258: ['Financial Accounting']: 0.0189
- 8. 147: ['Legal Consulting']: 0.0179
- 9. 115: ['Data Visualization']: 0.0167
- 10. 293: ['Intuit QuickBooks']: 0.0151
- 11. 105: ['Statistics']: 0.0134
- 12. 393: ['SAP']: 0.0133
- 13. 264: ['Financial Modeling']: 0.0129
- 14. 609: ['Immigration Law']: 0.0128
- 15. 329: ['Bank Reconciliation']: 0.0125
- 16. 222: ['Legal Research']: 0.0122
- 17. 112: ['Tax Law']: 0.0117
- 18. 731: ['AI Chatbot']: 0.0112
- 19. 284: ['Analytics']: 0.0111
- 20. 141: ['R']: 0.0106

Authority (HITS) - Top 5:

- 1. 26: ['Microsoft Excel']: 0.1118
- 2. 46: ['Data Analysis']: 0.0992
- 3. 105: ['Statistics']: 0.0846
- 4. 106: ['Data Science']: 0.0735
- 5. 141: ['R']: 0.0699
- 6. 231: ['Quantitative Analysis']: 0.0646
- 7. 229: ['SAS']: 0.0646
- 8. 115: ['Data Visualization']: 0.0310
- 9. 119: ['Financial Analysis']: 0.0290
- 10. 107: ['Accounting']: 0.0233
- 11. 180: ['Google Docs']: 0.0215
- 12. 184: ['Spreadsheet Software']: 0.0215
- 13. 264: ['Financial Modeling']: 0.0154
- 14. 128: ['Bookkeeping']: 0.0152
- 15. 284: ['Analytics']: 0.0147
- 16. 416: ['Qualitative Research']: 0.0142
- 17. 379: ['Report Writing']: 0.0142
- 18. 290: ['Microsoft Power BI']: 0.0138
- 19. 357: ['Business Analysis']: 0.0117
- 20. 372: ['IBM SPSS']: 0.0115

Eigenvector Centrality - Top 5:

- 1. 26: ['Microsoft Excel']: 0.4420
- 2. 107: ['Accounting']: 0.2908
- 3. 128: ['Bookkeeping']: 0.2335
- 4. 119: ['Financial Analysis']: 0.2264
- 5. 258: ['Financial Accounting']: 0.2131
- 6. 46: ['Data Analysis']: 0.1970
- 7. 293: ['Intuit QuickBooks']: 0.1779
- 8. 329: ['Bank Reconciliation']: 0.1674

9. 264: ['Financial Modeling']: 0.1550
10. 613: ['Accounting Basics']: 0.1418
11. 438: ['Business Plan']: 0.1411
12. 115: ['Data Visualization']: 0.1404
13. 105: ['Statistics']: 0.1313
14. 396: ['Tax Preparation']: 0.1269
15. 357: ['Business Analysis']: 0.1165
16. 1175: ['Budget Management']: 0.1147
17. 306: ['Business Intelligence']: 0.1068
18. 112: ['Tax Law']: 0.1048
19. 141: ['R']: 0.1036
20. 290: ['Microsoft Power BI']: 0.1023

Degree Centrality - Top 5:

1. 26: ['Microsoft Excel']: 0.4215
2. 107: ['Accounting']: 0.2231
3. 46: ['Data Analysis']: 0.1901
4. 119: ['Financial Analysis']: 0.1570
5. 128: ['Bookkeeping']: 0.1570
6. 135: ['Legal']: 0.1570
7. 258: ['Financial Accounting']: 0.1488
8. 147: ['Legal Consulting']: 0.1322
9. 115: ['Data Visualization']: 0.1240
10. 293: ['Intuit QuickBooks']: 0.1157
11. 105: ['Statistics']: 0.0992
12. 264: ['Financial Modeling']: 0.0992
13. 329: ['Bank Reconciliation']: 0.0992
14. 112: ['Tax Law']: 0.0909
15. 222: ['Legal Research']: 0.0909
16. 393: ['SAP']: 0.0826
17. 396: ['Tax Preparation']: 0.0826
18. 609: ['Immigration Law']: 0.0744
19. 613: ['Accounting Basics']: 0.0744
20. 141: ['R']: 0.0744

Betweenness Centrality - Top 5:

1. 26: ['Microsoft Excel']: 0.5456
2. 135: ['Legal']: 0.2906
3. 258: ['Financial Accounting']: 0.2355
4. 107: ['Accounting']: 0.2152
5. 46: ['Data Analysis']: 0.0908
6. 393: ['SAP']: 0.0799
7. 731: ['AI Chatbot']: 0.0645
8. 609: ['Immigration Law']: 0.0642
9. 119: ['Financial Analysis']: 0.0637
10. 128: ['Bookkeeping']: 0.0479
11. 112: ['Tax Law']: 0.0408
12. 284: ['Analytics']: 0.0388
13. 293: ['Intuit QuickBooks']: 0.0348
14. 687: ['Romanian']: 0.0328
15. 378: ['Inventory Management']: 0.0328
16. 618: ['Employer-Sponsored Visa']: 0.0314
17. 115: ['Data Visualization']: 0.0305

18. 335: ['Intellectual Property Protection']: 0.0246
19. 291: ['Intellectual Property Law']: 0.0239
20. 147: ['Legal Consulting']: 0.0199

Closeness Centrality - Top 5:

1. 26: ['Microsoft Excel']: 0.5193
2. 107: ['Accounting']: 0.4840
3. 258: ['Financial Accounting']: 0.4783
4. 119: ['Financial Analysis']: 0.4384
5. 128: ['Bookkeeping']: 0.4337
6. 293: ['Intuit QuickBooks']: 0.4172
7. 329: ['Bank Reconciliation']: 0.4144
8. 438: ['Business Plan']: 0.4074
9. 1175: ['Budget Management']: 0.4033
10. 264: ['Financial Modeling']: 0.4007
11. 613: ['Accounting Basics']: 0.3993
12. 46: ['Data Analysis']: 0.3967
13. 135: ['Legal']: 0.3954
14. 1178: ['Excel Formula']: 0.3891
15. 1180: ['Excel Macros']: 0.3891
16. 1181: ['Finance']: 0.3891
17. 1184: ['Budget Planning']: 0.3891
18. 260: ['Mathematics']: 0.3891
19. 112: ['Tax Law']: 0.3781
20. 396: ['Tax Preparation']: 0.3769

PageRank - Top 5:

1. 26: ['Microsoft Excel']: 0.0570
2. 107: ['Accounting']: 0.0275
3. 46: ['Data Analysis']: 0.0259
4. 135: ['Legal']: 0.0206
5. 119: ['Financial Analysis']: 0.0199
6. 128: ['Bookkeeping']: 0.0190
7. 258: ['Financial Accounting']: 0.0189
8. 147: ['Legal Consulting']: 0.0179
9. 115: ['Data Visualization']: 0.0167
10. 293: ['Intuit QuickBooks']: 0.0151
11. 105: ['Statistics']: 0.0134
12. 393: ['SAP']: 0.0133
13. 264: ['Financial Modeling']: 0.0129
14. 609: ['Immigration Law']: 0.0128
15. 329: ['Bank Reconciliation']: 0.0125
16. 222: ['Legal Research']: 0.0122
17. 112: ['Tax Law']: 0.0117
18. 731: ['AI Chatbot']: 0.0112
19. 284: ['Analytics']: 0.0111
20. 141: ['R']: 0.0106

Authority (HITS) - Top 5:

1. 26: ['Microsoft Excel']: 0.1118
2. 46: ['Data Analysis']: 0.0992
3. 105: ['Statistics']: 0.0846
4. 106: ['Data Science']: 0.0735
5. 141: ['R']: 0.0699
6. 229: ['SAS']: 0.0646
7. 231: ['Quantitative Analysis']: 0.0646

8. 115: ['Data Visualization']: 0.0310
9. 119: ['Financial Analysis']: 0.0290
10. 107: ['Accounting']: 0.0233
11. 180: ['Google Docs']: 0.0215
12. 184: ['Spreadsheet Software']: 0.0215
13. 264: ['Financial Modeling']: 0.0154
14. 128: ['Bookkeeping']: 0.0152
15. 284: ['Analytics']: 0.0147
16. 416: ['Qualitative Research']: 0.0142
17. 379: ['Report Writing']: 0.0142
18. 290: ['Microsoft Power BI']: 0.0138
19. 357: ['Business Analysis']: 0.0117
20. 372: ['IBM SPSS']: 0.0115

Eigenvector Centrality - Top 5:

1. 26: ['Microsoft Excel']: 0.4420
2. 107: ['Accounting']: 0.2908
3. 128: ['Bookkeeping']: 0.2335
4. 119: ['Financial Analysis']: 0.2264
5. 258: ['Financial Accounting']: 0.2131
6. 46: ['Data Analysis']: 0.1970
7. 293: ['Intuit QuickBooks']: 0.1779
8. 329: ['Bank Reconciliation']: 0.1674
9. 264: ['Financial Modeling']: 0.1550
10. 613: ['Accounting Basics']: 0.1418
11. 438: ['Business Plan']: 0.1411
12. 115: ['Data Visualization']: 0.1404
13. 105: ['Statistics']: 0.1313
14. 396: ['Tax Preparation']: 0.1269
15. 357: ['Business Analysis']: 0.1165
16. 1175: ['Budget Management']: 0.1147
17. 306: ['Business Intelligence']: 0.1068
18. 112: ['Tax Law']: 0.1048
19. 141: ['R']: 0.1036
20. 290: ['Microsoft Power BI']: 0.1023

Cluster 4 : Mobile Application Development and Platform Integration Ecosystem

Degree Centrality - Top 5:

1. 27: ['Mobile App Development']: 0.7727
2. 36: ['Flutter']: 0.5455
3. 39: ['iOS']: 0.5227
4. 38: ['Android']: 0.5000
5. 63: ['Android App Development']: 0.4773
6. 108: ['iOS Development']: 0.3636
7. 121: ['In-App Purchases']: 0.3409
8. 80: ['React Native']: 0.2955
9. 270: ['Smartphone']: 0.2727
10. 538: ['User Authentication']: 0.2727
11. 174: ['Firebase']: 0.2273
12. 201: ['Tablet']: 0.2273
13. 207: ['Dart']: 0.2273
14. 155: ['Kotlin']: 0.2045
15. 183: ['Swift']: 0.1818
16. 209: ['Google Cloud Platform']: 0.1818
17. 915: ['Mobx']: 0.1591

18. 593: ['Social Media Account Integration']: 0.1591
19. 228: ['Hybrid App']: 0.1591
20. 220: ['Firebase Cloud Firestore']: 0.1364

Betweenness Centrality - Top 5:

1. 27: ['Mobile App Development']: 0.3346
2. 121: ['In-App Purchases']: 0.1772
3. 36: ['Flutter']: 0.1150
4. 38: ['Android']: 0.1065
5. 63: ['Android App Development']: 0.0919
6. 209: ['Google Cloud Platform']: 0.0890
7. 39: ['iOS']: 0.0715
8. 538: ['User Authentication']: 0.0451
9. 108: ['iOS Development']: 0.0195
10. 593: ['Social Media Account Integration']: 0.0129
11. 80: ['React Native']: 0.0106
12. 174: ['Firebase']: 0.0101
13. 207: ['Dart']: 0.0092
14. 270: ['Smartphone']: 0.0085
15. 201: ['Tablet']: 0.0032
16. 155: ['Kotlin']: 0.0018
17. 183: ['Swift']: 0.0011
18. 220: ['Firebase Cloud Firestore']: 0.0002
19. 1155: ['Native App Development']: 0.0000
20. 1158: ['Mobile App Redesign']: 0.0000

Closeness Centrality - Top 5:

1. 27: ['Mobile App Development']: 0.8148
2. 36: ['Flutter']: 0.6875
3. 39: ['iOS']: 0.6567
4. 38: ['Android']: 0.6471
5. 63: ['Android App Development']: 0.6377
6. 121: ['In-App Purchases']: 0.5867
7. 108: ['iOS Development']: 0.5714
8. 538: ['User Authentication']: 0.5641
9. 270: ['Smartphone']: 0.5432
10. 80: ['React Native']: 0.5366
11. 174: ['Firebase']: 0.5301
12. 201: ['Tablet']: 0.5301
13. 207: ['Dart']: 0.5301
14. 164: ['Offline Functionality']: 0.5176
15. 169: ['TV Set']: 0.5176
16. 155: ['Kotlin']: 0.5116
17. 1183: ['Payment Integration Framework']: 0.5116
18. 593: ['Social Media Account Integration']: 0.5116
19. 915: ['Mobx']: 0.5000
20. 209: ['Google Cloud Platform']: 0.5000

PageRank - Top 5:

1. 27: ['Mobile App Development']: 0.0808
2. 36: ['Flutter']: 0.0564
3. 39: ['iOS']: 0.0533
4. 38: ['Android']: 0.0525
5. 63: ['Android App Development']: 0.0502
6. 121: ['In-App Purchases']: 0.0425
7. 108: ['iOS Development']: 0.0372

8. 538: ['User Authentication']: 0.0316
9. 80: ['React Native']: 0.0308
10. 270: ['Smartphone']: 0.0287
11. 209: ['Google Cloud Platform']: 0.0254
12. 174: ['Firebase']: 0.0252
13. 207: ['Dart']: 0.0248
14. 201: ['Tablet']: 0.0240
15. 155: ['Kotlin']: 0.0216
16. 593: ['Social Media Account Integration']: 0.0212
17. 183: ['Swift']: 0.0199
18. 915: ['Mobx']: 0.0176
19. 228: ['Hybrid App']: 0.0175
20. 220: ['Firebase Cloud Firestore']: 0.0168

Authority (HITS) - Top 5:

1. 27: ['Mobile App Development']: 0.1269
2. 36: ['Flutter']: 0.1072
3. 39: ['iOS']: 0.0922
4. 38: ['Android']: 0.0702
5. 63: ['Android App Development']: 0.0667
6. 108: ['iOS Development']: 0.0546
7. 80: ['React Native']: 0.0463
8. 183: ['Swift']: 0.0380
9. 155: ['Kotlin']: 0.0366
10. 228: ['Hybrid App']: 0.0335
11. 121: ['In-App Purchases']: 0.0330
12. 169: ['TV Set']: 0.0326
13. 164: ['Offline Functionality']: 0.0326
14. 207: ['Dart']: 0.0321
15. 201: ['Tablet']: 0.0312
16. 174: ['Firebase']: 0.0290
17. 209: ['Google Cloud Platform']: 0.0209
18. 220: ['Firebase Cloud Firestore']: 0.0209
19. 202: ['Google Maps']: 0.0205
20. 270: ['Smartphone']: 0.0174

Eigenvector Centrality - Top 5:

1. 27: ['Mobile App Development']: 0.3695
2. 39: ['iOS']: 0.3063
3. 36: ['Flutter']: 0.3053
4. 38: ['Android']: 0.2839
5. 63: ['Android App Development']: 0.2740
6. 108: ['iOS Development']: 0.2417
7. 80: ['React Native']: 0.2078
8. 270: ['Smartphone']: 0.2004
9. 201: ['Tablet']: 0.1848
10. 155: ['Kotlin']: 0.1740
11. 207: ['Dart']: 0.1701
12. 121: ['In-App Purchases']: 0.1643
13. 174: ['Firebase']: 0.1625
14. 538: ['User Authentication']: 0.1520
15. 915: ['Mobx']: 0.1439
16. 183: ['Swift']: 0.1397
17. 228: ['Hybrid App']: 0.1341
18. 412: ['Privacy Policy']: 0.1135
19. 967: ['Objective-C']: 0.1002

20. 917: ['Mobile App']: 0.0992

Clustere 5: Digital Design, Engineering and Architectural Visualization Ecosystem

Degree Centrality - Top 5:

1. 90: ['3D Rendering']: 0.2600
2. 92: ['Architectural Design']: 0.2600
3. 117: ['Autodesk AutoCAD']: 0.2500
4. 57: ['3D Modeling']: 0.2300
5. 73: ['3D Design']: 0.2200
6. 100: ['Mechanical Engineering']: 0.2200
7. 300: ['Drafting']: 0.1600
8. 95: ['SolidWorks']: 0.1400
9. 251: ['Embedded System']: 0.1400
10. 299: ['Electrical Engineering']: 0.1300
11. 449: ['CAD']: 0.1300
12. 429: ['PCB Design']: 0.1200
13. 304: ['Structural Engineering']: 0.1100
14. 305: ['Structural Analysis']: 0.1100
15. 331: ['2D Design']: 0.1100
16. 358: ['Residential Design']: 0.1100
17. 431: ['Floor Plan Design']: 0.1100
18. 584: ['Interior Design']: 0.1000
19. 603: ['Photorealistic Rendering']: 0.0900
20. 138: ['Finite Element Analysis']: 0.0900

Betweenness Centrality - Top 5:

1. 100: ['Mechanical Engineering']: 0.4069
2. 73: ['3D Design']: 0.2272
3. 429: ['PCB Design']: 0.1794
4. 346: ['LEED Consulting']: 0.1655
5. 117: ['Autodesk AutoCAD']: 0.1579
6. 267: ['Arduino']: 0.1545
7. 92: ['Architectural Design']: 0.1191
8. 251: ['Embedded System']: 0.1190
9. 90: ['3D Rendering']: 0.1119
10. 341: ['HVAC System Design']: 0.0960
11. 57: ['3D Modeling']: 0.0942
12. 358: ['Residential Design']: 0.0532
13. 138: ['Finite Element Analysis']: 0.0486
14. 304: ['Structural Engineering']: 0.0460
15. 305: ['Structural Analysis']: 0.0460
16. 479: ['Firmware']: 0.0348
17. 299: ['Electrical Engineering']: 0.0316
18. 300: ['Drafting']: 0.0239
19. 449: ['CAD']: 0.0222
20. 342: ['MATLAB']: 0.0200

Closeness Centrality - Top 5:

1. 100: ['Mechanical Engineering']: 0.4762
2. 73: ['3D Design']: 0.4630
3. 117: ['Autodesk AutoCAD']: 0.4484
4. 90: ['3D Rendering']: 0.4386
5. 57: ['3D Modeling']: 0.4367
6. 449: ['CAD']: 0.4098
7. 92: ['Architectural Design']: 0.4082

8. 95: ['SolidWorks']: 0.4082
9. 411: ['3D Printing']: 0.3922
10. 429: ['PCB Design']: 0.3802
11. 358: ['Residential Design']: 0.3731
12. 138: ['Finite Element Analysis']: 0.3717
13. 267: ['Arduino']: 0.3704
14. 300: ['Drafting']: 0.3663
15. 176: ['Autodesk 3ds Max']: 0.3650
16. 331: ['2D Design']: 0.3623
17. 664: ['Design for Manufacturing']: 0.3610
18. 168: ['Flow Simulation']: 0.3597
19. 171: ['Machine Design']: 0.3597
20. 683: ['AutoCAD Civil 3D']: 0.3559

PageRank - Top 5:

1. 92: ['Architectural Design']: 0.0304
2. 90: ['3D Rendering']: 0.0289
3. 117: ['Autodesk AutoCAD']: 0.0279
4. 100: ['Mechanical Engineering']: 0.0269
5. 57: ['3D Modeling']: 0.0262
6. 73: ['3D Design']: 0.0251
7. 251: ['Embedded System']: 0.0186
8. 300: ['Drafting']: 0.0184
9. 299: ['Electrical Engineering']: 0.0171
10. 95: ['SolidWorks']: 0.0162
11. 429: ['PCB Design']: 0.0159
12. 341: ['HVAC System Design']: 0.0153
13. 449: ['CAD']: 0.0147
14. 304: ['Structural Engineering']: 0.0147
15. 305: ['Structural Analysis']: 0.0147
16. 346: ['LEED Consulting']: 0.0140
17. 358: ['Residential Design']: 0.0138
18. 331: ['2D Design']: 0.0130
19. 431: ['Floor Plan Design']: 0.0126
20. 267: ['Arduino']: 0.0124

Authority (HITS) - Top 5:

1. 57: ['3D Modeling']: 0.1262
2. 95: ['SolidWorks']: 0.1195
3. 100: ['Mechanical Engineering']: 0.1061
4. 138: ['Finite Element Analysis']: 0.0939
5. 171: ['Machine Design']: 0.0935
6. 168: ['Flow Simulation']: 0.0935
7. 73: ['3D Design']: 0.0514
8. 90: ['3D Rendering']: 0.0433
9. 117: ['Autodesk AutoCAD']: 0.0306
10. 92: ['Architectural Design']: 0.0248
11. 411: ['3D Printing']: 0.0213
12. 176: ['Autodesk 3ds Max']: 0.0198
13. 223: ['High-Quality Rendering']: 0.0132
14. 224: ['Corona']: 0.0132
15. 449: ['CAD']: 0.0109
16. 358: ['Residential Design']: 0.0072
17. 300: ['Drafting']: 0.0068
18. 664: ['Design for Manufacturing']: 0.0061
19. 267: ['Arduino']: 0.0054

20. 319: ['Arduino Firmware']: 0.0054

Eigenvector Centrality - Top 5:

1. 90: ['3D Rendering']: 0.3301
2. 117: ['Autodesk AutoCAD']: 0.3177
3. 92: ['Architectural Design']: 0.2934
4. 73: ['3D Design']: 0.2802
5. 57: ['3D Modeling']: 0.2747
6. 300: ['Drafting']: 0.2098
7. 449: ['CAD']: 0.2085
8. 100: ['Mechanical Engineering']: 0.1957
9. 95: ['SolidWorks']: 0.1875
10. 431: ['Floor Plan Design']: 0.1712
11. 584: ['Interior Design']: 0.1679
12. 331: ['2D Design']: 0.1566
13. 358: ['Residential Design']: 0.1539
14. 674: ['SketchUp']: 0.1499
15. 683: ['AutoCAD Civil 3D']: 0.1376
16. 411: ['3D Printing']: 0.1308
17. 176: ['Autodesk 3ds Max']: 0.1304
18. 603: ['Photorealistic Rendering']: 0.1297
19. 522: ['Exterior Rendering']: 0.1238
20. 495: ['Interior Design Consultation']: 0.1236

Global 2022
NetworkX modularity: 0.4910

Cluster 2 - Node : 721, Edge: 7239

Cluster 0 - Node : 1702, Edge: 26684

Cluster 1 - Node : 1321, Edge: 14284

Cluster 0 : A creative production ecosystem at the intersection of digital content production and multimedia applications

Degree Centrality - Top 5:

1. 0: ['Graphic Design']: 0.4653
2. 12: ['Adobe Photoshop']: 0.3472
3. 25: ['Adobe Illustrator']: 0.3278
4. 46: ['Illustration']: 0.2625
5. 72: ['3D Modeling']: 0.2069
6. 14: ['Video Editing']: 0.2042
7. 40: ['Logo Design']: 0.1875
8. 82: ['3D Design']: 0.1875
9. 91: ['3D Rendering']: 0.1750
10. 29: ['Video Production']: 0.1694
11. 62: ['Adobe After Effects']: 0.1597
12. 85: ['Animation']: 0.1528
13. 78: ['Photo Editing']: 0.1500
14. 137: ['Infographic']: 0.1500
15. 163: ['Character Design']: 0.1361
16. 129: ['Layout Design']: 0.1333
17. 132: ['Architectural Design']: 0.1292
18. 134: ['Adobe InDesign']: 0.1292

19. 96: ['Social Media Imagery']: 0.1278
20. 60: ['Video Post-Editing']: 0.1264

Betweenness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.2505
2. 12: ['Adobe Photoshop']: 0.1250
3. 25: ['Adobe Illustrator']: 0.0934
4. 72: ['3D Modeling']: 0.0761
5. 101: ['Audio Editing']: 0.0612
6. 14: ['Video Editing']: 0.0582
7. 46: ['Illustration']: 0.0527
8. 82: ['3D Design']: 0.0492
9. 161: ['Autodesk AutoCAD']: 0.0462
10. 91: ['3D Rendering']: 0.0360
11. 29: ['Video Production']: 0.0350
12. 62: ['Adobe After Effects']: 0.0296
13. 132: ['Architectural Design']: 0.0244
14. 85: ['Animation']: 0.0238
15. 78: ['Photo Editing']: 0.0238
16. 40: ['Logo Design']: 0.0228
17. 314: ['Audio Production']: 0.0187
18. 344: ['Clothing']: 0.0182
19. 76: ['Videography']: 0.0159
20. 279: ['Product Design']: 0.0155

Closeness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.6344
2. 12: ['Adobe Photoshop']: 0.5878
3. 25: ['Adobe Illustrator']: 0.5774
4. 46: ['Illustration']: 0.5357
5. 72: ['3D Modeling']: 0.5248
6. 82: ['3D Design']: 0.5210
7. 14: ['Video Editing']: 0.5191
8. 91: ['3D Rendering']: 0.5128
9. 62: ['Adobe After Effects']: 0.5088
10. 40: ['Logo Design']: 0.5067
11. 29: ['Video Production']: 0.5010
12. 84: ['Motion Graphics']: 0.5000
13. 78: ['Photo Editing']: 0.4986
14. 85: ['Animation']: 0.4972
15. 163: ['Character Design']: 0.4891
16. 176: ['3D Animation']: 0.4888
17. 96: ['Social Media Imagery']: 0.4885
18. 137: ['Infographic']: 0.4875
19. 134: ['Adobe InDesign']: 0.4842
20. 60: ['Video Post-Editing']: 0.4819

PageRank - Top 5:

1. 0: ['Graphic Design']: 0.0210
2. 12: ['Adobe Photoshop']: 0.0145
3. 25: ['Adobe Illustrator']: 0.0142
4. 46: ['Illustration']: 0.0110
5. 72: ['3D Modeling']: 0.0102
6. 14: ['Video Editing']: 0.0093
7. 82: ['3D Design']: 0.0087
8. 91: ['3D Rendering']: 0.0080

9. 40: ['Logo Design']: 0.0078
10. 29: ['Video Production']: 0.0075
11. 161: ['Autodesk AutoCAD']: 0.0068
12. 62: ['Adobe After Effects']: 0.0065
13. 78: ['Photo Editing']: 0.0065
14. 85: ['Animation']: 0.0064
15. 132: ['Architectural Design']: 0.0062
16. 101: ['Audio Editing']: 0.0061
17. 137: ['Infographic']: 0.0059
18. 129: ['Layout Design']: 0.0058
19. 76: ['Videography']: 0.0054
20. 163: ['Character Design']: 0.0053

Authority (HITS) - Top 5:

1. 0: ['Graphic Design']: 0.0784
2. 12: ['Adobe Photoshop']: 0.0643
3. 25: ['Adobe Illustrator']: 0.0610
4. 40: ['Logo Design']: 0.0376
5. 46: ['Illustration']: 0.0354
6. 14: ['Video Editing']: 0.0340
7. 29: ['Video Production']: 0.0285
8. 62: ['Adobe After Effects']: 0.0259
9. 63: ['Adobe Premiere Pro']: 0.0234
10. 97: ['Brand Identity & Guidelines']: 0.0228
11. 60: ['Video Post-Editing']: 0.0225
12. 96: ['Social Media Imagery']: 0.0212
13. 134: ['Adobe InDesign']: 0.0179
14. 78: ['Photo Editing']: 0.0172
15. 84: ['Motion Graphics']: 0.0155
16. 137: ['Infographic']: 0.0147
17. 150: ['Print Design']: 0.0129
18. 85: ['Animation']: 0.0129
19. 129: ['Layout Design']: 0.0124
20. 76: ['Videography']: 0.0120

Eigenvector Centrality - Top 5:

1. 0: ['Graphic Design']: 0.2568
2. 12: ['Adobe Photoshop']: 0.2275
3. 25: ['Adobe Illustrator']: 0.2138
4. 46: ['Illustration']: 0.1856
5. 40: ['Logo Design']: 0.1423
6. 137: ['Infographic']: 0.1319
7. 85: ['Animation']: 0.1263
8. 62: ['Adobe After Effects']: 0.1202
9. 163: ['Character Design']: 0.1190
10. 14: ['Video Editing']: 0.1181
11. 96: ['Social Media Imagery']: 0.1175
12. 78: ['Photo Editing']: 0.1128
13. 82: ['3D Design']: 0.1123
14. 175: ['Drawing']: 0.1113
15. 134: ['Adobe InDesign']: 0.1089
16. 72: ['3D Modeling']: 0.1061
17. 127: ['2D Animation']: 0.1058
18. 29: ['Video Production']: 0.1047
19. 129: ['Layout Design']: 0.1043
20. 455: ['Contemporary Style']: 0.1021

Cluster 1 : Content Production and Digital Assistant Ecosystem

Degree Centrality - Top 5:

1. 2: ['English']: 0.3645
2. 1: ['Content Writing']: 0.3128
3. 8: ['Researcher']: 0.2934
4. 9: ['Data Entry']: 0.2598
5. 19: ['Virtual Assistant']: 0.2363
6. 43: ['Administrative Support']: 0.2081
7. 5: ['Social Media Marketing']: 0.1969
8. 33: ['Communications']: 0.1964
9. 10: ['Lead Generation']: 0.1928
10. 22: ['Marketing Strategy']: 0.1905
11. 20: ['Writing']: 0.1887
12. 18: ['Copywriting']: 0.1723
13. 44: ['Email Communication']: 0.1705
14. 55: ['Market Research']: 0.1693
15. 11: ['Search Engine Optimization']: 0.1658
16. 26: ['Creative Writing']: 0.1611
17. 32: ['Microsoft Excel']: 0.1605
18. 49: ['Customer Service']: 0.1540
19. 37: ['Translation']: 0.1446
20. 24: ['Social Media Management']: 0.1317

Betweenness Centrality - Top 5:

1. 2: ['English']: 0.1514
2. 1: ['Content Writing']: 0.0777
3. 8: ['Researcher']: 0.0676
4. 9: ['Data Entry']: 0.0592
5. 19: ['Virtual Assistant']: 0.0380
6. 37: ['Translation']: 0.0323
7. 32: ['Microsoft Excel']: 0.0304
8. 55: ['Market Research']: 0.0295
9. 10: ['Lead Generation']: 0.0293
10. 20: ['Writing']: 0.0290
11. 33: ['Communications']: 0.0281
12. 43: ['Administrative Support']: 0.0243
13. 22: ['Marketing Strategy']: 0.0229
14. 5: ['Social Media Marketing']: 0.0227
15. 26: ['Creative Writing']: 0.0196
16. 44: ['Email Communication']: 0.0188
17. 11: ['Search Engine Optimization']: 0.0187
18. 18: ['Copywriting']: 0.0174
19. 49: ['Customer Service']: 0.0166
20. 155: ['Education']: 0.0161

Closeness Centrality - Top 5:

1. 2: ['English']: 0.6002
2. 1: ['Content Writing']: 0.5839
3. 8: ['Researcher']: 0.5770
4. 9: ['Data Entry']: 0.5676
5. 19: ['Virtual Assistant']: 0.5568
6. 43: ['Administrative Support']: 0.5435
7. 33: ['Communications']: 0.5373

8. 10: ['Lead Generation']: 0.5359
9. 20: ['Writing']: 0.5342
10. 5: ['Social Media Marketing']: 0.5322
11. 22: ['Marketing Strategy']: 0.5314
12. 18: ['Copywriting']: 0.5279
13. 32: ['Microsoft Excel']: 0.5270
14. 44: ['Email Communication']: 0.5268
15. 11: ['Search Engine Optimization']: 0.5242
16. 55: ['Market Research']: 0.5226
17. 26: ['Creative Writing']: 0.5203
18. 49: ['Customer Service']: 0.5167
19. 13: ['Facebook']: 0.5120
20. 17: ['Instagram']: 0.5120

PageRank - Top 5:

1. 2: ['English']: 0.0121
2. 1: ['Content Writing']: 0.0091
3. 8: ['Researcher']: 0.0085
4. 9: ['Data Entry']: 0.0076
5. 19: ['Virtual Assistant']: 0.0065
6. 43: ['Administrative Support']: 0.0056
7. 20: ['Writing']: 0.0056
8. 33: ['Communications']: 0.0055
9. 10: ['Lead Generation']: 0.0054
10. 5: ['Social Media Marketing']: 0.0052
11. 22: ['Marketing Strategy']: 0.0051
12. 37: ['Translation']: 0.0051
13. 55: ['Market Research']: 0.0049
14. 32: ['Microsoft Excel']: 0.0048
15. 18: ['Copywriting']: 0.0047
16. 44: ['Email Communication']: 0.0047
17. 26: ['Creative Writing']: 0.0046
18. 11: ['Search Engine Optimization']: 0.0044
19. 49: ['Customer Service']: 0.0043
20. 59: ['Proofreading']: 0.0038

Authority (HITS) - Top 5:

1. 1: ['Content Writing']: 0.0527
2. 2: ['English']: 0.0329
3. 27: ['Article Writing']: 0.0322
4. 20: ['Writing']: 0.0306
5. 18: ['Copywriting']: 0.0306
6. 8: ['Researcher']: 0.0290
7. 5: ['Social Media Marketing']: 0.0272
8. 26: ['Creative Writing']: 0.0258
9. 35: ['Blog Content']: 0.0253
10. 45: ['Blog Writing']: 0.0235
11. 36: ['SEO Writing']: 0.0225
12. 11: ['Search Engine Optimization']: 0.0220
13. 17: ['Instagram']: 0.0202
14. 13: ['Facebook']: 0.0202
15. 65: ['Article']: 0.0184
16. 24: ['Social Media Management']: 0.0179
17. 22: ['Marketing Strategy']: 0.0162
18. 57: ['Social Media Content']: 0.0124
19. 10: ['Lead Generation']: 0.0119

20. 52: ['Social Media Content Creation']: 0.0118

Eigenvector Centrality - Top 5:

1. 1: ['Content Writing']: 0.1612
2. 8: ['Researcher']: 0.1557
3. 2: ['English']: 0.1525
4. 9: ['Data Entry']: 0.1368
5. 19: ['Virtual Assistant']: 0.1319
6. 43: ['Administrative Support']: 0.1244
7. 5: ['Social Media Marketing']: 0.1237
8. 10: ['Lead Generation']: 0.1201
9. 22: ['Marketing Strategy']: 0.1199
10. 18: ['Copywriting']: 0.1120
11. 11: ['Search Engine Optimization']: 0.1115
12. 20: ['Writing']: 0.1090
13. 33: ['Communications']: 0.1070
14. 44: ['Email Communication']: 0.1055
15. 57: ['Social Media Content']: 0.1048
16. 26: ['Creative Writing']: 0.1014
17. 13: ['Facebook']: 0.0973
18. 55: ['Market Research']: 0.0966
19. 52: ['Social Media Content Creation']: 0.0946
20. 17: ['Instagram']: 0.0926

Cluster 2: Multi-Tiered Web and Software Development Ecosystem

Degree Centrality - Top 5:

1. 4: ['JavaScript']: 0.3311
2. 3: ['Web Development']: 0.2439
3. 28: ['Python']: 0.2205
4. 15: ['HTML']: 0.2114
5. 31: ['API']: 0.2083
6. 53: ['Java']: 0.1985
7. 51: ['API Integration']: 0.1826
8. 21: ['PHP']: 0.1811
9. 16: ['CSS']: 0.1773
10. 93: ['SQL']: 0.1652
11. 6: ['Web Design']: 0.1621
12. 7: ['WordPress']: 0.1614
13. 69: ['Business with 1-9 Employees']: 0.1614
14. 64: ['Web Application']: 0.1583
15. 23: ['Android']: 0.1538
16. 74: ['MySQL']: 0.1462
17. 61: ['React']: 0.1455
18. 68: ['Node.js']: 0.1447
19. 67: ['HTML5']: 0.1326
20. 38: ['iOS']: 0.1242

Betweenness Centrality - Top 5:

1. 4: ['JavaScript']: 0.1185
2. 28: ['Python']: 0.0935
3. 3: ['Web Development']: 0.0566
4. 53: ['Java']: 0.0551

5. 165: ['C++']: 0.0510
6. 31: ['API']: 0.0461
7. 7: ['WordPress']: 0.0440
8. 23: ['Android']: 0.0394
9. 69: ['Business with 1-9 Employees']: 0.0354
10. 15: ['HTML']: 0.0343
11. 93: ['SQL']: 0.0340
12. 21: ['PHP']: 0.0333
13. 51: ['API Integration']: 0.0318
14. 6: ['Web Design']: 0.0296
15. 197: ['Linux']: 0.0294
16. 113: ['C#']: 0.0268
17. 16: ['CSS']: 0.0219
18. 192: ['Business with 10-99 Employees']: 0.0219
19. 74: ['MySQL']: 0.0208
20. 118: ['Automation']: 0.0202

Closeness Centrality - Top 5:

1. 4: ['JavaScript']: 0.5789
2. 3: ['Web Development']: 0.5375
3. 28: ['Python']: 0.5368
4. 31: ['API']: 0.5244
5. 53: ['Java']: 0.5232
6. 15: ['HTML']: 0.5181
7. 21: ['PHP']: 0.5136
8. 51: ['API Integration']: 0.5093
9. 16: ['CSS']: 0.5067
10. 7: ['WordPress']: 0.5048
11. 68: ['Node.js']: 0.5040
12. 69: ['Business with 1-9 Employees']: 0.5038
13. 74: ['MySQL']: 0.5025
14. 23: ['Android']: 0.5017
15. 192: ['Business with 10-99 Employees']: 0.5011
16. 64: ['Web Application']: 0.5008
17. 93: ['SQL']: 0.5004
18. 165: ['C++']: 0.4955
19. 113: ['C#']: 0.4951
20. 61: ['React']: 0.4920

PageRank - Top 5:

1. 4: ['JavaScript']: 0.0129
2. 28: ['Python']: 0.0096
3. 3: ['Web Development']: 0.0095
4. 31: ['API']: 0.0081
5. 53: ['Java']: 0.0080
6. 15: ['HTML']: 0.0080
7. 21: ['PHP']: 0.0071
8. 51: ['API Integration']: 0.0070
9. 7: ['WordPress']: 0.0068
10. 16: ['CSS']: 0.0066
11. 93: ['SQL']: 0.0064
12. 6: ['Web Design']: 0.0064
13. 69: ['Business with 1-9 Employees']: 0.0061
14. 23: ['Android']: 0.0060
15. 64: ['Web Application']: 0.0058
16. 61: ['React']: 0.0055

17. 74: ['MySQL']: 0.0055
18. 68: ['Node.js']: 0.0054
19. 165: ['C++']: 0.0048
20. 67: ['HTML5']: 0.0048

Authority (HITS) - Top 5:

1. 3: ['Web Development']: 0.0711
2. 4: ['JavaScript']: 0.0622
3. 16: ['CSS']: 0.0553
4. 15: ['HTML']: 0.0552
5. 7: ['WordPress']: 0.0532
6. 6: ['Web Design']: 0.0528
7. 21: ['PHP']: 0.0436
8. 67: ['HTML5']: 0.0241
9. 31: ['API']: 0.0197
10. 64: ['Web Application']: 0.0164
11. 23: ['Android']: 0.0158
12. 51: ['API Integration']: 0.0142
13. 56: ['Landing Page']: 0.0139
14. 74: ['MySQL']: 0.0139
15. 30: ['Mobile App Development']: 0.0136
16. 61: ['React']: 0.0133
17. 92: ['WooCommerce']: 0.0132
18. 41: ['Shopify']: 0.0131
19. 81: ['Website']: 0.0126
20. 39: ['Android App Development']: 0.0119

Eigenvector Centrality - Top 5:

1. 4: ['JavaScript']: 0.2104
2. 3: ['Web Development']: 0.1748
3. 15: ['HTML']: 0.1599
4. 31: ['API']: 0.1575
5. 51: ['API Integration']: 0.1452
6. 16: ['CSS']: 0.1448
7. 21: ['PHP']: 0.1448
8. 64: ['Web Application']: 0.1410
9. 53: ['Java']: 0.1384
10. 69: ['Business with 1-9 Employees']: 0.1359
11. 68: ['Node.js']: 0.1335
12. 28: ['Python']: 0.1321
13. 61: ['React']: 0.1317
14. 74: ['MySQL']: 0.1299
15. 6: ['Web Design']: 0.1287
16. 7: ['WordPress']: 0.1243
17. 67: ['HTML5']: 0.1204
18. 93: ['SQL']: 0.1181
19. 23: ['Android']: 0.1178
20. 192: ['Business with 10-99 Employees']: 0.1039

Global 2025

NetworkX modularity: 0.6394

Cluster 2 - Node : 180, Edge: 819
Cluster 1 - Node : 277, Edge: 1472
Cluster 0 - Node : 498, Edge: 2271
Cluster 4 - Node : 45, Edge: 188

Cluster 0: Visual Media and Digital Content Production Ecosystem

Degree Centrality - Top 5:

1. 0: ['Graphic Design']: 0.4358
2. 9: ['Adobe Photoshop']: 0.4078
3. 18: ['Adobe Illustrator']: 0.3687
4. 7: ['Video Editing']: 0.2682
5. 23: ['Adobe After Effects']: 0.1899
6. 35: ['Illustration']: 0.1899
7. 20: ['Adobe Premiere Pro']: 0.1788
8. 25: ['Video Production']: 0.1732
9. 83: ['Print Design']: 0.1676
10. 44: ['Logo Design']: 0.1564
11. 48: ['Brand Identity & Guidelines']: 0.1453
12. 85: ['Videography']: 0.1453
13. 29: ['Video Post-Editing']: 0.1341
14. 101: ['Canva']: 0.1117
15. 124: ['Explainer Video']: 0.1117
16. 56: ['Animation']: 0.1061
17. 93: ['Photo Editing']: 0.1061
18. 88: ['Adobe InDesign']: 0.1006
19. 97: ['Video Commercial']: 0.1006
20. 132: ['Infographic']: 0.1006

Betweenness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.3247
2. 9: ['Adobe Photoshop']: 0.1991
3. 18: ['Adobe Illustrator']: 0.1265
4. 126: ['Project Management']: 0.1077
5. 7: ['Video Editing']: 0.1058
6. 25: ['Video Production']: 0.0575
7. 125: ['Photography']: 0.0529
8. 56: ['Animation']: 0.0469
9. 23: ['Adobe After Effects']: 0.0438
10. 44: ['Logo Design']: 0.0407
11. 124: ['Explainer Video']: 0.0365
12. 185: ['Product Photography']: 0.0343
13. 35: ['Illustration']: 0.0322
14. 20: ['Adobe Premiere Pro']: 0.0317
15. 83: ['Print Design']: 0.0299
16. 101: ['Canva']: 0.0281
17. 87: ['Product Design']: 0.0260
18. 85: ['Videography']: 0.0254
19. 149: ['DaVinci Resolve']: 0.0222
20. 93: ['Photo Editing']: 0.0176

Closeness Centrality - Top 5:

1. 0: ['Graphic Design']: 0.6325
2. 9: ['Adobe Photoshop']: 0.6027
3. 18: ['Adobe Illustrator']: 0.5683
4. 7: ['Video Editing']: 0.5359
5. 23: ['Adobe After Effects']: 0.5188
6. 25: ['Video Production']: 0.5028
7. 20: ['Adobe Premiere Pro']: 0.5014
8. 35: ['Illustration']: 0.4918
9. 29: ['Video Post-Editing']: 0.4877

10. 124: ['Explainer Video']: 0.4799
11. 93: ['Photo Editing']: 0.4786
12. 99: ['Motion Graphics']: 0.4786
13. 44: ['Logo Design']: 0.4761
14. 56: ['Animation']: 0.4748
15. 101: ['Canva']: 0.4735
16. 94: ['2D Animation']: 0.4723
17. 185: ['Product Photography']: 0.4723
18. 132: ['Infographic']: 0.4674
19. 83: ['Print Design']: 0.4613
20. 196: ['YouTube']: 0.4613

PageRank - Top 5:

1. 0: ['Graphic Design']: 0.0425
2. 9: ['Adobe Photoshop']: 0.0385
3. 18: ['Adobe Illustrator']: 0.0342
4. 7: ['Video Editing']: 0.0260
5. 35: ['Illustration']: 0.0178
6. 23: ['Adobe After Effects']: 0.0173
7. 25: ['Video Production']: 0.0167
8. 20: ['Adobe Premiere Pro']: 0.0165
9. 83: ['Print Design']: 0.0159
10. 44: ['Logo Design']: 0.0153
11. 85: ['Videography']: 0.0147
12. 48: ['Brand Identity & Guidelines']: 0.0138
13. 126: ['Project Management']: 0.0129
14. 29: ['Video Post-Editing']: 0.0125
15. 124: ['Explainer Video']: 0.0115
16. 101: ['Canva']: 0.0109
17. 125: ['Photography']: 0.0109
18. 56: ['Animation']: 0.0108
19. 93: ['Photo Editing']: 0.0105
20. 97: ['Video Commercial']: 0.0099

Authority (HITS) - Top 5:

1. 7: ['Video Editing']: 0.1205
2. 20: ['Adobe Premiere Pro']: 0.1102
3. 23: ['Adobe After Effects']: 0.1003
4. 25: ['Video Production']: 0.0907
5. 29: ['Video Post-Editing']: 0.0812
6. 9: ['Adobe Photoshop']: 0.0475
7. 0: ['Graphic Design']: 0.0403
8. 18: ['Adobe Illustrator']: 0.0332
9. 97: ['Video Commercial']: 0.0244
10. 143: ['Final Cut Pro']: 0.0200
11. 153: ['CapCut']: 0.0195
12. 149: ['DaVinci Resolve']: 0.0179
13. 85: ['Videography']: 0.0162
14. 101: ['Canva']: 0.0147
15. 99: ['Motion Graphics']: 0.0141
16. 35: ['Illustration']: 0.0139
17. 124: ['Explainer Video']: 0.0138
18. 196: ['YouTube']: 0.0114
19. 93: ['Photo Editing']: 0.0101
20. 44: ['Logo Design']: 0.0091

Eigenvector Centrality - Top 5:

1. 9: ['Adobe Photoshop']: 0.3260
2. 0: ['Graphic Design']: 0.3218
3. 18: ['Adobe Illustrator']: 0.2904
4. 7: ['Video Editing']: 0.2026
5. 23: ['Adobe After Effects']: 0.1978
6. 35: ['Illustration']: 0.1854
7. 20: ['Adobe Premiere Pro']: 0.1741
8. 25: ['Video Production']: 0.1517
9. 44: ['Logo Design']: 0.1458
10. 29: ['Video Post-Editing']: 0.1426
11. 83: ['Print Design']: 0.1376
12. 99: ['Motion Graphics']: 0.1357
13. 101: ['Canva']: 0.1316
14. 48: ['Brand Identity & Guidelines']: 0.1314
15. 124: ['Explainer Video']: 0.1268
16. 94: ['2D Animation']: 0.1255
17. 56: ['Animation']: 0.1200
18. 93: ['Photo Editing']: 0.1168
19. 88: ['Adobe InDesign']: 0.1159
20. 196: ['YouTube']: 0.1118

Cluster 1 : Modern Web and Application Development Ecosystem

Degree Centrality - Top 5:

1. 2: ['JavaScript']: 0.3587
2. 1: ['Web Development']: 0.3152
3. 3: ['Web Design']: 0.2862
4. 10: ['API']: 0.2536
5. 4: ['WordPress']: 0.2464
6. 8: ['PHP']: 0.2391
7. 11: ['HTML']: 0.2355
8. 24: ['Python']: 0.2101
9. 17: ['CSS']: 0.2065
10. 61: ['React']: 0.1957
11. 43: ['API Integration']: 0.1667
12. 52: ['Node.js']: 0.1486
13. 104: ['Web Application']: 0.1377
14. 21: ['Shopify']: 0.1304
15. 151: ['PostgreSQL']: 0.1304
16. 242: ['RESTful API']: 0.1196
17. 40: ['Landing Page']: 0.1087
18. 96: ['SQL']: 0.1014
19. 109: ['Java']: 0.1014
20. 129: ['Website']: 0.1014

Betweenness Centrality - Top 5:

1. 2: ['JavaScript']: 0.1954
2. 3: ['Web Design']: 0.1593
3. 1: ['Web Development']: 0.1162
4. 10: ['API']: 0.1110
5. 24: ['Python']: 0.0845
6. 4: ['WordPress']: 0.0682
7. 11: ['HTML']: 0.0678
8. 8: ['PHP']: 0.0675

9. 21: ['Shopify']: 0.0534
10. 61: ['React']: 0.0454
11. 43: ['API Integration']: 0.0445
12. 96: ['SQL']: 0.0379
13. 604: ['Usability Testing']: 0.0352
14. 17: ['CSS']: 0.0345
15. 82: ['Shopify Theme']: 0.0343
16. 151: ['PostgreSQL']: 0.0322
17. 551: ['Microsoft Azure']: 0.0261
18. 60: ['Automation']: 0.0257
19. 76: ['Artificial Intelligence']: 0.0253
20. 104: ['Web Application']: 0.0225

Closeness Centrality - Top 5:

1. 2: ['JavaScript']: 0.5860
2. 1: ['Web Development']: 0.5598
3. 10: ['API']: 0.5444
4. 8: ['PHP']: 0.5349
5. 3: ['Web Design']: 0.5298
6. 11: ['HTML']: 0.5247
7. 4: ['WordPress']: 0.5227
8. 61: ['React']: 0.5159
9. 17: ['CSS']: 0.5102
10. 43: ['API Integration']: 0.5027
11. 24: ['Python']: 0.4946
12. 104: ['Web Application']: 0.4902
13. 151: ['PostgreSQL']: 0.4902
14. 52: ['Node.js']: 0.4894
15. 21: ['Shopify']: 0.4783
16. 145: ['Next.js']: 0.4623
17. 242: ['RESTful API']: 0.4608
18. 91: ['HTML5']: 0.4600
19. 109: ['Java']: 0.4600
20. 129: ['Website']: 0.4554

PageRank - Top 5:

1. 2: ['JavaScript']: 0.0286
2. 3: ['Web Design']: 0.0251
3. 1: ['Web Development']: 0.0242
4. 10: ['API']: 0.0204
5. 4: ['WordPress']: 0.0198
6. 11: ['HTML']: 0.0188
7. 8: ['PHP']: 0.0185
8. 24: ['Python']: 0.0178
9. 17: ['CSS']: 0.0155
10. 61: ['React']: 0.0150
11. 43: ['API Integration']: 0.0132
12. 21: ['Shopify']: 0.0118
13. 52: ['Node.js']: 0.0113
14. 104: ['Web Application']: 0.0106
15. 151: ['PostgreSQL']: 0.0105
16. 242: ['RESTful API']: 0.0092
17. 96: ['SQL']: 0.0088
18. 40: ['Landing Page']: 0.0087
19. 76: ['Artificial Intelligence']: 0.0084
20. 60: ['Automation']: 0.0080

Authority (HITS) - Top 5:

1. 1: ['Web Development']: 0.0900
2. 2: ['JavaScript']: 0.0786
3. 8: ['PHP']: 0.0711
4. 4: ['WordPress']: 0.0693
5. 11: ['HTML']: 0.0687
6. 3: ['Web Design']: 0.0639
7. 17: ['CSS']: 0.0560
8. 10: ['API']: 0.0462
9. 21: ['Shopify']: 0.0233
10. 43: ['API Integration']: 0.0206
11. 55: ['MySQL']: 0.0198
12. 24: ['Python']: 0.0184
13. 40: ['Landing Page']: 0.0156
14. 113: ['Custom PHP']: 0.0149
15. 91: ['HTML5']: 0.0135
16. 61: ['React']: 0.0128
17. 74: ['Shopify Templates']: 0.0120
18. 103: ['cPanel']: 0.0117
19. 52: ['Node.js']: 0.0114
20. 96: ['SQL']: 0.0098

Eigenvector Centrality - Top 5:

1. 2: ['JavaScript']: 0.2685
2. 1: ['Web Development']: 0.2604
3. 11: ['HTML']: 0.2184
4. 8: ['PHP']: 0.2158
5. 17: ['CSS']: 0.2098
6. 4: ['WordPress']: 0.2097
7. 3: ['Web Design']: 0.2033
8. 10: ['API']: 0.2010
9. 61: ['React']: 0.1892
10. 52: ['Node.js']: 0.1618
11. 24: ['Python']: 0.1607
12. 43: ['API Integration']: 0.1586
13. 104: ['Web Application']: 0.1557
14. 91: ['HTML5']: 0.1262
15. 129: ['Website']: 0.1248
16. 109: ['Java']: 0.1206
17. 21: ['Shopify']: 0.1164
18. 242: ['RESTful API']: 0.1155
19. 194: ['CSS 3']: 0.1132
20. 55: ['MySQL']: 0.1129

Cluster 2 : Digital Marketing and Content-Driven Growth Ecosystem

Degree Centrality - Top 5:

1. 16: ['English']: 0.2274
2. 5: ['Lead Generation']: 0.1911
3. 13: ['Data Entry']: 0.1529
4. 19: ['Content Writing']: 0.1388
5. 6: ['Social Media Marketing']: 0.1328
6. 26: ['Microsoft Excel']: 0.1247
7. 15: ['Marketing Strategy']: 0.1147

8. 14: ['Search Engine Optimization']: 0.1127
9. 31: ['Administrative Support']: 0.0946
10. 41: ['Sales']: 0.0885
11. 45: ['Communications']: 0.0865
12. 33: ['Customer Service']: 0.0805
13. 22: ['Instagram']: 0.0785
14. 58: ['Email Marketing']: 0.0785
15. 65: ['Google Ads']: 0.0785
16. 28: ['SEO Keyword Research']: 0.0704
17. 12: ['Facebook']: 0.0664
18. 68: ['Translation']: 0.0664
19. 49: ['Market Research']: 0.0644
20. 77: ['German']: 0.0644

Betweenness Centrality - Top 5:

1. 16: ['English']: 0.2756
2. 5: ['Lead Generation']: 0.1217
3. 13: ['Data Entry']: 0.1112
4. 19: ['Content Writing']: 0.1078
5. 26: ['Microsoft Excel']: 0.0949
6. 14: ['Search Engine Optimization']: 0.0754
7. 31: ['Administrative Support']: 0.0646
8. 6: ['Social Media Marketing']: 0.0637
9. 45: ['Communications']: 0.0490
10. 77: ['German']: 0.0448
11. 49: ['Market Research']: 0.0376
12. 15: ['Marketing Strategy']: 0.0366
13. 65: ['Google Ads']: 0.0313
14. 33: ['Customer Service']: 0.0309
15. 32: ['Data Scraping']: 0.0274
16. 41: ['Sales']: 0.0274
17. 520: ['Price & Quote Negotiation']: 0.0272
18. 139: ['Market Analysis']: 0.0272
19. 28: ['SEO Keyword Research']: 0.0244
20. 50: ['Google Analytics']: 0.0240

Closeness Centrality - Top 5:

1. 16: ['English']: 0.4965
2. 5: ['Lead Generation']: 0.4945
3. 13: ['Data Entry']: 0.4892
4. 19: ['Content Writing']: 0.4788
5. 45: ['Communications']: 0.4568
6. 14: ['Search Engine Optimization']: 0.4526
7. 31: ['Administrative Support']: 0.4518
8. 33: ['Customer Service']: 0.4494
9. 41: ['Sales']: 0.4453
10. 6: ['Social Media Marketing']: 0.4449
11. 49: ['Market Research']: 0.4422
12. 65: ['Google Ads']: 0.4337
13. 15: ['Marketing Strategy']: 0.4262
14. 12: ['Facebook']: 0.4198
15. 58: ['Email Marketing']: 0.4180
16. 72: ['Copywriting']: 0.4159
17. 26: ['Microsoft Excel']: 0.4152
18. 226: ['Sales & Marketing']: 0.4152
19. 53: ['Virtual Assistance']: 0.4107

20. 77: ['German']: 0.4067

PageRank - Top 5:

1. 16: ['English']: 0.0222
2. 5: ['Lead Generation']: 0.0170
3. 13: ['Data Entry']: 0.0140
4. 19: ['Content Writing']: 0.0125
5. 26: ['Microsoft Excel']: 0.0125
6. 6: ['Social Media Marketing']: 0.0121
7. 14: ['Search Engine Optimization']: 0.0105
8. 15: ['Marketing Strategy']: 0.0100
9. 31: ['Administrative Support']: 0.0091
10. 45: ['Communications']: 0.0080
11. 41: ['Sales']: 0.0078
12. 68: ['Translation']: 0.0073
13. 33: ['Customer Service']: 0.0072
14. 65: ['Google Ads']: 0.0072
15. 22: ['Instagram']: 0.0071
16. 58: ['Email Marketing']: 0.0068
17. 28: ['SEO Keyword Research']: 0.0065
18. 49: ['Market Research']: 0.0064
19. 77: ['German']: 0.0064
20. 107: ['Accounting']: 0.0063

Authority (HITS) - Top 5:

1. 6: ['Social Media Marketing']: 0.1018
2. 12: ['Facebook']: 0.0910
3. 22: ['Instagram']: 0.0751
4. 15: ['Marketing Strategy']: 0.0635
5. 30: ['Social Media Management']: 0.0567
6. 5: ['Lead Generation']: 0.0369
7. 54: ['Social Media Content Creation']: 0.0315
8. 19: ['Content Writing']: 0.0295
9. 14: ['Search Engine Optimization']: 0.0203
10. 81: ['Influencer Marketing']: 0.0192
11. 13: ['Data Entry']: 0.0175
12. 69: ['LinkedIn']: 0.0169
13. 37: ['List Building']: 0.0166
14. 127: ['Social Media Content']: 0.0151
15. 32: ['Data Scraping']: 0.0131
16. 72: ['Copywriting']: 0.0125
17. 28: ['SEO Keyword Research']: 0.0112
18. 41: ['Sales']: 0.0112
19. 216: ['Pinterest']: 0.0102
20. 31: ['Administrative Support']: 0.0093

Eigenvector Centrality - Top 5:

1. 5: ['Lead Generation']: 0.2824
2. 13: ['Data Entry']: 0.2396
3. 6: ['Social Media Marketing']: 0.2143
4. 16: ['English']: 0.2100
5. 19: ['Content Writing']: 0.2029
6. 15: ['Marketing Strategy']: 0.1964
7. 14: ['Search Engine Optimization']: 0.1754
8. 41: ['Sales']: 0.1664
9. 45: ['Communications']: 0.1578

10. 58: ['Email Marketing']: 0.1557
11. 33: ['Customer Service']: 0.1458
12. 65: ['Google Ads']: 0.1437
13. 31: ['Administrative Support']: 0.1429
14. 12: ['Facebook']: 0.1371
15. 72: ['Copywriting']: 0.1319
16. 30: ['Social Media Management']: 0.1217
17. 22: ['Instagram']: 0.1193
18. 49: ['Market Research']: 0.1181
19. 71: ['Email Communication']: 0.1145
20. 28: ['SEO Keyword Research']: 0.1127

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Cluster 3 Cross Platform Mobile Application Development Ecosystem

Degree Centrality - Top 5:

1. 27: ['Mobile App Development']: 0.7727
2. 36: ['Flutter']: 0.5455
3. 39: ['iOS']: 0.5227
4. 38: ['Android']: 0.5000
5. 63: ['Android App Development']: 0.4773
6. 108: ['iOS Development']: 0.3636
7. 121: ['In-App Purchases']: 0.3409
8. 80: ['React Native']: 0.2955
9. 270: ['Smartphone']: 0.2727
10. 538: ['User Authentication']: 0.2727
11. 174: ['Firebase']: 0.2273
12. 201: ['Tablet']: 0.2273
13. 207: ['Dart']: 0.2273
14. 155: ['Kotlin']: 0.2045
15. 183: ['Swift']: 0.1818
16. 209: ['Google Cloud Platform']: 0.1818
17. 915: ['Mobx']: 0.1591
18. 593: ['Social Media Account Integration']: 0.1591
19. 228: ['Hybrid App']: 0.1591
20. 220: ['Firebase Cloud Firestore']: 0.1364

Betweenness Centrality - Top 5:

1. 27: ['Mobile App Development']: 0.3346
2. 121: ['In-App Purchases']: 0.1772
3. 36: ['Flutter']: 0.1150
4. 38: ['Android']: 0.1065
5. 63: ['Android App Development']: 0.0919
6. 209: ['Google Cloud Platform']: 0.0890
7. 39: ['iOS']: 0.0715
8. 538: ['User Authentication']: 0.0451
9. 108: ['iOS Development']: 0.0195
10. 593: ['Social Media Account Integration']: 0.0129
11. 80: ['React Native']: 0.0106
12. 174: ['Firebase']: 0.0101
13. 207: ['Dart']: 0.0092
14. 270: ['Smartphone']: 0.0085
15. 201: ['Tablet']: 0.0032
16. 155: ['Kotlin']: 0.0018
17. 183: ['Swift']: 0.0011

- 18. 220: ['Firebase Cloud Firestore']: 0.0002
- 19. 1155: ['Native App Development']: 0.0000
- 20. 1158: ['Mobile App Redesign']: 0.0000

Closeness Centrality - Top 5:

- 1. 27: ['Mobile App Development']: 0.8148
- 2. 36: ['Flutter']: 0.6875
- 3. 39: ['iOS']: 0.6567
- 4. 38: ['Android']: 0.6471
- 5. 63: ['Android App Development']: 0.6377
- 6. 121: ['In-App Purchases']: 0.5867
- 7. 108: ['iOS Development']: 0.5714
- 8. 538: ['User Authentication']: 0.5641
- 9. 270: ['Smartphone']: 0.5432
- 10. 80: ['React Native']: 0.5366
- 11. 174: ['Firebase']: 0.5301
- 12. 201: ['Tablet']: 0.5301
- 13. 207: ['Dart']: 0.5301
- 14. 164: ['Offline Functionality']: 0.5176
- 15. 169: ['TV Set']: 0.5176
- 16. 155: ['Kotlin']: 0.5116
- 17. 1183: ['Payment Integration Framework']: 0.5116
- 18. 593: ['Social Media Account Integration']: 0.5116
- 19. 915: ['Mobx']: 0.5000
- 20. 209: ['Google Cloud Platform']: 0.5000

PageRank - Top 5:

- 1. 27: ['Mobile App Development']: 0.0808
- 2. 36: ['Flutter']: 0.0564
- 3. 39: ['iOS']: 0.0533
- 4. 38: ['Android']: 0.0525
- 5. 63: ['Android App Development']: 0.0502
- 6. 121: ['In-App Purchases']: 0.0425
- 7. 108: ['iOS Development']: 0.0372
- 8. 538: ['User Authentication']: 0.0316
- 9. 80: ['React Native']: 0.0308
- 10. 270: ['Smartphone']: 0.0287
- 11. 209: ['Google Cloud Platform']: 0.0254
- 12. 174: ['Firebase']: 0.0252
- 13. 207: ['Dart']: 0.0248
- 14. 201: ['Tablet']: 0.0240
- 15. 155: ['Kotlin']: 0.0216
- 16. 593: ['Social Media Account Integration']: 0.0212
- 17. 183: ['Swift']: 0.0199
- 18. 915: ['Mobx']: 0.0176
- 19. 228: ['Hybrid App']: 0.0175
- 20. 220: ['Firebase Cloud Firestore']: 0.0168

Authority (HITS) - Top 5:

- 1. 27: ['Mobile App Development']: 0.1269
- 2. 36: ['Flutter']: 0.1072
- 3. 39: ['iOS']: 0.0922
- 4. 38: ['Android']: 0.0702
- 5. 63: ['Android App Development']: 0.0667
- 6. 108: ['iOS Development']: 0.0546
- 7. 80: ['React Native']: 0.0463

8. 183: ['Swift']: 0.0380
9. 155: ['Kotlin']: 0.0366
10. 228: ['Hybrid App']: 0.0335
11. 121: ['In-App Purchases']: 0.0330
12. 164: ['Offline Functionality']: 0.0326
13. 169: ['TV Set']: 0.0326
14. 207: ['Dart']: 0.0321
15. 201: ['Tablet']: 0.0312
16. 174: ['Firebase']: 0.0290
17. 209: ['Google Cloud Platform']: 0.0209
18. 220: ['Firebase Cloud Firestore']: 0.0209
19. 202: ['Google Maps']: 0.0205
20. 270: ['Smartphone']: 0.0174

Eigenvector Centrality - Top 5:

1. 27: ['Mobile App Development']: 0.3695
2. 39: ['iOS']: 0.3063
3. 36: ['Flutter']: 0.3053
4. 38: ['Android']: 0.2839
5. 63: ['Android App Development']: 0.2740
6. 108: ['iOS Development']: 0.2417
7. 80: ['React Native']: 0.2078
8. 270: ['Smartphone']: 0.2004
9. 201: ['Tablet']: 0.1848
10. 155: ['Kotlin']: 0.1740
11. 207: ['Dart']: 0.1701
12. 121: ['In-App Purchases']: 0.1643
13. 174: ['Firebase']: 0.1625
14. 538: ['User Authentication']: 0.1520
15. 915: ['Mobx']: 0.1439
16. 183: ['Swift']: 0.1397
17. 228: ['Hybrid App']: 0.1341
18. 412: ['Privacy Policy']: 0.1135
19. 967: ['Objective-C']: 0.1002
20. 917: ['Mobile App']: 0.0992

Cluster 4 Design Oriented Engineering and Architectural Modeling Ecosystem

Degree Centrality - Top 5:

1. 90: ['3D Rendering']: 0.2600
2. 92: ['Architectural Design']: 0.2600
3. 117: ['Autodesk AutoCAD']: 0.2500
4. 57: ['3D Modeling']: 0.2300
5. 73: ['3D Design']: 0.2200
6. 100: ['Mechanical Engineering']: 0.2200
7. 300: ['Drafting']: 0.1600
8. 95: ['SolidWorks']: 0.1400
9. 251: ['Embedded System']: 0.1400
10. 299: ['Electrical Engineering']: 0.1300
11. 449: ['CAD']: 0.1300
12. 429: ['PCB Design']: 0.1200
13. 304: ['Structural Engineering']: 0.1100
14. 305: ['Structural Analysis']: 0.1100
15. 331: ['2D Design']: 0.1100

16. 358: ['Residential Design']: 0.1100
17. 431: ['Floor Plan Design']: 0.1100
18. 584: ['Interior Design']: 0.1000
19. 603: ['Photorealistic Rendering']: 0.0900
20. 138: ['Finite Element Analysis']: 0.0900

Betweenness Centrality - Top 5:

1. 100: ['Mechanical Engineering']: 0.4069
2. 73: ['3D Design']: 0.2272
3. 429: ['PCB Design']: 0.1794
4. 346: ['LEED Consulting']: 0.1655
5. 117: ['Autodesk AutoCAD']: 0.1579
6. 267: ['Arduino']: 0.1545
7. 92: ['Architectural Design']: 0.1191
8. 251: ['Embedded System']: 0.1190
9. 90: ['3D Rendering']: 0.1119
10. 341: ['HVAC System Design']: 0.0960
11. 57: ['3D Modeling']: 0.0942
12. 358: ['Residential Design']: 0.0532
13. 138: ['Finite Element Analysis']: 0.0486
14. 304: ['Structural Engineering']: 0.0460
15. 305: ['Structural Analysis']: 0.0460
16. 479: ['Firmware']: 0.0348
17. 299: ['Electrical Engineering']: 0.0316
18. 300: ['Drafting']: 0.0239
19. 449: ['CAD']: 0.0222
20. 342: ['MATLAB']: 0.0200

Closeness Centrality - Top 5:

1. 100: ['Mechanical Engineering']: 0.4762
2. 73: ['3D Design']: 0.4630
3. 117: ['Autodesk AutoCAD']: 0.4484
4. 90: ['3D Rendering']: 0.4386
5. 57: ['3D Modeling']: 0.4367
6. 449: ['CAD']: 0.4098
7. 92: ['Architectural Design']: 0.4082
8. 95: ['SolidWorks']: 0.4082
9. 411: ['3D Printing']: 0.3922
10. 429: ['PCB Design']: 0.3802
11. 358: ['Residential Design']: 0.3731
12. 138: ['Finite Element Analysis']: 0.3717
13. 267: ['Arduino']: 0.3704
14. 300: ['Drafting']: 0.3663
15. 176: ['Autodesk 3ds Max']: 0.3650
16. 331: ['2D Design']: 0.3623
17. 664: ['Design for Manufacturing']: 0.3610
18. 168: ['Flow Simulation']: 0.3597
19. 171: ['Machine Design']: 0.3597
20. 683: ['AutoCAD Civil 3D']: 0.3559

PageRank - Top 5:

1. 92: ['Architectural Design']: 0.0304
2. 90: ['3D Rendering']: 0.0289
3. 117: ['Autodesk AutoCAD']: 0.0279
4. 100: ['Mechanical Engineering']: 0.0269
5. 57: ['3D Modeling']: 0.0262

6. 73: ['3D Design']: 0.0251
7. 251: ['Embedded System']: 0.0186
8. 300: ['Drafting']: 0.0184
9. 299: ['Electrical Engineering']: 0.0171
10. 95: ['SolidWorks']: 0.0162
11. 429: ['PCB Design']: 0.0159
12. 341: ['HVAC System Design']: 0.0153
13. 449: ['CAD']: 0.0147
14. 304: ['Structural Engineering']: 0.0147
15. 305: ['Structural Analysis']: 0.0147
16. 346: ['LEED Consulting']: 0.0140
17. 358: ['Residential Design']: 0.0138
18. 331: ['2D Design']: 0.0130
19. 431: ['Floor Plan Design']: 0.0126
20. 267: ['Arduino']: 0.0124

Authority (HITS) - Top 5:

1. 57: ['3D Modeling']: 0.1262
2. 95: ['SolidWorks']: 0.1195
3. 100: ['Mechanical Engineering']: 0.1061
4. 138: ['Finite Element Analysis']: 0.0939
5. 171: ['Machine Design']: 0.0935
6. 168: ['Flow Simulation']: 0.0935
7. 73: ['3D Design']: 0.0514
8. 90: ['3D Rendering']: 0.0433
9. 117: ['Autodesk AutoCAD']: 0.0306
10. 92: ['Architectural Design']: 0.0248
11. 411: ['3D Printing']: 0.0213
12. 176: ['Autodesk 3ds Max']: 0.0198
13. 224: ['Corona']: 0.0132
14. 223: ['High-Quality Rendering']: 0.0132
15. 449: ['CAD']: 0.0109
16. 358: ['Residential Design']: 0.0072
17. 300: ['Drafting']: 0.0068
18. 664: ['Design for Manufacturing']: 0.0061
19. 267: ['Arduino']: 0.0054
20. 318: ['Laser Cutting & Engraving Software']: 0.0054

Eigenvector Centrality - Top 5:

1. 90: ['3D Rendering']: 0.3301
2. 117: ['Autodesk AutoCAD']: 0.3177
3. 92: ['Architectural Design']: 0.2934
4. 73: ['3D Design']: 0.2802
5. 57: ['3D Modeling']: 0.2747
6. 300: ['Drafting']: 0.2098
7. 449: ['CAD']: 0.2085
8. 100: ['Mechanical Engineering']: 0.1957
9. 95: ['SolidWorks']: 0.1875
10. 431: ['Floor Plan Design']: 0.1712
11. 584: ['Interior Design']: 0.1679
12. 331: ['2D Design']: 0.1566
13. 358: ['Residential Design']: 0.1539
14. 674: ['SketchUp']: 0.1499
15. 683: ['AutoCAD Civil 3D']: 0.1376
16. 411: ['3D Printing']: 0.1308
17. 176: ['Autodesk 3ds Max']: 0.1304

18. 603: ['Photorealistic Rendering']: 0.1297
19. 522: ['Exterior Rendering']: 0.1238
20. 495: ['Interior Design Consultation']: 0.1236

Generative AI

America : GenAI 208 / Total 14289----- Frekans 0.0146
 Asia : GenAI 235 / Total 15371----- Frekans 0.0153
 Africa : GenAI 243 / Total 19900 -----Frekans 0.0122
 Europe : GenAI 240/ Total 19687 ----- Frekans 0.0122

Europe 2025 &GenAI

Degree Centrality - Top 5:

1. 89: ['GenAI 2025']: 1.0952
2. 24: ['Python']: 0.5238
3. 0: ['Graphic Design']: 0.4762
4. 62: ['React']: 0.4286
5. 8: ['Adobe Illustrator']: 0.3810
6. 90: ['PostgreSQL']: 0.3810
7. 512: ['Tailwind CSS']: 0.3333
8. 7: ['JavaScript']: 0.3333
9. 521: ['Websockets']: 0.3333
10. 209: ['Redis']: 0.3333
11. 474: ['RESTful API']: 0.3333
12. 6: ['Adobe Photoshop']: 0.2857
13. 30: ['Figma']: 0.2857
14. 427: ['YouTube']: 0.2381
15. 587: ['YouTube Thumbnail']: 0.2381
16. 213: ['Artificial Intelligence']: 0.2381
17. 406: ['CRM Automation']: 0.1429
18. 703: ['Stable Diffusion']: 0.1429
19. 110: ['n8n']: 0.1429
20. 640: ['Data Transformation']: 0.0952
21. 527: ['Character Animation']: 0.0952
22. 622: ['Omnichannel Automation']: 0.0952

Betweenness Centrality - Top 5:

1. 89: ['GenAI 2025']: 0.5994
2. 24: ['Python']: 0.0476
3. 0: ['Graphic Design']: 0.0387
4. 7: ['JavaScript']: 0.0172
5. 62: ['React']: 0.0168
6. 8: ['Adobe Illustrator']: 0.0144
7. 30: ['Figma']: 0.0079
8. 90: ['PostgreSQL']: 0.0048
9. 213: ['Artificial Intelligence']: 0.0032
10. 6: ['Adobe Photoshop']: 0.0024
11. 406: ['CRM Automation']: 0.0024
12. 110: ['n8n']: 0.0024
13. 512: ['Tailwind CSS']: 0.0000
14. 640: ['Data Transformation']: 0.0000

15. 521: ['Websockets']: 0.0000
16. 527: ['Character Animation']: 0.0000
17. 427: ['YouTube']: 0.0000
18. 703: ['Stable Diffusion']: 0.0000
19. 587: ['YouTube Thumbnail']: 0.0000
20. 209: ['Redis']: 0.0000
21. 474: ['RESTful API']: 0.0000
22. 622: ['Omnichannel Automation']: 0.0000

Closeness Centrality - Top 5:

1. 89: ['GenAI 2025']: 1.0000
2. 24: ['Python']: 0.6774
3. 0: ['Graphic Design']: 0.6562
4. 62: ['React']: 0.6364
5. 8: ['Adobe Illustrator']: 0.6176
6. 90: ['PostgreSQL']: 0.6176
7. 512: ['Tailwind CSS']: 0.6000
8. 7: ['JavaScript']: 0.6000
9. 521: ['Websockets']: 0.6000
10. 209: ['Redis']: 0.6000
11. 474: ['RESTful API']: 0.6000
12. 6: ['Adobe Photoshop']: 0.5833
13. 30: ['Figma']: 0.5833
14. 427: ['YouTube']: 0.5676
15. 587: ['YouTube Thumbnail']: 0.5676
16. 213: ['Artificial Intelligence']: 0.5676
17. 406: ['CRM Automation']: 0.5385
18. 703: ['Stable Diffusion']: 0.5385
19. 110: ['n8n']: 0.5385
20. 640: ['Data Transformation']: 0.5250
21. 527: ['Character Animation']: 0.5250
22. 622: ['Omnichannel Automation']: 0.5250

PageRank - Top 5:

1. 89: ['GenAI 2025']: 0.1463
2. 24: ['Python']: 0.0680
3. 0: ['Graphic Design']: 0.0659
4. 62: ['React']: 0.0563
5. 8: ['Adobe Illustrator']: 0.0532
6. 90: ['PostgreSQL']: 0.0505
7. 7: ['JavaScript']: 0.0475
8. 209: ['Redis']: 0.0447
9. 474: ['RESTful API']: 0.0447
10. 512: ['Tailwind CSS']: 0.0447
11. 521: ['Websockets']: 0.0447
12. 6: ['Adobe Photoshop']: 0.0416
13. 30: ['Figma']: 0.0407
14. 213: ['Artificial Intelligence']: 0.0358
15. 427: ['YouTube']: 0.0357
16. 587: ['YouTube Thumbnail']: 0.0357
17. 110: ['n8n']: 0.0298
18. 406: ['CRM Automation']: 0.0298
19. 703: ['Stable Diffusion']: 0.0242
20. 622: ['Omnichannel Automation']: 0.0209
21. 640: ['Data Transformation']: 0.0209
22. 527: ['Character Animation']: 0.0182

Authority (HITS) - Top 5:

1. 0: ['Graphic Design']: 0.3113
2. 6: ['Adobe Photoshop']: 0.2949
3. 8: ['Adobe Illustrator']: 0.2727
4. 30: ['Figma']: 0.0546
5. 89: ['GenAI 2025']: 0.0159
6. 7: ['JavaScript']: 0.0109
7. 24: ['Python']: 0.0106
8. 587: ['YouTube Thumbnail']: 0.0100
9. 427: ['YouTube']: 0.0100
10. 213: ['Artificial Intelligence']: 0.0036
11. 62: ['React']: 0.0018
12. 90: ['PostgreSQL']: 0.0008
13. 474: ['RESTful API']: 0.0005
14. 209: ['Redis']: 0.0005
15. 521: ['Websockets']: 0.0004
16. 512: ['Tailwind CSS']: 0.0004
17. 527: ['Character Animation']: 0.0004
18. 703: ['Stable Diffusion']: 0.0002
19. 406: ['CRM Automation']: 0.0001
20. 622: ['Omnichannel Automation']: 0.0001
21. 110: ['n8n']: 0.0001
22. 640: ['Data Transformation']: 0.0001

Eigenvector Centrality - Top 5:

1. 89: ['GenAI 2025']: 0.4801
2. 24: ['Python']: 0.3191
3. 62: ['React']: 0.2750
4. 90: ['PostgreSQL']: 0.2561
5. 0: ['Graphic Design']: 0.2428
6. 512: ['Tailwind CSS']: 0.2343
7. 521: ['Websockets']: 0.2343
8. 209: ['Redis']: 0.2343
9. 474: ['RESTful API']: 0.2343
10. 7: ['JavaScript']: 0.2114
11. 8: ['Adobe Illustrator']: 0.2105
12. 30: ['Figma']: 0.1822
13. 6: ['Adobe Photoshop']: 0.1614
14. 213: ['Artificial Intelligence']: 0.1560
15. 427: ['YouTube']: 0.1426
16. 587: ['YouTube Thumbnail']: 0.1426
17. 703: ['Stable Diffusion']: 0.1013
18. 527: ['Character Animation']: 0.0797
19. 406: ['CRM Automation']: 0.0708
20. 110: ['n8n']: 0.0708
21. 640: ['Data Transformation']: 0.0635
22. 622: ['Omnichannel Automation']: 0.0635

America 2025 & GenAI

Degree Centrality - Top 5:

1. 60: ['GenAI 2025']: 1.1667
2. 3: ['JavaScript']: 0.5833
3. 35: ['React']: 0.5833

4. 1: ['Web Development']: 0.5000
5. 172: ['PostgreSQL']: 0.5000
6. 48: ['Artificial Intelligence']: 0.5000
7. 87: ['React Native']: 0.5000
8. 94: ['Next.js']: 0.5000
9. 139: ['AI Mobile App Development']: 0.3333
10. 140: ['Chatbot']: 0.3333
11. 141: ['AI App Development']: 0.3333
12. 219: ['QA Testing']: 0.1667
13. 30: ['English']: 0.1667

Betweenness Centrality - Top 5:

1. 60: ['GenAI 2025']: 0.5505
2. 87: ['React Native']: 0.0455
3. 3: ['JavaScript']: 0.0202
4. 35: ['React']: 0.0202
5. 1: ['Web Development']: 0.0000
6. 139: ['AI Mobile App Development']: 0.0000
7. 140: ['Chatbot']: 0.0000
8. 141: ['AI App Development']: 0.0000
9. 172: ['PostgreSQL']: 0.0000
10. 48: ['Artificial Intelligence']: 0.0000
11. 94: ['Next.js']: 0.0000
12. 219: ['QA Testing']: 0.0000
13. 30: ['English']: 0.0000

Closeness Centrality - Top 5:

1. 60: ['GenAI 2025']: 1.0000
2. 3: ['JavaScript']: 0.7059
3. 35: ['React']: 0.7059
4. 1: ['Web Development']: 0.6667
5. 172: ['PostgreSQL']: 0.6667
6. 48: ['Artificial Intelligence']: 0.6667
7. 87: ['React Native']: 0.6667
8. 94: ['Next.js']: 0.6667
9. 139: ['AI Mobile App Development']: 0.6000
10. 140: ['Chatbot']: 0.6000
11. 141: ['AI App Development']: 0.6000
12. 219: ['QA Testing']: 0.5455
13. 30: ['English']: 0.5455

PageRank - Top 5:

1. 60: ['GenAI 2025']: 0.1717
2. 3: ['JavaScript']: 0.0890
3. 35: ['React']: 0.0890
4. 87: ['React Native']: 0.0826
5. 1: ['Web Development']: 0.0772
6. 172: ['PostgreSQL']: 0.0772
7. 48: ['Artificial Intelligence']: 0.0772
8. 94: ['Next.js']: 0.0772
9. 139: ['AI Mobile App Development']: 0.0599
10. 140: ['Chatbot']: 0.0599
11. 141: ['AI App Development']: 0.0599
12. 219: ['QA Testing']: 0.0396
13. 30: ['English']: 0.0396

Authority (HITS) - Top 5:

1. 3: ['JavaScript']: 0.2484
2. 1: ['Web Development']: 0.2361
3. 35: ['React']: 0.1902
4. 94: ['Next.js']: 0.1114
5. 60: ['GenAI 2025']: 0.0518
6. 48: ['Artificial Intelligence']: 0.0489
7. 172: ['PostgreSQL']: 0.0435
8. 87: ['React Native']: 0.0267
9. 140: ['Chatbot']: 0.0115
10. 139: ['AI Mobile App Development']: 0.0115
11. 141: ['AI App Development']: 0.0115
12. 219: ['QA Testing']: 0.0041
13. 30: ['English']: 0.0041

Eigenvector Centrality - Top 5:

1. 60: ['GenAI 2025']: 0.4857
2. 3: ['JavaScript']: 0.3423
3. 35: ['React']: 0.3423
4. 1: ['Web Development']: 0.3113
5. 172: ['PostgreSQL']: 0.3113
6. 48: ['Artificial Intelligence']: 0.3113
7. 94: ['Next.js']: 0.3113
8. 87: ['React Native']: 0.2409
9. 139: ['AI Mobile App Development']: 0.1526
10. 140: ['Chatbot']: 0.1526
11. 141: ['AI App Development']: 0.1526
12. 219: ['QA Testing']: 0.0843
13. 30: ['English']: 0.0843

Asia 2025 & GenAI

Degree Centrality - Top 5:

1. 31: ['GenAI 2025']: 1.1053
2. 40: ['Python']: 0.5263
3. 34: ['Artificial Intelligence']: 0.4211
4. 9: ['Graphic Design']: 0.2632
5. 13: ['Facebook']: 0.2632
6. 28: ['API']: 0.2632
7. 29: ['Machine Learning']: 0.2632
8. 30: ['Adobe Illustrator']: 0.2632
9. 32: ['Deep Learning']: 0.2632
10. 35: ['Instagram']: 0.2632
11. 347: ['Twilio API']: 0.2632
12. 351: ['Chatbot Development']: 0.2632
13. 110: ['Brand Identity & Guidelines']: 0.2632
14. 339: ['FastAPI']: 0.2105
15. 87: ['AI Agent Development']: 0.2105
16. 90: ['MongoDB']: 0.2105
17. 242: ['Stable Diffusion']: 0.2105
18. 1: ['Video Editing']: 0.1579
19. 276: ['AI-Generated Art']: 0.1579
20. 278: ['Runway']: 0.1579

Betweenness Centrality - Top 5:

1. 31: ['GenAI 2025']: 0.6944
2. 40: ['Python']: 0.0643
3. 34: ['Artificial Intelligence']: 0.0219
4. 29: ['Machine Learning']: 0.0015
5. 32: ['Deep Learning']: 0.0015
6. 1: ['Video Editing']: 0.0000
7. 9: ['Graphic Design']: 0.0000
8. 13: ['Facebook']: 0.0000
9. 276: ['AI-Generated Art']: 0.0000
10. 278: ['Runway']: 0.0000
11. 28: ['API']: 0.0000
12. 30: ['Adobe Illustrator']: 0.0000
13. 35: ['Instagram']: 0.0000
14. 339: ['FastAPI']: 0.0000
15. 87: ['AI Agent Development']: 0.0000
16. 90: ['MongoDB']: 0.0000
17. 347: ['Twilio API']: 0.0000
18. 351: ['Chatbot Development']: 0.0000
19. 110: ['Brand Identity & Guidelines']: 0.0000
20. 242: ['Stable Diffusion']: 0.0000

Closeness Centrality - Top 5:

1. 31: ['GenAI 2025']: 1.0000
2. 40: ['Python']: 0.6786
3. 34: ['Artificial Intelligence']: 0.6333
4. 9: ['Graphic Design']: 0.5758
5. 13: ['Facebook']: 0.5758
6. 28: ['API']: 0.5758
7. 29: ['Machine Learning']: 0.5758
8. 30: ['Adobe Illustrator']: 0.5758
9. 32: ['Deep Learning']: 0.5758
10. 35: ['Instagram']: 0.5758
11. 347: ['Twilio API']: 0.5758
12. 351: ['Chatbot Development']: 0.5758
13. 110: ['Brand Identity & Guidelines']: 0.5758
14. 339: ['FastAPI']: 0.5588
15. 87: ['AI Agent Development']: 0.5588
16. 90: ['MongoDB']: 0.5588
17. 242: ['Stable Diffusion']: 0.5588
18. 1: ['Video Editing']: 0.5429
19. 276: ['AI-Generated Art']: 0.5429
20. 278: ['Runway']: 0.5429

PageRank - Top 5:

1. 31: ['GenAI 2025']: 0.1632
2. 40: ['Python']: 0.0818
3. 34: ['Artificial Intelligence']: 0.0658
4. 9: ['Graphic Design']: 0.0451
5. 13: ['Facebook']: 0.0451
6. 30: ['Adobe Illustrator']: 0.0451
7. 35: ['Instagram']: 0.0451
8. 110: ['Brand Identity & Guidelines']: 0.0451
9. 29: ['Machine Learning']: 0.0435
10. 32: ['Deep Learning']: 0.0435
11. 28: ['API']: 0.0430
12. 347: ['Twilio API']: 0.0430

13. 351: ['Chatbot Development']: 0.0430
14. 339: ['FastAPI']: 0.0372
15. 87: ['AI Agent Development']: 0.0372
16. 90: ['MongoDB']: 0.0372
17. 242: ['Stable Diffusion']: 0.0362
18. 1: ['Video Editing']: 0.0333
19. 276: ['AI-Generated Art']: 0.0333
20. 278: ['Runway']: 0.0333

Authority (HITS) - Top 5:

1. 29: ['Machine Learning']: 0.1735
2. 32: ['Deep Learning']: 0.1696
3. 34: ['Artificial Intelligence']: 0.1658
4. 31: ['GenAI 2025']: 0.1096
5. 40: ['Python']: 0.0799
6. 242: ['Stable Diffusion']: 0.0591
7. 1: ['Video Editing']: 0.0206
8. 278: ['Runway']: 0.0206
9. 276: ['AI-Generated Art']: 0.0206
10. 28: ['API']: 0.0190
11. 9: ['Graphic Design']: 0.0190
12. 30: ['Adobe Illustrator']: 0.0188
13. 351: ['Chatbot Development']: 0.0171
14. 347: ['Twilio API']: 0.0171
15. 87: ['AI Agent Development']: 0.0169
16. 90: ['MongoDB']: 0.0169
17. 339: ['FastAPI']: 0.0169
18. 35: ['Instagram']: 0.0137
19. 13: ['Facebook']: 0.0137
20. 110: ['Brand Identity & Guidelines']: 0.0114

Eigenvector Centrality - Top 5:

1. 31: ['GenAI 2025']: 0.5490
2. 40: ['Python']: 0.3275
3. 34: ['Artificial Intelligence']: 0.2891
4. 347: ['Twilio API']: 0.2180
5. 351: ['Chatbot Development']: 0.2180
6. 28: ['API']: 0.2180
7. 29: ['Machine Learning']: 0.2107
8. 32: ['Deep Learning']: 0.2107
9. 242: ['Stable Diffusion']: 0.1715
10. 9: ['Graphic Design']: 0.1641
11. 13: ['Facebook']: 0.1641
12. 30: ['Adobe Illustrator']: 0.1641
13. 35: ['Instagram']: 0.1641
14. 110: ['Brand Identity & Guidelines']: 0.1641
15. 339: ['FastAPI']: 0.1640
16. 87: ['AI Agent Development']: 0.1640
17. 90: ['MongoDB']: 0.1640
18. 1: ['Video Editing']: 0.1027
19. 276: ['AI-Generated Art']: 0.1027
20. 278: ['Runway']: 0.1027

Africa 2025 & GenAI

Degree Centrality - Top 5:

1. 147: ['AI Speech-to-Text']: 1.0000

2. 148: ['Colossyan']: 1.0000
3. 149: ['Deepbrain AI']: 1.0000
4. 150: ['Descript']: 1.0000
5. 151: ['Elai']: 1.0000
6. 152: ['InVideo']: 1.0000
7. 153: ['Kaiber']: 1.0000
8. 154: ['Lumen5']: 1.0000
9. 155: ['GenAI 2025']: 1.0000

Betweenness Centrality - Top 5:

1. 147: ['AI Speech-to-Text']: 0.0000
2. 148: ['Colossyan']: 0.0000
3. 149: ['Deepbrain AI']: 0.0000
4. 150: ['Descript']: 0.0000
5. 151: ['Elai']: 0.0000
6. 152: ['InVideo']: 0.0000
7. 153: ['Kaiber']: 0.0000
8. 154: ['Lumen5']: 0.0000
9. 155: ['GenAI 2025']: 0.0000

Closeness Centrality - Top 5:

1. 147: ['AI Speech-to-Text']: 1.0000
2. 148: ['Colossyan']: 1.0000
3. 149: ['Deepbrain AI']: 1.0000
4. 150: ['Descript']: 1.0000
5. 151: ['Elai']: 1.0000
6. 152: ['InVideo']: 1.0000
7. 153: ['Kaiber']: 1.0000
8. 154: ['Lumen5']: 1.0000
9. 155: ['GenAI 2025']: 1.0000

PageRank - Top 5:

1. 147: ['AI Speech-to-Text']: 0.1111
2. 148: ['Colossyan']: 0.1111
3. 149: ['Deepbrain AI']: 0.1111
4. 150: ['Descript']: 0.1111
5. 151: ['Elai']: 0.1111
6. 152: ['InVideo']: 0.1111
7. 153: ['Kaiber']: 0.1111
8. 154: ['Lumen5']: 0.1111
9. 155: ['GenAI 2025']: 0.1111

Authority (HITS) - Top 5:

1. 151: ['Elai']: 0.1111
2. 149: ['Deepbrain AI']: 0.1111
3. 154: ['Lumen5']: 0.1111
4. 155: ['GenAI 2025']: 0.1111
5. 148: ['Colossyan']: 0.1111
6. 150: ['Descript']: 0.1111
7. 153: ['Kaiber']: 0.1111
8. 147: ['AI Speech-to-Text']: 0.1111
9. 152: ['InVideo']: 0.1111

Eigenvector Centrality - Top 5:

1. 147: ['AI Speech-to-Text']: 0.3333
2. 148: ['Colossyan']: 0.3333

3. 149: ['Deepbrain AI']: 0.3333
4. 150: ['Descript']: 0.3333
5. 151: ['Elai']: 0.3333
6. 152: ['InVideo']: 0.3333
7. 153: ['Kaiber']: 0.3333
8. 154: ['Lumen5']: 0.3333
9. 155: ['GenAI 2025']: 0.3333

Global 2025 & GenAI

Degree Centrality - Top 5:

1. 49: ['GenAI 2025']: 1.0392
2. 2: ['JavaScript']: 0.4118
3. 24: ['Python']: 0.4118
4. 1: ['Web Development']: 0.3725
5. 0: ['Graphic Design']: 0.3529
6. 18: ['Adobe Illustrator']: 0.3137
7. 62: ['React']: 0.2941
8. 77: ['Artificial Intelligence']: 0.2941
9. 10: ['API']: 0.2745
10. 9: ['Adobe Photoshop']: 0.2549
11. 152: ['PostgreSQL']: 0.2549
12. 7: ['Video Editing']: 0.1961
13. 16: ['English']: 0.1961
14. 63: ['Figma']: 0.1961
15. 87: ['Machine Learning']: 0.1961
16. 399: ['Tailwind CSS']: 0.1961
17. 242: ['RESTful API']: 0.1961
18. 12: ['Facebook']: 0.1765
19. 146: ['Next.js']: 0.1765
20. 197: ['YouTube']: 0.1765
21. 252: ['MongoDB']: 0.1765
22. 271: ['AI Speech-to-Text']: 0.1569
23. 272: ['Colossyan']: 0.1569
24. 273: ['Deepbrain AI']: 0.1569
25. 275: ['Descript']: 0.1569
26. 276: ['Elai']: 0.1569
27. 277: ['InVideo']: 0.1569
28. 278: ['Kaiber']: 0.1569
29. 279: ['Lumen5']: 0.1569
30. 546: ['Redis']: 0.1569
31. 182: ['AI Agent Development']: 0.1569
32. 81: ['React Native']: 0.1569
33. 22: ['Instagram']: 0.1373
34. 48: ['Brand Identity & Guidelines']: 0.1373
35. 978: ['Websockets']: 0.1373
36. 943: ['YouTube Thumbnail']: 0.1176
37. 360: ['AI-Generated Art']: 0.1176
38. 621: ['CRM Automation']: 0.1176
39. 112: ['n8n']: 0.1176
40. 244: ['AI App Development']: 0.1176
41. 121: ['Chatbot Development']: 0.1176
42. 147: ['Deep Learning']: 0.0980
43. 539: ['Stable Diffusion']: 0.0980
44. 860: ['Twilio API']: 0.0980

45. 803: ['FastAPI']: 0.0784
46. 575: ['Runway']: 0.0784
47. 451: ['AI Mobile App Development']: 0.0784
48. 452: ['Chatbot']: 0.0784
49. 1106: ['Omnichannel Automation']: 0.0392
50. 596: ['QA Testing']: 0.0392
51. 1123: ['Data Transformation']: 0.0392
52. 868: ['Character Animation']: 0.0392

Betweenness Centrality - Top 5:

1. 49: ['GenAI 2025']: 0.6577
2. 24: ['Python']: 0.0315
3. 2: ['JavaScript']: 0.0304
4. 1: ['Web Development']: 0.0238
5. 0: ['Graphic Design']: 0.0179
6. 18: ['Adobe Illustrator']: 0.0112
7. 77: ['Artificial Intelligence']: 0.0100
8. 10: ['API']: 0.0090
9. 62: ['React']: 0.0072
10. 16: ['English']: 0.0056
11. 9: ['Adobe Photoshop']: 0.0049
12. 81: ['React Native']: 0.0042
13. 182: ['AI Agent Development']: 0.0040
14. 7: ['Video Editing']: 0.0034
15. 63: ['Figma']: 0.0034
16. 152: ['PostgreSQL']: 0.0032
17. 252: ['MongoDB']: 0.0025
18. 87: ['Machine Learning']: 0.0022
19. 621: ['CRM Automation']: 0.0021
20. 244: ['AI App Development']: 0.0020
21. 112: ['n8n']: 0.0019
22. 399: ['Tailwind CSS']: 0.0013
23. 12: ['Facebook']: 0.0011
24. 242: ['RESTful API']: 0.0010
25. 197: ['YouTube']: 0.0009
26. 146: ['Next.js']: 0.0008
27. 360: ['AI-Generated Art']: 0.0007
28. 48: ['Brand Identity & Guidelines']: 0.0004
29. 22: ['Instagram']: 0.0004
30. 546: ['Redis']: 0.0002
31. 147: ['Deep Learning']: 0.0002
32. 539: ['Stable Diffusion']: 0.0002
33. 121: ['Chatbot Development']: 0.0002
34. 575: ['Runway']: 0.0001
35. 943: ['YouTube Thumbnail']: 0.0001
36. 271: ['AI Speech-to-Text']: 0.0000
37. 272: ['Colossyan']: 0.0000
38. 273: ['Deepbrain AI']: 0.0000
39. 275: ['Descript']: 0.0000
40. 276: ['Elai']: 0.0000
41. 277: ['InVideo']: 0.0000
42. 278: ['Kaiber']: 0.0000
43. 279: ['Lumen5']: 0.0000
44. 803: ['FastAPI']: 0.0000
45. 451: ['AI Mobile App Development']: 0.0000
46. 452: ['Chatbot']: 0.0000

47. 1106: ['Omnichannel Automation']: 0.0000
48. 978: ['Websockets']: 0.0000
49. 596: ['QA Testing']: 0.0000
50. 860: ['Twilio API']: 0.0000
51. 1123: ['Data Transformation']: 0.0000
52. 868: ['Character Animation']: 0.0000

Closeness Centrality - Top 5:

1. 49: ['GenAI 2025']: 1.0000
2. 2: ['JavaScript']: 0.6296
3. 24: ['Python']: 0.6296
4. 1: ['Web Development']: 0.6145
5. 0: ['Graphic Design']: 0.6071
6. 18: ['Adobe Illustrator']: 0.5930
7. 62: ['React']: 0.5862
8. 77: ['Artificial Intelligence']: 0.5862
9. 10: ['API']: 0.5795
10. 9: ['Adobe Photoshop']: 0.5730
11. 152: ['PostgreSQL']: 0.5730
12. 7: ['Video Editing']: 0.5543
13. 16: ['English']: 0.5543
14. 63: ['Figma']: 0.5543
15. 87: ['Machine Learning']: 0.5543
16. 399: ['Tailwind CSS']: 0.5543
17. 242: ['RESTful API']: 0.5543
18. 12: ['Facebook']: 0.5484
19. 146: ['Next.js']: 0.5484
20. 197: ['YouTube']: 0.5484
21. 252: ['MongoDB']: 0.5484
22. 271: ['AI Speech-to-Text']: 0.5426
23. 272: ['Colossyan']: 0.5426
24. 273: ['Deepbrain AI']: 0.5426
25. 275: ['Descript']: 0.5426
26. 276: ['Elai']: 0.5426
27. 277: ['InVideo']: 0.5426
28. 278: ['Kaiber']: 0.5426
29. 279: ['Lumen5']: 0.5426
30. 546: ['Redis']: 0.5426
31. 182: ['AI Agent Development']: 0.5426
32. 81: ['React Native']: 0.5426
33. 22: ['Instagram']: 0.5368
34. 48: ['Brand Identity & Guidelines']: 0.5368
35. 978: ['Websockets']: 0.5368
36. 943: ['YouTube Thumbnail']: 0.5312
37. 360: ['AI-Generated Art']: 0.5312
38. 621: ['CRM Automation']: 0.5312
39. 112: ['n8n']: 0.5312
40. 244: ['AI App Development']: 0.5312
41. 121: ['Chatbot Development']: 0.5312
42. 147: ['Deep Learning']: 0.5258
43. 539: ['Stable Diffusion']: 0.5258
44. 860: ['Twilio API']: 0.5258
45. 803: ['FastAPI']: 0.5204
46. 575: ['Runway']: 0.5204
47. 451: ['AI Mobile App Development']: 0.5204
48. 452: ['Chatbot']: 0.5204

49. 1106: ['Omnichannel Automation']: 0.5100
50. 596: ['QA Testing']: 0.5100
51. 1123: ['Data Transformation']: 0.5100
52. 868: ['Character Animation']: 0.5100

PageRank - Top 5:

1. 49: ['GenAI 2025']: 0.0985
2. 2: ['JavaScript']: 0.0382
3. 24: ['Python']: 0.0380
4. 1: ['Web Development']: 0.0342
5. 0: ['Graphic Design']: 0.0329
6. 18: ['Adobe Illustrator']: 0.0292
7. 77: ['Artificial Intelligence']: 0.0276
8. 62: ['React']: 0.0271
9. 10: ['API']: 0.0263
10. 9: ['Adobe Photoshop']: 0.0244
11. 152: ['PostgreSQL']: 0.0237
12. 16: ['English']: 0.0204
13. 7: ['Video Editing']: 0.0202
14. 87: ['Machine Learning']: 0.0192
15. 63: ['Figma']: 0.0190
16. 242: ['RESTful API']: 0.0189
17. 399: ['Tailwind CSS']: 0.0188
18. 197: ['YouTube']: 0.0178
19. 252: ['MongoDB']: 0.0178
20. 12: ['Facebook']: 0.0176
21. 271: ['AI Speech-to-Text']: 0.0175
22. 272: ['Colossyan']: 0.0175
23. 273: ['Deepbrain AI']: 0.0175
24. 275: ['Descript']: 0.0175
25. 276: ['Elai']: 0.0175
26. 277: ['InVideo']: 0.0175
27. 278: ['Kaiber']: 0.0175
28. 279: ['Lumen5']: 0.0175
29. 182: ['AI Agent Development']: 0.0173
30. 81: ['React Native']: 0.0172
31. 146: ['Next.js']: 0.0170
32. 546: ['Redis']: 0.0157
33. 22: ['Instagram']: 0.0144
34. 621: ['CRM Automation']: 0.0143
35. 112: ['n8n']: 0.0143
36. 48: ['Brand Identity & Guidelines']: 0.0142
37. 244: ['AI App Development']: 0.0142
38. 978: ['Websockets']: 0.0140
39. 360: ['AI-Generated Art']: 0.0130
40. 121: ['Chatbot Development']: 0.0126
41. 943: ['YouTube Thumbnail']: 0.0126
42. 539: ['Stable Diffusion']: 0.0111
43. 147: ['Deep Learning']: 0.0111
44. 860: ['Twilio API']: 0.0110
45. 451: ['AI Mobile App Development']: 0.0106
46. 452: ['Chatbot']: 0.0106
47. 575: ['Runway']: 0.0098
48. 803: ['FastAPI']: 0.0096
49. 1106: ['Omnichannel Automation']: 0.0065
50. 1123: ['Data Transformation']: 0.0065

51. 596: ['QA Testing']: 0.0062
52. 868: ['Character Animation']: 0.0060

Authority (HITS) - Top 5:

1. 0: ['Graphic Design']: 0.1673
2. 9: ['Adobe Photoshop']: 0.1585
3. 18: ['Adobe Illustrator']: 0.1419
4. 1: ['Web Development']: 0.0937
5. 2: ['JavaScript']: 0.0930
6. 10: ['API']: 0.0632
7. 48: ['Brand Identity & Guidelines']: 0.0369
8. 24: ['Python']: 0.0361
9. 63: ['Figma']: 0.0269
10. 12: ['Facebook']: 0.0266
11. 62: ['React']: 0.0195
12. 121: ['Chatbot Development']: 0.0140
13. 22: ['Instagram']: 0.0139
14. 7: ['Video Editing']: 0.0131
15. 49: ['GenAI 2025']: 0.0119
16. 197: ['YouTube']: 0.0113
17. 146: ['Next.js']: 0.0086
18. 16: ['English']: 0.0068
19. 360: ['AI-Generated Art']: 0.0067
20. 77: ['Artificial Intelligence']: 0.0063
21. 152: ['PostgreSQL']: 0.0052
22. 112: ['n8n']: 0.0047
23. 242: ['RESTful API']: 0.0047
24. 87: ['Machine Learning']: 0.0042
25. 399: ['Tailwind CSS']: 0.0041
26. 252: ['MongoDB']: 0.0036
27. 81: ['React Native']: 0.0023
28. 943: ['YouTube Thumbnail']: 0.0017
29. 147: ['Deep Learning']: 0.0016
30. 182: ['AI Agent Development']: 0.0011
31. 860: ['Twilio API']: 0.0009
32. 244: ['AI App Development']: 0.0008
33. 277: ['InVideo']: 0.0006
34. 272: ['Colossyan']: 0.0006
35. 279: ['Lumen5']: 0.0006
36. 273: ['Deepbrain AI']: 0.0006
37. 278: ['Kaiber']: 0.0006
38. 271: ['AI Speech-to-Text']: 0.0006
39. 275: ['Descript']: 0.0006
40. 276: ['Elai']: 0.0006
41. 575: ['Runway']: 0.0006
42. 803: ['FastAPI']: 0.0005
43. 868: ['Character Animation']: 0.0005
44. 546: ['Redis']: 0.0005
45. 621: ['CRM Automation']: 0.0004
46. 539: ['Stable Diffusion']: 0.0004
47. 978: ['Websockets']: 0.0004
48. 451: ['AI Mobile App Development']: 0.0003
49. 452: ['Chatbot']: 0.0003
50. 596: ['QA Testing']: 0.0002
51. 1123: ['Data Transformation']: 0.0000
52. 1106: ['Omnichannel Automation']: 0.0000

Eigenvector Centrality - Top 5:

1. 49: ['GenAI 2025']: 0.4395
2. 2: ['JavaScript']: 0.2593
3. 24: ['Python']: 0.2482
4. 1: ['Web Development']: 0.2431
5. 0: ['Graphic Design']: 0.2133
6. 77: ['Artificial Intelligence']: 0.2023
7. 62: ['React']: 0.2010
8. 18: ['Adobe Illustrator']: 0.2008
9. 152: ['PostgreSQL']: 0.1840
10. 10: ['API']: 0.1840
11. 9: ['Adobe Photoshop']: 0.1613
12. 87: ['Machine Learning']: 0.1543
13. 399: ['Tailwind CSS']: 0.1497
14. 63: ['Figma']: 0.1484
15. 242: ['RESTful API']: 0.1445
16. 146: ['Next.js']: 0.1444
17. 16: ['English']: 0.1316
18. 252: ['MongoDB']: 0.1259
19. 12: ['Facebook']: 0.1198
20. 546: ['Redis']: 0.1148
21. 197: ['YouTube']: 0.1117
22. 7: ['Video Editing']: 0.1075
23. 978: ['Websockets']: 0.1064
24. 81: ['React Native']: 0.1050
25. 121: ['Chatbot Development']: 0.1018
26. 48: ['Brand Identity & Guidelines']: 0.0988
27. 182: ['AI Agent Development']: 0.0968
28. 22: ['Instagram']: 0.0927
29. 943: ['YouTube Thumbnail']: 0.0886
30. 360: ['AI-Generated Art']: 0.0845
31. 860: ['Twilio API']: 0.0844
32. 147: ['Deep Learning']: 0.0806
33. 621: ['CRM Automation']: 0.0786
34. 539: ['Stable Diffusion']: 0.0783
35. 112: ['n8n']: 0.0775
36. 244: ['AI App Development']: 0.0703
37. 803: ['FastAPI']: 0.0654
38. 271: ['AI Speech-to-Text']: 0.0634
39. 272: ['Colossyan']: 0.0634
40. 273: ['Deepbrain AI']: 0.0634
41. 275: ['Descript']: 0.0634
42. 276: ['Elai']: 0.0634
43. 277: ['InVideo']: 0.0634
44. 278: ['Kaiber']: 0.0634
45. 279: ['Lumen5']: 0.0634
46. 575: ['Runway']: 0.0548
47. 868: ['Character Animation']: 0.0502
48. 451: ['AI Mobile App Development']: 0.0476
49. 452: ['Chatbot']: 0.0476
50. 596: ['QA Testing']: 0.0410
51. 1106: ['Omnichannel Automation']: 0.0372
52. 1123: ['Data Transformation']: 0.0371